SALES FORECASTING USING A STACKED ENSEMBLE MODEL

By

TAKUDZWA TREVOR GANDA

H200148X

HIT400 Capstone project Submitted in Partial Fulfillment of the

Requirements of the degree of

Bachelor of Technology

In

Software Engineering

In the

School of Information Sciences and Technology

Harare Institute of Technology

Zimbabwe

Supervisor

……..

Month/Year

ABSTRACT

Zimbabwean retail businesses face a challenging economic landscape characterized by volatility and a heavy reliance on imports. Intuition-based planning methods prevalent in the sector often lead to significant losses due to overstocking or understocking. I propose a sales forecasting system that uses a stacked ensemble machine learning model to forecast future sales to address this challenge. The system leverages historical sales data to identify trends, seasonality, and patterns. Stacked ensemble models combine the strengths of multiple models, leading to improved generalizability and robustness. By providing accurate sales predictions, this system empowers businesses to anticipate demand, identify growth opportunities, and mitigate risks associated with economic volatility. This research contributes to the field of sales forecasting by demonstrating the effectiveness of stacked ensemble models in a real-world, economically volatile environment, particularly for Zimbabwean retail businesses.

PREFACE

This document acts as an overview of Sales Forecasting using a stacked ensemble model, a comprehensive project designed to predict future sales and future sales trends. The document describes the processes and tasks undertaken to complete the development of the system. The intended audience is the Harare Institute of Technology. The cases described and the research conducted in this document help to describe the problem being solved and the implementation of solutions to solve the problem.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who have contributed to the successful completion of this Sales forecasting project.

First and foremost, I would like to thank God Almighty, whose has helped me from the formulation of this idea through the development phase and the completion of this project.

I am also eternally grateful to my supervisor, Ms. Chibaya, for her priceless guidance, expert advice, and unwavering support throughout the project. Her dedication and profound insights have been instrumental to this project since the very start. She has been the light throughout the whole project showing all the loose threads, points to improve and even mental support to put me in the right attitude to carry out the project.

Furthermore, I would like to express my gratitude to the staff of the Software Engineering department for their assistance, reviews and insights throughout the duration of the project. The department's resources and facilities have been crucial in carrying out the research and development tasks.

I would like to thank my dear parents for fighting for me to have a quality education and also their encouragement and motivation. I would also like to thank the rest of my family and friends for their unwavering support, encouragement, and love. Their constant belief in my abilities has been a driving force in overcoming challenges and achieving success.

Finally, I would like to thank all the participants who volunteered their time and efforts to provide valuable data for this project. Their cooperation and contribution have been instrumental in the successful completion of this project.

This is to certify that HIT 400 Project entitled "**Sales Forecasting using a stacked ensemble model" has** been completed by **Takudzwa Trevor Ganda** (H200148X) for partial fulfilment of the requirements for the award of **Bachelor of Technology** degree in **Software Engineering**. This work is carried out by **him** under my supervision and has not been submitted earlier for the award of any other degree or diploma in any university to the best of my knowledge.

Supervisor Name	**Approved/Not Approved**
Project Supervisor	Project Coordinator
Signature: ……	**Signature: ……**
Date: ……	**Date: …….**

Certificate of Declaration

This is to certify that work entitled "Sales Forecasting using a stacked ensemble model "is *submitted in partial fulfillment of the requirements for the award of Bachelor of Technology (Hons) in Software Engineering, Harare Institute of Technology. It is further certified that no part of research has been submitted to any university for the award of any other degree.*

(Supervisor)	Signature……..	Date…….
(Mentor)	Signature……..	Date……
(Chairman)	Signature……..	Date……..

Table of Contents

ABSTRACT ii
PREFACE iii
ACKNOWLEDGEMENT iv
CHAPTER 1: INTRODUCTION 1
1.1 BACKGROUND 1
1.2 PROBLEM STATEMENT 1
1.3 OBJECTIVES 1
1.4 HYPOTHESIS 2
1.5 JUSTIFICATION 2
1.6 PROPOSED TOOLS 2
1.7 FEASIBILITY STUDY 3
1.7.1 TECHNICAL FEASIBILITY 3
1.7.2 ECONOMIC FEASIBILITY 4
1.7.3 OPERATIONAL FEASIBILITY 4
1.8 PROJECT PLAN 5
1.9 GANNT CHART 6
CHAPTER 2: LITERATURE REVIEW 7
2.1 INTRODUCTION 7
2.2 SALES FORECASTING 7
2.3 ENSEMBLE MODELS 8
2.4 MODEL PERFORMANCE EVALUATION METRICS 8
2.4 RELATED WORK 10
2.3 CONCLUSION 17
CHAPTER 3: ANALYSIS 19
3.1 DESCRIPTION OF SYSTEM 19
3.2 INFORMATION GATHERING TOOLS 19
3.2.1 INTERVIEWS 19
3.2.2 QUESTIONNAIRES 20
3.3 DATA ANALYSIS 20
3.3.1 CONTROL FLOW DIAGRAM OF EXISTING SYSTEM 20
3.4 EVALUATION OF ALTERNATIVE SYSTEMS 21
3.5 FUNCTIONAL ANALYSIS OF PROPOSED SYSTEM 22
3.5.1 FUNCTIONAL REQUIREMENTS 22
3.5.2 NON-FUNCTIONAL REQUIREMENTS 22
3.6 USE CASE DIAGRAM 23

CHAPTER 4: DESIGN 24
4.1 SYSTEM DIAGRAMS 24
4.2 INTERFACE DESIGN 33
CHAPTER 5: IMPLEMENTATION AND TESTING 38
5.1 PSEUDO CODE 38
5.2 SOFTWARE TESTING 43
5.2.1 UNIT TESTING 43
5.2.2 INTEGRATION TESTING 47
5.2.3 SYSTEM TESTING 49
5.2.3 ACCEPTANCE TESTING 52
CHAPTER 6: CONCLUSIONS AND RECOMMENDATIONS 55
6.1 ANALYSIS OF RESULTS 55
6.2 RESULTS AND SUMMARY 61
6.3 RECOMMENDATIONS 62
6.4 FUTURE WORKS 63
REFERENCES 64
APPENDICES 66

TABLE OF FIGURES

Figure 1: Gannt Chart 6
Figure 2: Example of a stacked ensemble model 8
Figure 3: Blended model for airline tickets forecasting 13
Figure 4: Stacked ensemble model for SWB prediction 14
Figure 5: Stacked ensemble model framework for forecasting PM2.5 concentration in Beijing 16
Figure 6: Trend for the forecasting of Rossmann Store Sales 17
Figure 7: Forecasting errors of different models of Rossmann Store Sales 17
Figure 8: Control Flow for existing system 21
Figure 9:Use case for proposed system 24
Figure 10: Control Flow Diagram 24
Figure 11: Process Flow Diagram 25
Figure 12: Context level diagram 26
Figure 13: Data Flow Diagram 26
Figure 14:Activity Diagram 27
Figure 15: Sequence Diagram 28
Figure 16: Class Diagram 29
Figure 17: ER Diagram 30
Figure 18: Package Diagram 31
Figure 19: Architecture Diagram 32
Figure 20: Login Page Design 33
Figure 21: Dashboard Design 33
Figure 22: Upload Sales Data Form 34
Figure 23: Products Analysis Results Visualization Design 34
Figure 24: Choosing Product to Analyze 35
Figure 25: Choosing type of analysis 35
Figure 26: Single Product Analysis Results Visualization 36
Figure 27: Product Forecasting Input Form 36
Figure 28: Product Forecasting Results 37
Figure 29: Unit tests results 47
Figure 30: Integration test results 49
Figure 31: Login Failed 49
Figure 32: On login success 50
Figure 33: Failed single product analysis 50
Figure 34: On successful single product analysis 51
Figure 35: Failed product forecasting 51
Figure 36: On successful product forecasting 52
Figure 37: System responsiveness 52
Figure 38: System Performance 53
Figure 39: Ease of use of system 53
Figure 40: User friendliness of system 54
Figure 41: Exchange Rate Distribution 55
Figure 42: Actual sales vs Predicted sales for Consumables Masvingo Toyota 56
Figure 43: Actual sales vs Predicted sales for Valvoline 57
Figure 44: Actual Sales vs Predicted Sales for Engen Premium 58
Figure 45: Actual Sales vs Predicted Sales for Engen Brake fluid 59

Figure 46: Actual Sales vs Predicted Sales for Diesel 50 60
Figure 47: Actual Sales vs Predicted Sales for Degreaser 750ML 61

LIST OF TABLES

Table 1: Project Plan 5
Table 2: Metrics for evaluating forecasting truck components 10
Table 3: Naive forecasting vs Linear Regression 11
Table 4: Spring onion demand forecasting 12
Table 5: Comparison of accuracy of models 12
Table 6: Blending vs Individual Algorithms 13
Table 7: Stacking vs Individual Algorithms in forecasting CGSS 14
Table 8: Stacking vs Individual Algorithms for steel demand forecasting 15
Table 9: Ensemble model vs Individual Algorithms for forecasting of PM2.5 concentrations 16
Table 10: Evaluation metrics for Consumables Masvingo Toyota 56
Table 11: Evaluation metrics for Valvoline 57
Table 12: Evaluation metrics for Engen Premium 58
Table 13: Evaluation metrics for Engen Brake fluid: 59
Table 14: Evaluation metrics for Diesel 50 60
Table 15: : Evaluation metrics for Degreaser 750ML 61

CHAPTER 1: INTRODUCTION

1.1 BACKGROUND

The Sales Forecasting project aims to develop a robust and accurate system for predicting future sales figures based on historical data, market trends, and economic indicators in Zimbabwe. The project is driven by the need for local Zimbabwe retail businesses to make informed decisions, optimize inventory management, allocate resources effectively and plan sales and marketing strategies in an economically volatile environment. Accurate sales predictions help organizations anticipate demand, identify growth opportunities and effectively handle potential risks. Business planning strategies have mostly been based on intuition and this might lead to huge losses for business without figurative backing. The project will leverage historical sales data from previous periods to identify trends, patterns, and seasonality. The system will also factor in external factors that influence sales such as seasonality, holidays, exchange rates and inflation rates in coming up with an accurate sales forecasting report. Machine learning techniques have recently emerged as powerful tools for forecasting sales accurately [12] thereby improving sales of companies. The use of an ensemble model for prediction allows for the combination of several base estimators to improve the robustness and generalizability over a single estimator [14]. Combining the strengths of multiple machine learning techniques for sales forecasting will allow to produce more accurate results.

1.2 PROBLEM STATEMENT

The Zimbabwean retail industry, a significant contributor to the national economy, suffers from substantial financial losses due to inefficiencies in inventory management and marketing strategies. These inefficiencies largely stem from a reliance on intuition-based decision making in a highly competitive and economically volatile environment. Traditional planning methods fail to account for critical factors like import dependency, fluctuating exchange rates, inflation, and seasonal trends. This results in overstocking (leading to losses of $471.9 billion globally [13]) or understocking (costing $34.1 billion globally [13]), hindering profitability. Local Zimbabwean retail businesses lack the necessary data-driven strategies to optimize inventory management, resource allocation, and marketing campaigns in this challenging landscape.

This highlights the critical need for a robust and accurate sales forecasting system specifically tailored to the Zimbabwean retail sector. Such a system would empower businesses to make informed decisions, mitigate risks associated with economic volatility, and achieve sustainable growth.

1.3 OBJECTIVES

1. To develop a web-based sales forecasting system that predicts future sales by utilizing a machine learning model trained on historical sales data.
2. To develop a web-based sales forecasting system that accurately predicts sales trends over a specific timeframe using historical sales data.
3. To develop a web-based sales forecasting system to that analyzes sales data

1.4 HYPOTHESIS

A stacked ensemble model will improve the accuracy and reliability of sales forecasts compared to the traditional instinct based forecasting methods. The stacked ensemble model is expected to outperform single forecasting models by achieving lower MAE, RMSE and MFE values and a higher R-squared value by at least 5%. The model will be trained on 80% of the data and evaluated on the remaining 20% to ensure generalizability. The stacked ensemble model will comprise of a mix of algorithms like Gradient Boosting Machines (GBM), time series algorithms and decision trees to enable the model to capture a wider range of patterns and relationships in the data. The stacked ensemble model will then be integrated to a web application to allow the business to use the model for prediction on future sales values and trends. While the stacked ensemble approach might require more computational resources, the accuracy gains outweigh this cost for sales forecasting in the retail industry, where capturing seasonal patterns and having an insight to future sales is very important.

1.5 JUSTIFICATION

Instinct based forecasting has its shortcomings as it cannot capture the underlying complex relationships between features in the sales data. Each individual algorithm in the ensemble model will brings its own strengths and capabilities in the forecasting of sales. The combination of these algorithms in a stacked ensemble model leads to the leveraging of the complementary strengths of each algorithm to improve the overall accuracy of the sales forecasts. The ensemble model can capture different aspects of the sales data such as seasonality and complex relationships between other features and sales leading to more accurate predictions. The chosen evaluation metrics R-squared, Mean Absolute Error, Root Mean Squared Error and the Mean Forecasting Error will provide a deeper insight in the performance of the stacked ensemble model. R-squared measures the variance to how much the model can explain the sales data. The Mean Absolute Error measures average absolute difference between the predicted and the actual sales. The Root Mean Squared Error penalizes larger errors in forecasting more than the Mean Absolute Error through squaring.

1.6 PROPOSED TOOLS

- **Python**
  This is a popular and powerful programming language used by many developers to build and develop various projects that involve machine learning and deep learning. It has a wide variety of extensive libraries that can be used to develop a sales forecasting system.

- **Google Colab**
  This is a cloud-based IDE that simplifies sales forecasting tasks. It supports importing and preprocessing sales data, visualizing patterns, and implementing forecasting models. It has GPU/TPU support which helps accelerate training of complex models.

- **Django**
  It is a powerful Python web framework suitable for developing a web-based sales forecasting system. It allows the integration with machine learning models for accurate predictions, user authentication and permission management, and frontend development capabilities for intuitive visualization of sales data and forecasts.

- **MySQL**
  It is a popular open-source relational database management system (RDBMS) used for storage of data. It will be used to store user credentials for logging to the system, storing sales data excel files and storing reports for analysis and forecasts.

- **Machine Learning Libraries**
  These libraries like scikit-learn (Python), Visualization libraries (matplotlib, plotly) and Pickle (serialization of the stacked ensemble model for integration to the web application) provide the algorithms and tools to build and train forecasting models.

1.7 FEASIBILITY STUDY

The feasibility study is aimed at assessing the feasibility of developing the sales forecasting system using a stacked ensemble model. This study aims to identify potential risks, challenges and opportunities associated with the development of the sales forecasting system.

1.7.1 TECHNICAL FEASIBILITY

Data Availability and Quality

The availability of historical sales data is crucial for training and evaluating the ensemble model. It is important to ensure that the data is comprehensive, accurate, and representative of the sales patterns and dynamics. Adequate data preprocessing and cleaning techniques may be required to handle missing values, outliers, or other data quality issues.

Ensemble Framework Design

Designing the ensemble framework involves determining the combination strategy for aggregating the predictions from the individual models. Stacking will be used to combine the strengths of the algorithms which also helps counter overfitting.

Computational Resources

The computational resources required to train and run the sales forecasting model should be considered. Complex models or large datasets may demand significant computational power, memory, and storage. Access to suitable hardware, such as powerful CPUs or GPUs, and sufficient storage capabilities is essential and hence the use of Google Colab.

Performance Evaluation

Evaluating the performance of the sales forecasting model is crucial to assess its effectiveness for sales forecasting. Appropriate evaluation metrics, such as R-Squared, Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Mean Forecasting Error (MFE) can be used to compare the ensemble's performance against individual models or other forecasting approaches.

Integration with Visualization Tools

The use of visualization libraries like matplotlib, plotly and seaborn will allow for the presentation of the graphical representation of the analytics and forecast results. The use of the Django framework will also ensure the presentation of these results in a more interactive way for the user.

1.7.2 ECONOMIC FEASIBILITY

Cost of Implementation

Assessing the costs associated with implementing the sales forecasting model is crucial. This includes the expenses related to acquiring or accessing the necessary data, employing skilled data scientists or analysts, investing in computational resources, and potentially purchasing or licensing software tools or libraries. It is important to evaluate whether the benefits gained from the sales forecasting model outweigh the costs of its implementation.

Return on Investment (ROI)

Evaluating the potential return on investment is an important aspect of economic feasibility. Organizations should consider the potential financial gains that can be achieved by improving the accuracy of sales forecasts. For example, a more accurate sales forecast can lead to better inventory management, improved production planning, optimized resource allocation, and enhanced decision-making, which can ultimately result in cost savings or increased revenue.

Competitive Advantage

Consider the potential competitive advantage that can be gained by implementing a sales forecasting model for sales forecasting. Accurate forecasts can enable better decision-making, enhanced customer satisfaction, and improved market responsiveness. This can help organizations gain a competitive edge, increase market share, and generate economic benefits through increased sales and customer loyalty.

Cost of Errors

Inaccurate sales forecasts can lead to missed sales opportunities, excess inventory, stockouts, inefficient resource allocation, and suboptimal decision-making. Quantifying the potential costs of these errors can help assess the economic feasibility of implementing a sales forecasting model that aims to improve forecast accuracy.

Sensitivity Analysis

Conducting sensitivity analysis can help assess the economic feasibility under different scenarios and assumptions. Evaluating how variations in key parameters, such as market conditions, or model accuracy, impact the economic outcomes will be essential. This analysis can provide insights into the robustness of the economic feasibility of the sales forecasting model.

1.7.3 OPERATIONAL FEASIBILITY

Data Availability and Accessibility

Ensuring the availability and accessibility of historical sales data is essential for sales forecasting. The historical data should be accessible in a timely manner, and there should be a process in place to collect and update the data regularly. If the data is fragmented or difficult to obtain, it may hinder the operational feasibility of the forecasting approach.

Scalability and Performance

Assessing the scalability and performance of the sales forecasting model is crucial for operational feasibility. The model should be able to handle larger datasets and increasing forecasting demands without significant degradation in performance. If the model's computational requirements or processing time increase disproportionately with data size, it may affect the operational feasibility.

Integration with Existing Systems

Integrating the sales forecasting model into existing systems and workflows is an important consideration. If the integration requires substantial modifications or disruptions to existing processes, it may pose operational challenges. Seamless integration with other systems, such as CRM or ERP systems, can enhance the operational feasibility by streamlining data flows and automating the forecasting process.

Interpretability

The interpretability of the sales forecasting model's forecasts is important for operational feasibility. Sales managers may require insights into the factors driving the forecasts and the reasoning behind them. Ensuring that the sales forecasting model provides interpretable and explainable outputs can enhance its operational feasibility.

1.8 PROJECT PLAN

Task	Start date	End date	Duration
Requirements Gathering	Oct 1, 2023	Oct 7, 2023	7 days
Literature Review	**Oct 8, 2023**	**Oct 21, 2023**	**14 days**
System Research	Oct 22, 2023	Nov 6, 2023	16 days
Data Preparation and pre-processing	Nov 7, 2023	Nov 27, 2023	21 days
Model Development	Nov 28, 2023	Mar 1, 2024	95 days
Model testing	Feb 22, 2024	Mar 6, 2024	14 days
Web Application Development	Mar 7, 2024	Mar 20, 2024	14 days
Integrating model to web application	Mar 21, 2024	Apr 3, 2024	14 days
Testing system and fixing bugs	**Apr 4, 2024**	**Apr 30, 2024**	**26 days**

Table 1: Project Plan

1.9 GANNT CHART

ID	Task Name	Duration
1	Requirements gathering	1 w. 0 d.
2	Literature Review	2 w. 0 d.
3	System Research	2 w. 2 d.
4	Data Preparation and preprocessing	3 w. 0 d.
5	Model Developement	13 w. 4 d.
6	Model Testing	2 w. 0 d.
7	Web Application Development	2 w. 0 d.
8	Intergrating model to web application	2 w. 0 d.
9	Testing system and fixing bugs	3 w. 6 d.

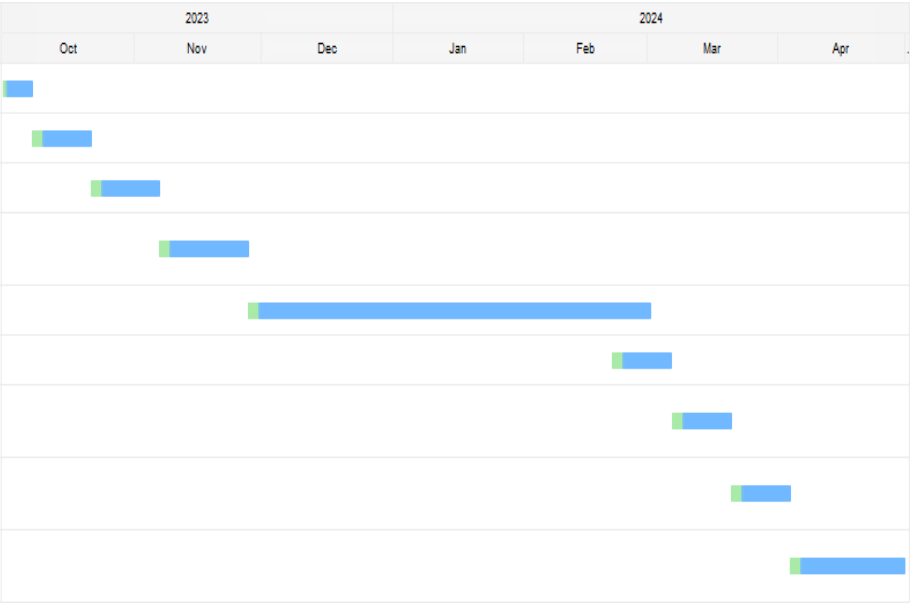


Figure 1: Gannt Chart

CHAPTER 2: LITERATURE REVIEW

2.1 INTRODUCTION

Sales forecasting is a critical component of decision-making for businesses operating in the retail industry. Accurate and timely sales predictions enable retailers to optimize inventory management, plan marketing strategies, and allocate resources effectively. Traditionally, forecasting methods relied on historical sales data and statistical techniques. However, with the advent of machine learning, retailers now have access to advanced predictive models that can analyze vast amounts of data and uncover complex patterns to improve forecasting accuracy.

This literature review aims to explore the application of machine learning techniques in sales forecasting across different domains and industries. The retail industry in Zimbabwe is characterized by its volatility, influenced by many various factors such as seasonality, consumer behavior, unemployment rates, fluctuating exchange rates and inflation rates. The current employed methods for forecasting sales therefore will render less effective considering the various factors that may affect sales for companies. Machine learning algorithms, on the other hand, factors in these factors in coming up with a more accurate and robust sales forecast report.

By reviewing the existing literature on sales forecasting using machine learning, this study aims to provide insights into the current state of research, identify gaps in knowledge, and highlight potential avenues for future research. The findings will not only contribute to the academic understanding of sales forecasting but also provide practical guidance for retailers seeking to leverage machine learning to enhance their forecasting capabilities and gain a competitive edge in the dynamic retail landscape.

2.2 SALES FORECASTING

Sales forecasting is the process of estimating future sales revenue for a given time period [16]. A sales forecast is a prediction based on past sales performance and an analysis of expected market conditions [19]. In the realm of retail supply chain management, accurate forecasting is paramount for informed decision making, as it directly impacts business operations and profitability [18]. The need for sales forecasting spans centuries when merchants used manual methods like intuitive judgment and trend analysis to predict future sales. These methods helped in anticipating the demand of products and also help in deciding the supply of products. [20] clearly laid out that matching the demand of products with the correct amount of supply mitigates the risk of sales losses due to lack of inventory or extra expenses due to overstocking. Sales forecasting is very vital to the efficient operation of firms as they can aid managers in decision making processes and inventory management [19]. Sales forecasting uses patterns from historical data to predict future sales which allows for informed courses of action such as allocating or diverting existing inventory, or increasing or decreasing future production [20]. The idea of businesses to carry out sales forecasting is to help them have control over their business and also give them the power to determine the trajectory of the businesses.

[21] describes some methods used in the forecasting of sales including extrapolation methods (which use only historical data on the series of data), rule-based forecasting and causal approaches to sales forecasting. Machine learning in sales forecasting leverages data to uncover patterns and relationships in data that may not be apparent to human analysis [22]. Manual forecasting methods mentioned above like intuitive judgment could lead to drastic errors leading to poor management of the organization and it is also time consuming as mentioned in [17]. The ability of machine learning algorithms to analyze data from various

sources such as market trends, promotions, holidays, economic indicators and consumer behavior would significantly optimize sales forecasts and marketing strategies [23].

2.3 ENSEMBLE MODELS

Ensemble techniques combine different algorithms into an individual method, where each algorithm could be more sensitive under varying conditions and stacking is an ensemble method in which various machine learning algorithms predict and act as input to the next level predictor [22]. An ensemble model is one in which numerous base models are constructed to address the same problem, with each model learning the dataset's feature attributes and making a prediction [4]. There's always a degree of uncertainty in sales forecasting and no algorithm can work perfectly but using a combination of methods and regularly adjusting forecasts, organizations can make more accurate sales predictions [24]. Ensemble forecasting is utilized in the retail sector to overcome uncertainties in data, parameters and models and to decrease the risks of relying on a single model [23]. The combination of multiple different algorithms helps to combine the strengths of the different algorithms to come up with a more robust and accurate forecast. Ensemble methods will reduce the impact of individual weaknesses of the algorithms on the final forecasted values. The overall structure of ensemble learning is to produce a group of individual learners and then combine them with some strategies and because the individual learner needs to meet diversity and accuracy, its decision-making is often more effective and its algorithm performance is better [3]. Stacking is used for model fusion [3]. Bagging, boosting and gradient boosting are the other types ensemble models. Bagging combines multiple instances of the same model trained on different subsets of the data. Boosting involves combing multiple models where each subsequent model focuses on the errors of the previous model. Gradient Boosting combines multiple decision trees to optimize the prediction. Popular ensemble models include Random Forests, LightGBM and CatBoost.

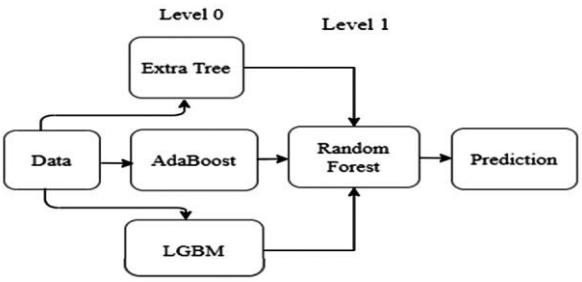


Figure 2: Example of a stacked ensemble model

2.4 MODEL PERFORMANCE EVALUATION METRICS

These are the metrics used to evaluate the accuracy of forecasting models.

2.4.1 Coefficient of Determination (R-Squared, R^2)

This is used to evaluate the efficacy of a linear regression model. An input - independent variable can be used to forecast the amount of output - dependent attribute change. Based on the percentage of total deviation of the model's reported results, it is used to determine how effectively the model's observed outcomes are reproduced. Its value ranges from 0 to 1. The closer the numbers are to 1, the higher the accuracy of the model. It shows how much of the actual values can be predicted by the model. It is calculated as follows:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

, where SS_{res} is called a residual sum of squares and is given by: -

$$SS_{res} = \sum_i (y_i - a_i)^2$$

, where y_i represents the predicted values and a_i represents the actual values given in the dataset, and SS_{tot} is called the total sum of squares given by: -

$$SS_{tot} = \sum_i (a_i - \bar{a})^2$$

, where a is the mean of the actual data and is given by: -

$$\bar{a} = \frac{1}{n}\sum_{i=1}^{n} a_i$$

…… [5]

2.4.2 Mean Absolute Error (MAE)

It is a performance metric used for regression models. This is calculated by taking the average of the absolute difference between predicted values of the model and actual values. Every predictive error is the difference between the predicted value and the actual value. The accuracy of the model is higher when the MAE values are close to zero i.e. the lower the MAE value, the more accurate the model is. MAE gives equal importance to all the individual errors which makes it an appropriate measure for calculating average model performance [10]. MAE is calculated as follows:

$$MAE(y, x) = \left(\frac{1}{n_{samples}}\right) \sum_{i=0}^{n_{samples}-1} |y_i - x_i|$$

…….[5], where y_i represent the actual values, x_i represent the predicted values and $n_{samples}$ represent the feature variables [10].

2.4.3 Root Mean Squared Error (RMSE)

This is the square root of the difference between the predicted values of the model and the actual values. The value of RMSE indicates the fit of the models on a particular dataset. Values close to zero implies a better fit and thereby reducing the impact of outliers. RMSE increases as the variance associated with the frequency distribution of error magnitudes increases. The lower the RMSE value, the higher the accuracy of the model. RMSE is calculated as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

…… [5], where n represents the feature variables, y_i represents actual values and $\hat{y}_i$ represents the predicted or forecasted value [10].

2.4 RELATED WORK

Sales forecasting has been an area of interest ever since the rise of machine learning, big data analytics and artificial intelligence. The need to assist organizations to have figurative backing in making business decisions and inventory management has led to more research on the need for sales forecasting using machine learning.

2.4.1 Sales Forecasting of Truck Components using Neural networks

[10] highlighted that quotations for trucks were made based on the customer's choice and different customers would choose different parts based on factors such as customer preference and region. Therefore, because of such diversity, it is difficult to identify which components the customers will select and maintain the stock level [10]. The researcher then suggested a Sales Forecasting of Volvo truck components so as help the company maintain sufficient stock levels of truck components and optimize truck sales processes. This enables the organization to optimize the truck sales process by addressing issues related to stock level and delivery time [10]. The objective of the research carried out was to investigate appropriate machine learning algorithms for forecasting the sales of truck components and then conduct experiments to forecast sales with the selected machine learning algorithms and to evaluate the performances of the models using performance metrics (MAE and RMSE). In order to determine the best algorithm to use, Long Short-Term Memory (LSTM), Multilayer Perception neural network and the Recurrent Neural Network were studied so as to determine the suitability of those algorithms. The dataset was obtained from Digital Transformation department of Volvo Trucks group which consisted of order management information of variants such as engine brake, front rims and instrument cluster display of a Volvo truck which was collected weekly from 2014 to 2019 and also consisted of 20 components. Two evaluation metrics were used for the project which ere MAE and RMSE.

Algorithms	Mean Absolute Error	Root Mean Square Error
Multilayer Perceptron	0.12944	0.22514
Recurrent Neural Network	0.0570	0.1507
Long Short-Term Memory	0.03974	0.10559

Table 2: Metrics for evaluating forecasting truck components

The LSTM model was eventually chosen for the project as it had the least Mean Absolute Error and the least Root Mean Square.

2.4.2 A study on the effect of sales forecasting on the enterprises

[2] appreciated the need for businesses to carry out sales forecasting to help them in making smart and strategic sales decisions. They also pointed out the use of sales predictions by company executives to plan for enterprise expansion and evaluation of future company performance. Sales forecasting is primarily used by businesses to evaluate two things. First, to determine the present consumer needs for the product and service. Second, to forecast future need for a firm's products or services. Future sales plans help with higher production use, planning, transportation, and stock control. This, in turn, leads to increased customer satisfaction and a reduction in production costs [2]. The researchers highlighted that the major reason that an industry creates a forecast is to balance marketing and promotional strategy against supply capacity organizing. The research was carried out as an attempt at the prediction of sales of a retail business through the use of Linear Regression and Naïve Forecasting. A retail dataset of the global superstore was used.

Linear Regression and Naïve Forecasting were both implemented on the retail dataset in order to determine the best approach to use so as to come up with more accurate results. Linear Regression is used to generate insights on consumer behavior, enterprise understanding and factors impacting revenue and profits, and is also employed to interpret trends and make predictions [2]. Naïve forecasting uses the previous period's sales data to estimate the subsequent without making any presumptions or altering the variables and is primarily based on historical sales or other variables such as cashflows and revenues [2]. The Mean Absolute Error and the Root Mean Squared Error were the evaluation metrics used to determine the performance of the models. Below shows the results of the forecasting of sales using Linear Regression and Naïve forecasting: -

Name	RMSE	MAE
Linear Regression	625.4	272.7
Naive Forecasting	874.8	337.1

Table 3: Naive forecasting vs Linear Regression

Linear Regression approach, performs better based the results above. Naive Forecasting method had the most significant RMSE and MAE of 874.8, 337.1 respectively and the method with least RMSE and MAE value was Linear regression having 625.4, 272.7 respectively. Hence, the conclusion according to the obtained results was that the linear regression is the most reliable predictor for the whole set because it had the lowest RMSE and MAE values.

2.4.3 Spring onion seed demand forecasting using a hybrid Holt-Winters and support vector machine model

[8] presents a hybrid forecasting model combining the Holt-Winters and the Support Vector Machine algorithms for the prediction of the demand of spring onion seeds. The main objective of this paper was to compare the performances of statistical-based methods (ARIMA and Holt-Winters), advanced machine learning methods (ANN, random forest and SVM) and the hybrid model in forecasting the demand for spring onion seeds based on dynamic factors including historical sales, seed inventory, spring onion price and weather data. The hybrid model made use of different dynamic factors, including historical seed sales, seed inventory, spring onion crop market price and weather data, as inputs to forecast spring onion seed demand as they have effects on the demand of spring onion seeds. Seed inventory, spring onion crop market price and historical seed sales were the most important dynamic factors, among which seed inventory had short-term influence while other two had mid-term influence on seed demand forecasting. The absolute minimum temperature was the only factor having long-term influence [8]. [8] suggested that accurate demand forecasting is crucial for inventory control and reduction of operational costs and ensure market share growth for seed companies, as well as ensuring sufficient seed supply for customers and growers. The hybrid model of the Holt-Winters and Support Vector Machine combined the strengths of the two algorithms hence producing more accurate results. In the hybrid forecasting model, the monthly sales training data was inputted into the Holt-Winters model and the results reflected the linearity and seasonal trends in the historical data and then prepared for the SVM model as variables. The evaluation metrics used included Mean Absolute Error (MAE), Mean Squared Error (MSE) and the Mean Absolute Percentage Error (MAPE). Below shows the comparison of demand forecasting performance on the testing set among different models used: -

	Error type	ARIMA	mul-HW	ANN	RF	Proposed hybrid model
Variety A [a]	MAPE	67.70%	36.90%	30.27%	32.68%	17.65%
	MAE	289.962	222.280	117.157	177.280	83.2094
	MSE	131271	117031	19305.2	64329.7	11524.3
Variety B	MAPE	446.00%	221.80%	120.92%	104.74%	49.83%
	MAE	98.6893	104.778	39.7028	65.9704	16.7287
	MSE	15583.4	20054.7	1985.59	14737.5	447.259
Variety B (Sep.—Dec. 2016) [b]	MAPE	39.84%	42.28%	20.32%	18.14%	17.88%
	MAE	114.991	120.396	43.7411	77.3880	18.4626
	MSE	29513.2	24621.7	2343.33	18318.8	535.098
Variety C	MAPE	55.30%	38.60%	35.82%	26.22%	13.35%
	MAE	217.845	184.033	136.930	131.409	64.2900
	MSE	51321.1	48359.8	19578.6	28913.1	6062.61

Table 4: Spring onion demand forecasting

The proposed hybrid model had very low error measurement values than the mul-HW model, indicating the forecasting accuracy of the proposed hybrid model is promising for agricultural production planning and supply chain management in terms of MAPE. Compared to the three advanced machine learning models (ANN, RF, SVM), the proposed hybrid SVM model had the best forecasting performance in terms of all three error measurements

2.4.4 Sales Prediction Analysis

[1] carried out a research project for sales prediction analysis using machine learning to improve on the traditional forecasting methods which are more prone to underestimating and overestimating which leads to loss in profits for companies. The aim of the research was to come up with a more accurate sales forecast using Machine Learning techniques which would help analyze big data. [1] iterated the need for organizations to do sales forecasting as it enables the effective management of workforce, budget and resources which helps the determination of how much products should be manufactured, how much revenue can be expected and requirements of employees, investments and equipment. For the research, a Black Friday dataset was used and Linear Regression, XGBoost, Random Forest, Gradient Boosting and extremely Randomized Trees (Extra Trees) Classifiers algorithms were implemented on the dataset to analyze their performance to suggest a suitable predictive algorithm.

ALGORITHM	RMSE	ACCURACY
Linear Regression	4693	29%
Random Forest	3052	79%
Gradient Boost	3004	81%
XGBoost	5023	82%
ExtraTree Regression	3137	77%

Table 5: Comparison of accuracy of models

The final conclusion was that XGBoost and Gradient Boost were considered to have best fit and better performance as compared to the other algorithms as they had the higher accuracy percentages.

2.4.5 A Blending Model Combined DNN and LightGBM for Forecasting the Sales of Airline Tickets

[3] carried out research to forecast sales of airline tickets in time series affected by factors such as different flights, date features and historical sales. Deep Neural Network (DNN) and Light Gradient Boosting Machine (LightGBM) were combined in a blended model to forecast the sales of airline tickets. According to [3], domestic and foreign research shows that sales forecasting accuracy increases by 10%, revenue can increase by 0.5-3.0%. The researchers understood the need to use ensemble approach in prediction requirements to improve the effect of machine learning algorithms. In the experiment, the scheme of automatic adjustment of model parameters is adopted, and more suitable Super-parameters are found which improves the effect of the model. The data used was obtained from the order database of an airline company in China over a period from 2013 to 2018. Blending method which involves using the output of each learner as the input of the new model was used and the weight of each learner is obtained by training. Three models were implemented on the data, DNN, LightGBM and a blend of the DNN and LightGBM. Below shows the diagram for the blending model used for the prediction of the airline tickets.

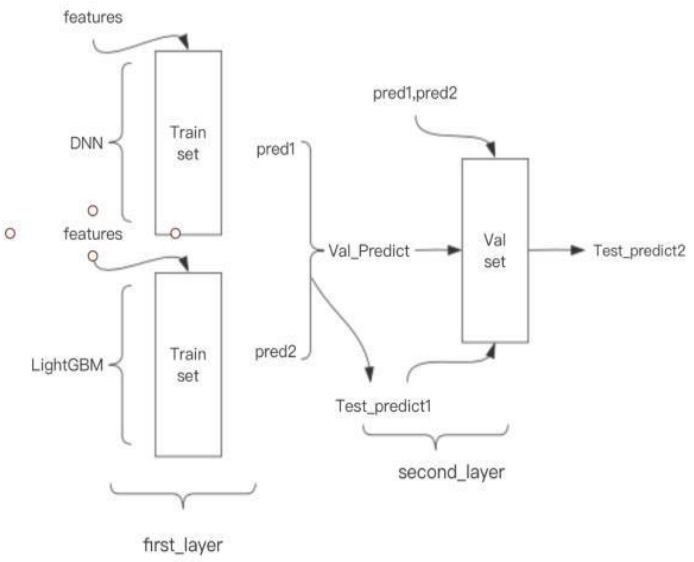


Figure 3: Blended model for airline tickets forecasting

To verify the effectiveness of the developed models, the models were used on the ticket sales of 16590 flight sales data sets and the data was split with 80% of the data was for training, 10% for verification and 10% for tests. The evaluation metrics used are Mean Absolute Error (MAE), Mean Squared Error (MSE), R-squared (R2) and Explained Variance Score (EVS). Below shows the results of the forecasts using the different models.

Model	MAE	MSE	R2	EVS
LightGBM	6.36	110.16	0.78	0.79
DNN	6.58	115.08	0.72	0.72
Blending	6.16	100.91	0.77	0.79

Table 6: Blending vs Individual Algorithms

From the above table, the blended model performed better as it had the lower MAE and MSE values.

2.4.6 Stacking Model for Optimizing Subjective Well-Being Predictions Based on the CGSS Database

[27] used a stacked model based on ANN, XGBoost, Logistic Regression, Catboost and LightGBM to predict the Subjective Well-Being (SWB) of Chinese residents using the Chinese General Social Survey (CGSS) datasets from 2011-2015 and 2017. The feature importance of tree models was used to reveal change in the important factors affecting SWB. The below diagram shows the model used in the prediction of SWB:

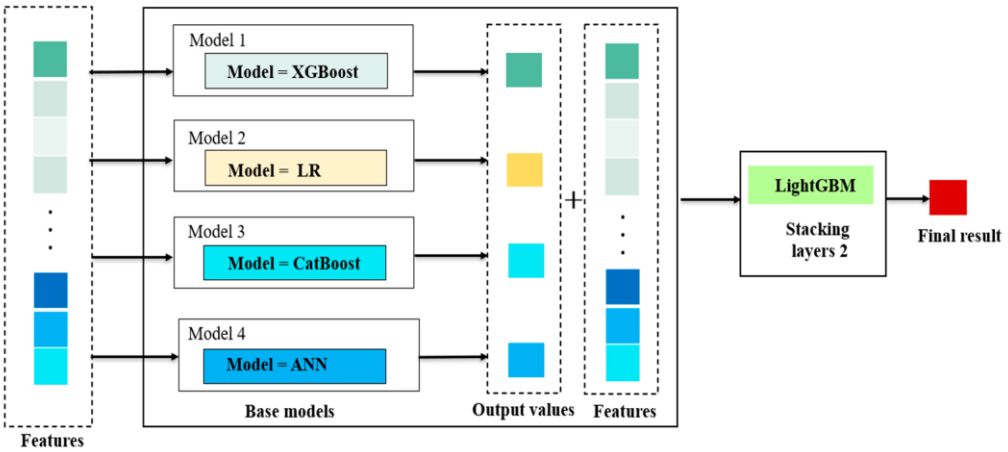


Figure 4: Stacked ensemble model for SWB prediction

The features that affect residents' SWB are put into the Logistic Regression, Catboost, ANN and XGBoost algorithms then the predictions of these models together with the original features are made inputs to the LightGBM model. The dataset was split using the 8:2 ratio for train set and test set. The evaluation indexes were Accuracy, precision, F1 score, Area Under ROC Curve (AUC) and Kolmogorov-Smirnov. The below diagram shows the results of the forecasting using different algorithms and the AUC metric.

Model/Year	2011	2013	2015	2017
LR	0.7458	0.7542	0.8163	0.7397
XGBoost	0.8561	0.8608	0.8971	0.8584
CatBoost	0.8478	0.8341	0.8754	0.8289
LightGBM	0.8587	0.8441	0.8845	0.8645
ANN	0.8186	0.8413	0.8482	0.8446
CNN	0.8263	0.8338	0.8989	0.8438
Stacking model (Meta model: LightGBM)	0.8675	0.8957	0.9154	0.8913

Table 7: Stacking vs Individual Algorithms in forecasting CGSS

The stacked model had the largest values of AUC which indicates that it was the most accurate model for forecasting SWB than the other singular algorithms.

2.4.7 An Approach for Demand Forecasting in Steel Industries Using Ensemble Learning

[4] noted that the steel industry in Bangladesh was a fast-growing industry with industries producing large amounts of steel for the local and international markets.[4] also stressed out that producing large volumes of products without proper forecasting could lead to various problems, with this phenomenon supported by [20]. This led to [4] to introduce a framework for forecasting demand that involved feature selection, cross validation and regression ensemble framework. Bagging (random forest regression (RFR)), boosting (gradient boosting regression (GBR) and extreme gradient boosting regression (XGBR) and stacking (STACK) were employed as ensemble models. Support vector regression (SVR), extreme learning machine (ELM), and multilayer perceptron neural network (MLP) were adopted as reference models. To maximize

the determination coefficient (R2) value and reduce the root mean square error (RMSE), hyperparameters were set using the grid search method. A steel industry dataset was used. The algorithms used for the different stacks were $STACK_1$ (ELM + GBR + XGBR-SVR) and $STACK_2$ (ELM + GBR + XGBR-LASSO) models. The evaluation metrics used were RMSE, MAPE, R^2 and MAE. Below shows the results of the demand forecasting:

Models	R_2	MAE	RMSE	MAPE
SVR	0.931	0.202	0.246	0.902
ELM	0.942	0.183	0.226	0.880
MLP	0.961	0.149	0.186	0.569
RFR	0.966	0.133	0.172	0.517
GBR	0.969	0.125	0.164	0.401
XGBR	0.974	0.120	0.151	0.619
$STACK_2$	**0.977**	**0.112**	**0.143**	**0.463**
$STACK_1$	**0.977**	**0.112**	**0.144**	**0.445**

Table 8: Stacking vs Individual Algorithms for steel demand forecasting

The conclusion was that the stacked models produced the best results based on the evaluation metrics highlighted in bold.

2.4.8 Development of a stacked ensemble model for forecasting and analyzing daily average PM2.5 concentrations in Beijing, China

A stacked ensemble model was developed for the forecasting and analyzing of daily concentrations of fine particulate matter in Beijing, China. Feature extraction procedures like transformation and combination were conducted to identify significant features based on exploratory data analysis. Single models including LASSO, Adaboost, XGBoost and multi-layer perceptron optimized by the genetic algorithm (GA-MLP) were established in the level 0 space and were then integrated by support vector regression (SVR) in the level 1 space via stacked generalization. Feature importance analysis was undertaken and nitrogen dioxide (NO_2) and carbon monoxide (CO) concentrations from Zhangjiakou city were regarded as the most important elements of pollution factors for forecasting $PM_{2.5}$ concentrations. Local extreme wind speeds and maximum wind speeds were considered to extend most effects of meteorological factors to the cross-regional transportation of contaminants [14]. Pollutants found in the cities of Zhangjiakou and Chengde had a stronger impact on air quality in Beijing than other surrounding factors [14]. The meteorological data used ranged from December 2013 to June 2017 and it was split with 80% train data and 20% test. A higher level (level 1) meta-model is trained on out-of-fold predictions of base models (level 0), generating a final prediction of the overall ensemble model. Below is the diagram showing the framework of the stacked ensemble model used:

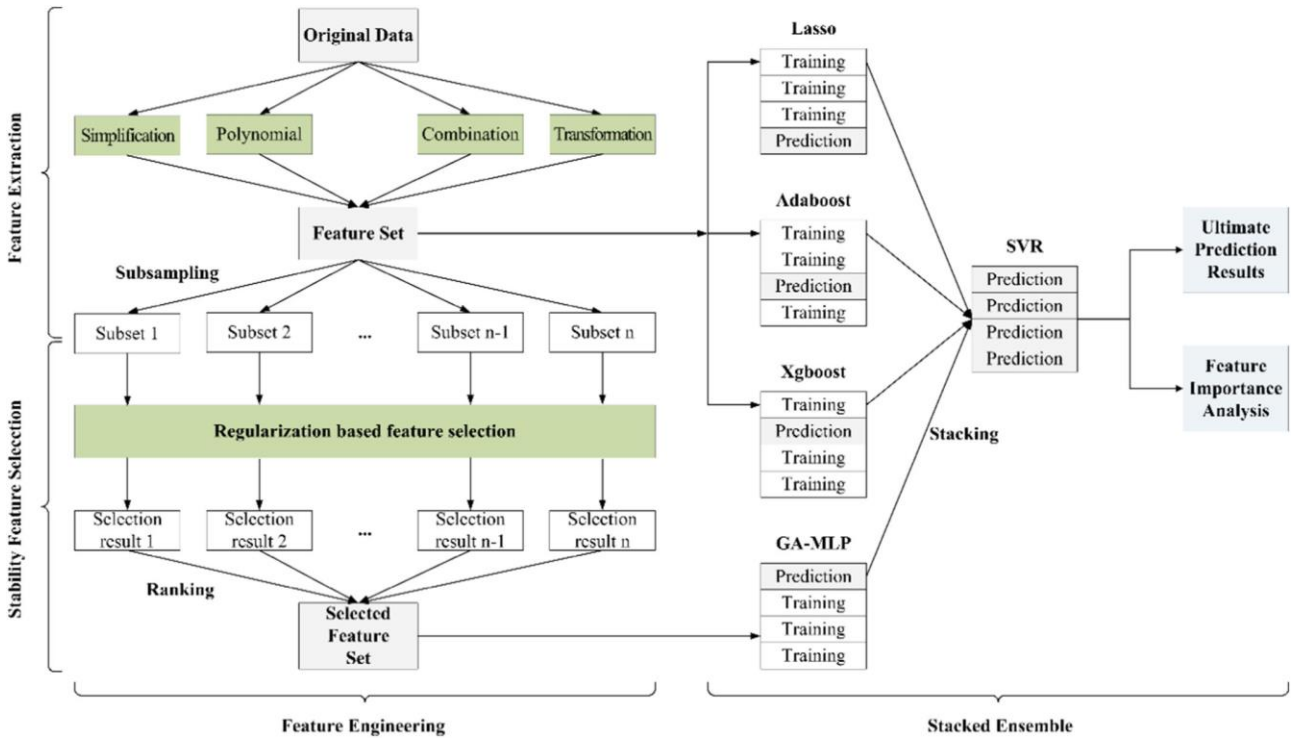


Figure 5: Stacked ensemble model framework for forecasting PM2.5 concentration in Beijing

Below shows the results of the forecasting of $PM_{2.5}$ concentrations:

	LASSO	Adaboost	XGBoost	GA-MLP	SVR	Ensemble
Training set R_2	0.87	0.91	0.90	0.88	0.91	0.90
IA	0.96	0.97	0.97	0.96	0.97	0.97
Bias (μg/m³)	4.17	1.98	11.54	3.22	0.47	2.07
MAE (μg/m³)	15.81	13.20	12.21	15.80	13.00	15.16
MAPE (%)	28.67	19.07	12.75	28.50	25.96	30.06
RMSE (μg/m³)	22.68	19.21	20.44	22.04	19.29	20.72
Test set R_2	0.84	0.78	0.82	0.87	0.85	0.90
IA	0.95	0.92	0.94	0.96	0.95	0.97
Bias (μg/m³)	2.89	8.92	11.73	2.01	10.18	1.73
MAE (μg/m³)	16.67	21.58	16.96	17.81	18.92	14.43
MAPE (%)	29.30	38.23	27.07	32.84	28.64	27.01
RMSE (μg/m³)	28.37	34.00	26.81	25.45	27.95	23.69

Table 9: Ensemble model vs Individual Algorithms for forecasting of PM2.5 concentrations

The results show that the ensemble model generally performs 3.45% better in determining R^2 values, 13.93% better in determining bias values, 18.98% better in determining MAE values and 6.91% better in determining RMSE values than the best single model.

2.4.9 Machine-Learning Models for Sales Time Series Forecasting

This paper explored the study of the usage of machine learning models for sales time series forecasting. A stacking approach for building regression ensemble of single models was studied. According to [6], time series models such as Holt-Winters, SARIMA, SARIMAX and ARIMA are problematic to use in sales prediction due to the need to have historical data for a long period of time in order to capture seasonality, complicated seasonality of sale (intra-day, intra-week, annual), lots of outliers and missing data in sales data and also the need to take into account exogeneous factors which have impact on sales. Sales prediction is rather a regression problem than a time series problem [6]. The analysis was carried out using historical

store sales data from Rossmann Store Sales Kaggle competition. Stacking was used for building ensemble of predictive models. The historical data was split on the training and validation sets using time splitting such that the training data will lie in the first time period and the validation set in the next. The below figure shows the time series forecasting on the validation sets obtained using different diagrams: -

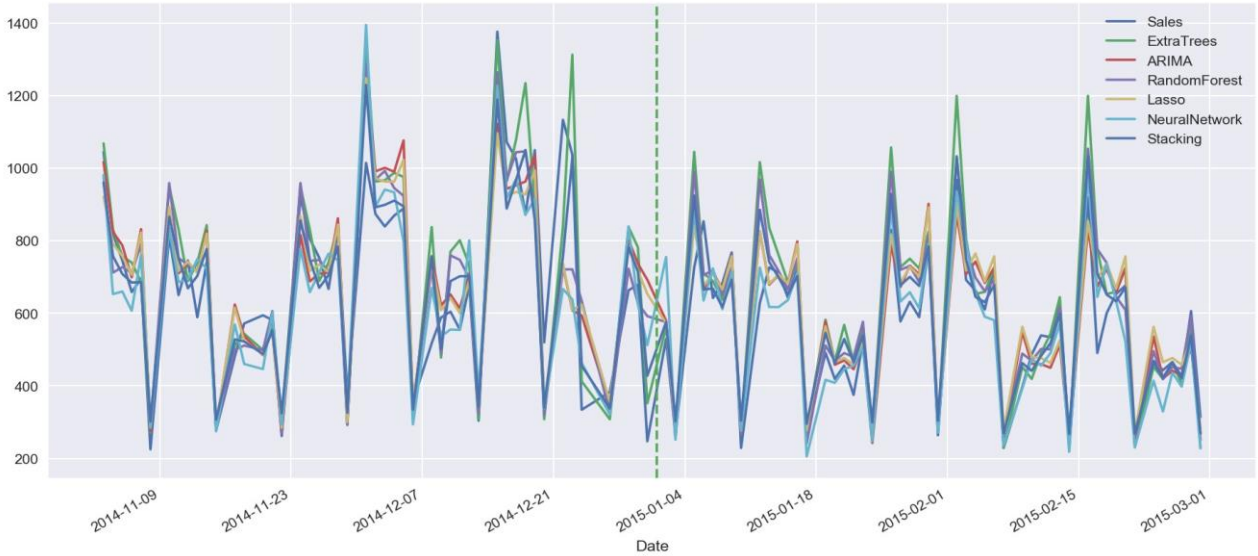


Figure 6: Trend for the forecasting of Rossmann Store Sales

Below shows the forecasting error table for the various models used on the dataset: -

Model	Validation error	Out-of-sample error
ExtraTree	14.6%	13.9%
ARIMA	13.8%	11.4%
RandomForest	13.6%	11.9%
Lasso	13.4%	11.5%
Neural Network	13.6%	11.3%
Stacking	12.6%	10.2%

Figure 7: Forecasting errors of different models of Rossmann Store Sales

From the table, stacking produced the least percentage of forecasting errors as compared to the other models.

2.3 CONCLUSION

Researches on the effect of Machine Learning on sales forecasting have proved its use to be more effective. Traditional sales forecasting techniques have been proven to be less effective in the current age and less accurate as compared to the use of Machine Learning models. Even though the use of Machine Learning models increases the accuracy of sales forecasting predictions, there are more accurate algorithms and more effective techniques that may be implemented on data to come up with more accurate models. From

literature review conducted, it has been made clear that the implementation of an individual algorithm on data will produce less accurate results as compared to a combination of multiple algorithms as in hybrid models, ensemble models, bagging, boosting, blending and stacking. An advantage of the implementation of machine learning models for sales forecasting is the ability of the machine learning algorithms to factor external values that affect sales such as seasonality, holidays, temperature, inflation rates, unemployment rates and exchange rates. Machine learning algorithms are capable of capturing complex and non-linear relationships in data that humans cannot be able to identify. The incorporation of these factors in a machine learning model helps increase the accuracy of the sales predictions as they mimic the exact operation of the real world and produces realistic results based on prevailing factors.

Sales Forecasting has also been proven to be a very important tool in business management and [1] supported this fact by pointing to the use of sales forecasting in management of resources, budget and the workforce. Sales forecasting is also important in inventory management [2]. These revelations in terms of sales forecasting have rightly and effectively strengthened the need for computational sales forecasting methods over traditional forecasting methods in Zimbabwe.

CHAPTER 3: ANALYSIS

3.1 DESCRIPTION OF SYSTEM

The sales forecasting system aims to predict future sales based on historical data using a pre-trained stacked ensemble model.

The dataset will first be preprocessed using python libraries like pandas and NumPy to remove missing values, remove outliers, remove duplicate values and also prepare the data for time series forecasting. The preprocessed dataset will then be split into training and test data sets using 80% for training set and 20% for the test set. The machine learning model will utilize the SARIMA time series algorithm and LightGBM as the base learners and Random Forest as the meta learner.

The factors that affect sales (features) will then be made inputs to the LightGBM algorithm to come up with the LightGBM predictions. The SARIMA algorithm will analyze the trend and seasonality of the time series data to come up with the SARIMA predictions. The predictions from both the SARIMA model and the LightGBM are combined with the original sales features to form the input into the Random Forest meta-learner. The Random Forest meta-learner will then process the input to come up with the final sales forecast.

The LightGBM, SARIMA and Random Forest models are saved in a dictionary as a pickle file to be used in the web application for the forecasting of future sales values using historical sales data.

The web application will then be integrated with the pretrained model in order to create an interactive way for users to access the functionality of the forecasting model. The web application will allow the user to log in to the system to access all the functionalities of the system. The system will have sales analysis functionalities and data visualization capabilities that will allow users to view the sales analytics using graphs making the analytics more understandable. Users will access a forecasting page that will enable the users to enter the parameters for forecasting based on the features that would affect sales. The system will prepare the input data for forecasting by matching it with the format used during model training. The stacked ensemble model, stored as a pickle file, is then used to generate predictions with the input data provided by the user as the inputs. The results of the forecast are presented to the user through visual graphs and a tabular format for interpretability, leveraging the Django framework for web-based display.

The use of the Django framework allows for the interactive usage of the sales forecasting system by the users. The graphical presentation allows for the easy interpretability of the results of the forecasting.

3.2 INFORMATION GATHERING TOOLS

Information gathering is one of the most important activities in the duration of the project. Important and helpful information like functional requirements, formulation of non-functional requirements, user requirements and system requirements are gathered and are cleaned in order to come up with an effective system. The requirements for the proposed system were gathered using interviews and questionnaires.

3.2.1 INTERVIEWS

An interview is an information gathering method where the researcher asks a series of questions to another party in order to obtain valuable information on a particular topic. The interviews were conducted targeted at sales managers in order to obtain information on how they manage their stock levels, how they come up with marketing strategies and how important a sales forecasting system would help their businesses. The

approach used for the interviews was the adoption of a semi-structured interview session in which the questions asked could be flexible to allow to solicit more information from the sales managers.

ADVANTAGES OF INTERVIEWS

- There was immediate response from the interviewees
- More questions could be asked to add clarity to some aspects of the topic which led to a deeper understanding of the topic at hand
- The body language of the respondents is easy to monitor to get a deeper understanding of the topic through the way the respondents carry themselves when explaining a concept

DISADVANTAGES OF INTERVIEWS

- Some respondents fail to show up for the interview
- The interviewer may fail to interpret the information correctly

3.2.2 QUESTIONNAIRES

A questionnaire is a list of questions used to gather information from respondents about their experiences and opinions on a particular topic. The questionnaires were directed at sales managers and some other members of the business who took part in the formulation of business strategies and important decisions. The questionnaire included questions on how the business made marketing strategies, when they'd carry out promotional activities and the importance of forecasting or predicting sales. The flexibility of the questionnaire to be answered at any time was essential in obtaining many filled up questionnaires.

ADVANTAGES OF QUESTIONNAIRES

- Respondents answer more accurately due to the anonymous nature of the questionnaires
- Information is obtained straight from the desired people
- Respondents can answer at any time they feel free

DISADVANTAGES OF QUESTIONNAIRES

- There may be bias in the answers as some respondents can answer questions without taking note of the question as there is no supervision
- The information gathered may not give a clear understanding of the data
- The questions are not flexible and cannot be changed

3.3 DATA ANALYSIS

3.3.1 CONTROL FLOW DIAGRAM OF EXISTING SYSTEM

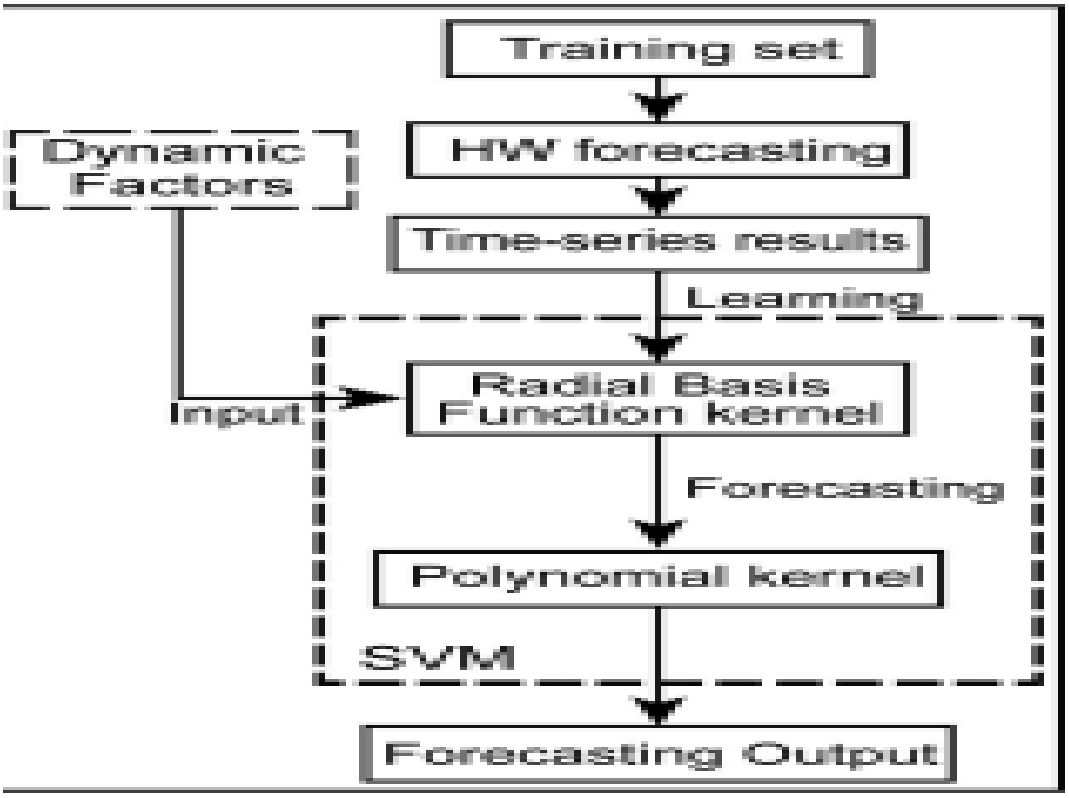


Figure 8: Control Flow for existing system

3.4 EVALUATION OF ALTERNATIVE SYSTEMS

There are several alternative systems that provides sales forecasting capabilities. The following is a description of some of these systems.

3.4.1 PIPEDRIVE

This is a cloud-based Customer Relationship Management System (CRM) tool used by sales teams for sales pipelines. Pipedrive leverages Artificial Intelligence (AI), historical data and automation to help sales teams to construct correctly defined and accurate forecasts. It offers the Insights Revenue Forecast used for projecting future revenue.

3.4.2 BOOSTUP

BoostUp.ai offers predictive analysis capabilities using an Artificial Intelligence engine for business revenue forecasts. The forecasting software utilizes past data to analyze sales trends, make informed decisions and spot patterns in data. One of the features of BoostUp is that it leverages the use of AI to predict forecasting. It keeps track of real-time trends and analyze changes instantly for forecasting purposes.

3.4.3 ANAPLAN

This is a cloud-based system that gives sales leaders the ability to deliver forecasts that are reliable, dynamic and that allow them to make updates on the fly based on current market conditions. It also allows the real-time tracking of sales pipelines and forecasts by connecting data from various systems. Anaplan allows the generation of accurate, data-driven revenue predictions in real-time that improve decision making throughout the company. Sales can be forecasted by geography or product line. It also combines third-party

data with historical sales patterns including trend and seasonality which helps create a more accurate forecast.

3.4.4 SPOTIO

This is a cloud-based sales tracking solution designed to assist businesses with resource tracking and sales pipeline management. SPOTIO captures and tracks all sales data that the business requires to obtain complete and accurate sales forecasts. It allows monitoring critical sales activities, reviewing pipeline metrics and track revenue data. It can be integrated with CRMs and ERPs in order to obtain accurate sales forecasts and track progress against sales goals.

3.5 FUNCTIONAL ANALYSIS OF PROPOSED SYSTEM

3.5.1 FUNCTIONAL REQUIREMENTS

- **Forecasting Sales**
  The system will predict future sales of products based on historical data over a period of time.

- **Scenario Analysis**
  The system will allow adjusting the input of various external factors that will affect the sales of products such as promotions, exchange rates, inflation rates and unemployment rates. This allows the comparison of various scenarios and variables that will produce a better sales prediction which then allows the implementation of effective marketing strategies that leads to profit for the organization.

- **Forecast Visualization**
  The system will provide an interactive dashboard that allows the visualization of sales forecast. The forecast data will be displayed in form of bar graphs and trend graphs which allows for the easy interpretation of the sales data.

- **Sales Analysis**
  The web application system will allow the user to make various analysis on the data to get a better understanding of the forecast data. The system will allow the user to check the accuracy of the forecast, check the most selling products and check the overall trend of the sales data which is essential in enabling the business to make informed decisions on how to structure and plan their marketing strategies.

3.5.2 NON-FUNCTIONAL REQUIREMENTS

- **Accuracy**
  The sales forecasting system should produce highly accurate and reliable forecast results.

- **Interpretability**
  The sales forecast reports generated by the system should be easily interpreted by the user. The use of visualizations such as flow charts, bar charts and trend graphs would ensure the interpretability of the results.

- **Scalability**
  The system should be able to adapt to the increasing size of users on the system and the size of the input data.

- **Integration**
  The system should also support the integration with other system such as CRMs and ERPs.

3.6 USE CASE DIAGRAM

A use case diagram describes the functions and the scope of the system. Shown below is the use case diagram of the proposed system.

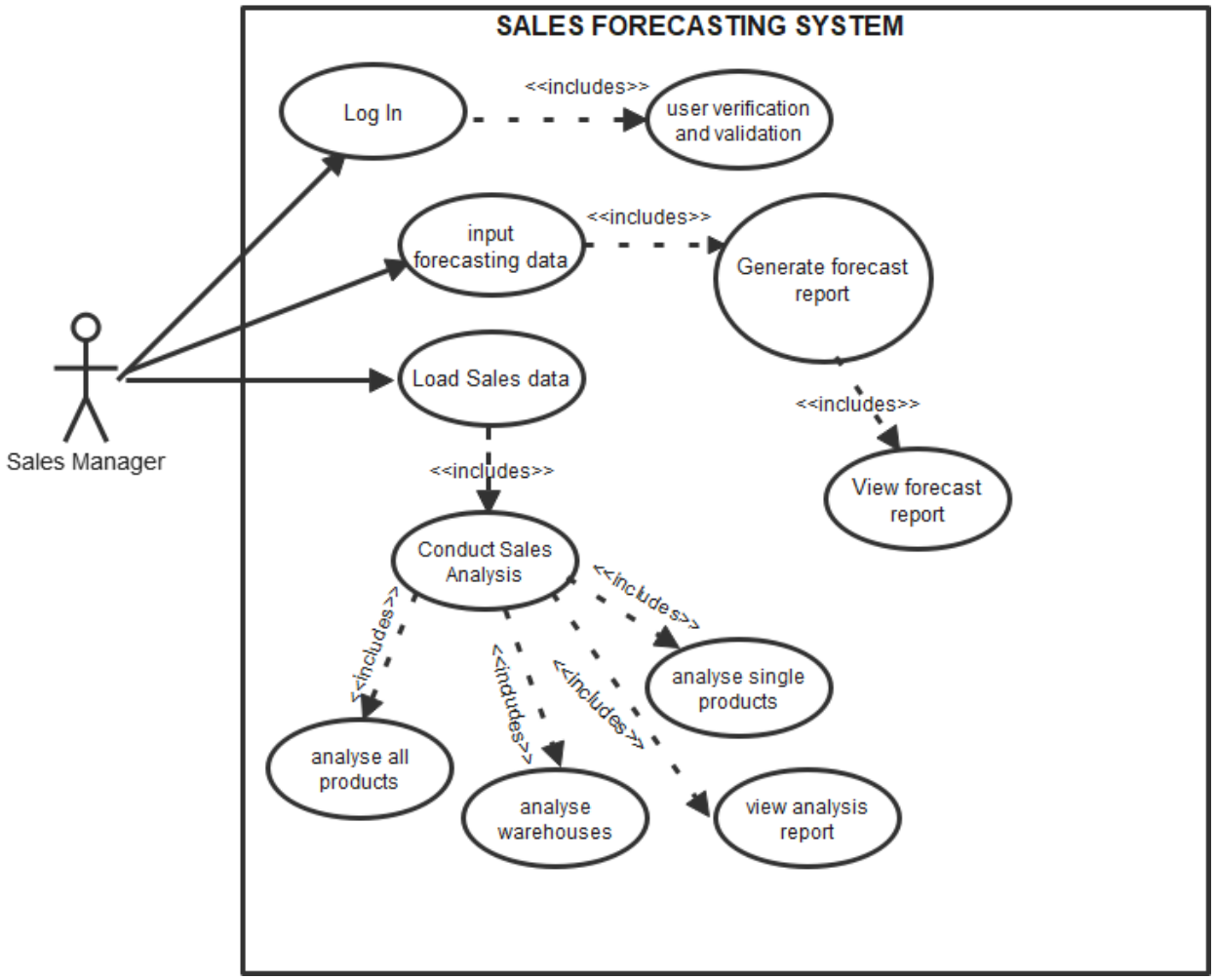


Figure 9:Use case for proposed system

CHAPTER 4: DESIGN

4.1 SYSTEM DIAGRAMS

4.1.1 CONTROL FLOW DIAGRAM

This diagram illustrates the flow of the training process of the stacked ensemble model and how the algorithms are joined together.

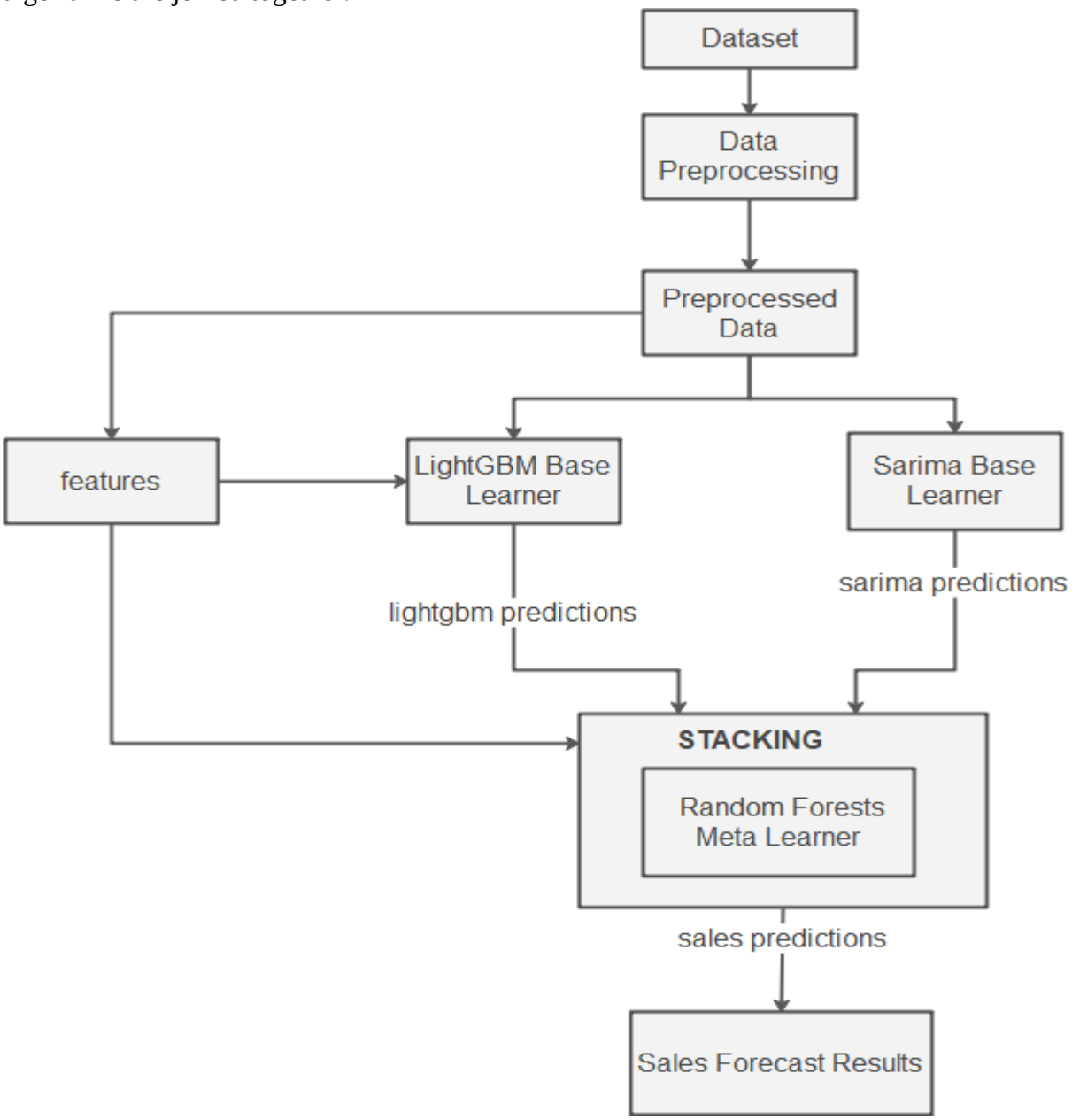


Figure 10: Control Flow Diagram

4.1.2 PROCESS FLOW DIAGRAM

This diagram illustrates the flow of data and processes within the sales forecasting system until the final sales predictions and the sales analysis results.

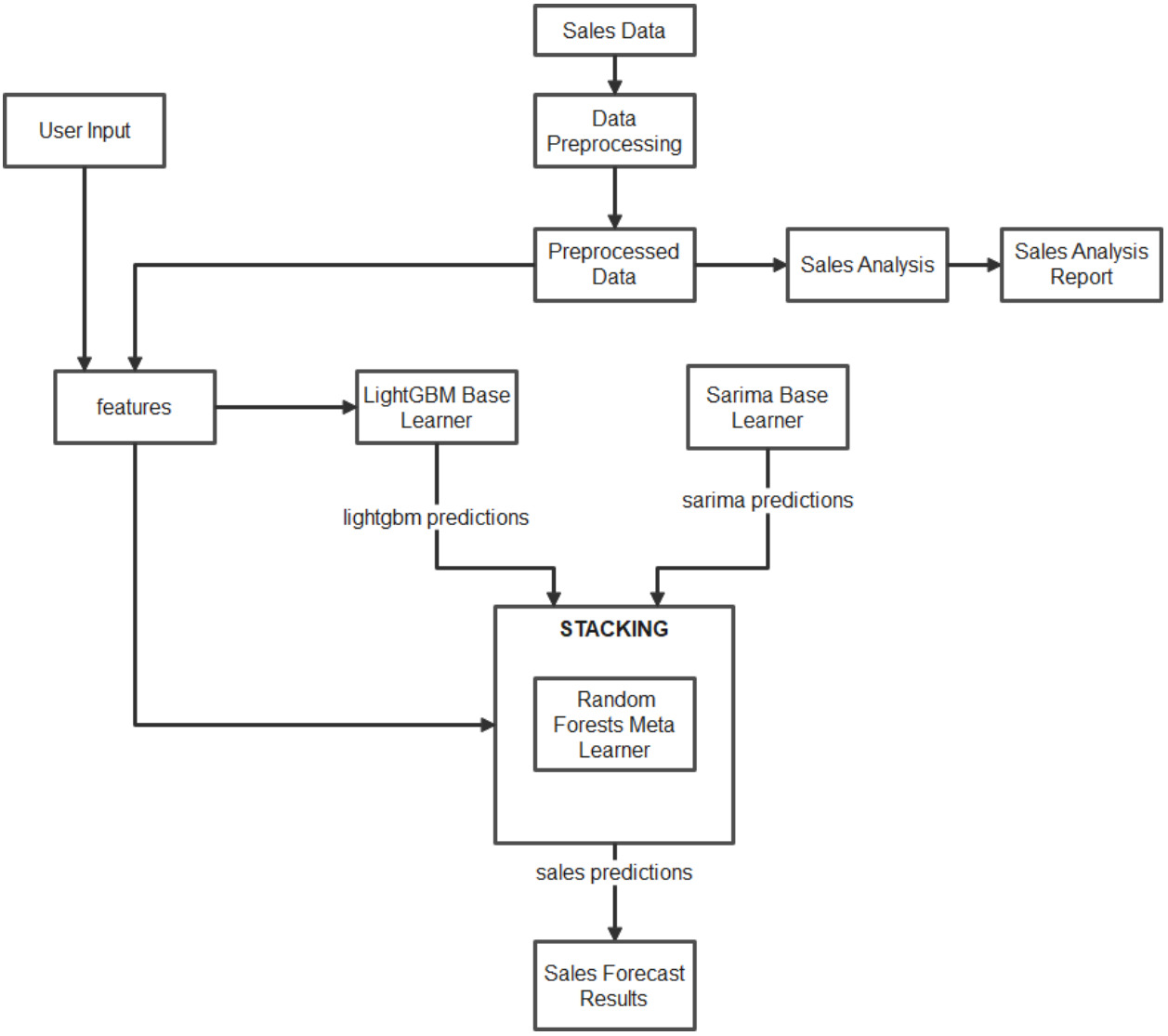


Figure 11: Process Flow Diagram

4.1.3 CONTEXT DIAGRAM

A context diagram provides a high-level view of the system and its interactions with external entities.

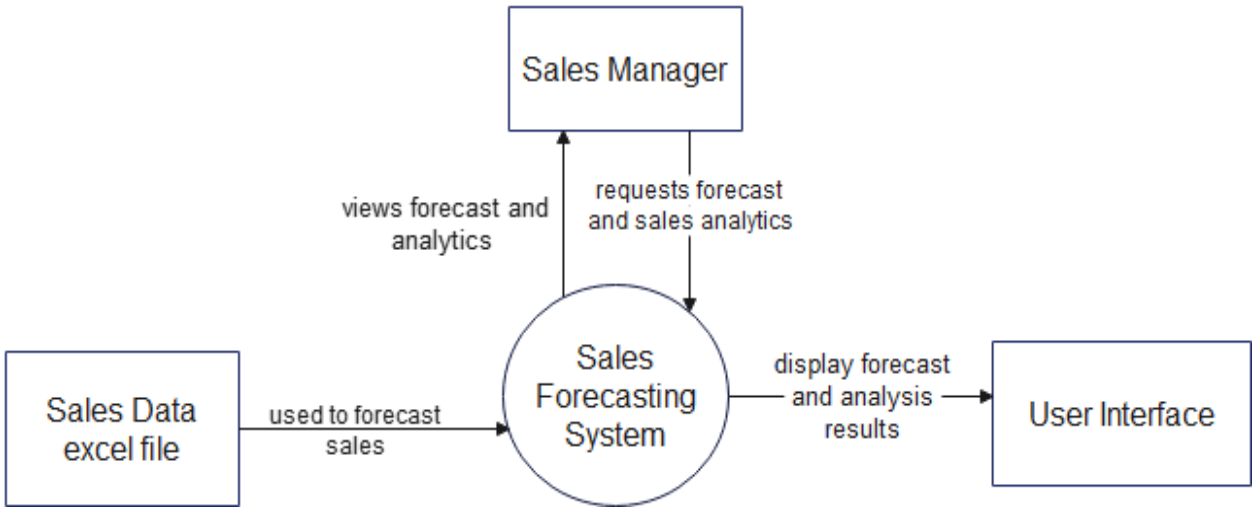


Figure 12: Context level diagram

4.1.4 DATA FLOW DIAGRAM

This illustrates the flow of information within the sales forecasting system.

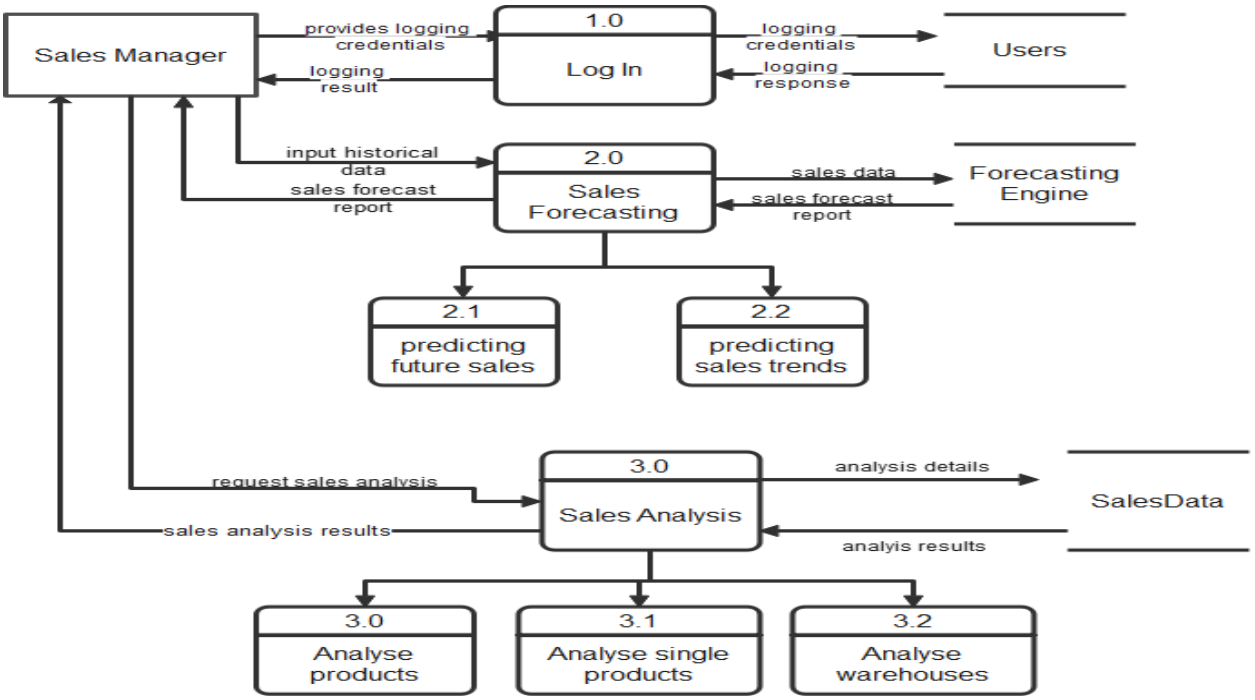


Figure 13: Data Flow Diagram

4.1.5 ACTIVITY DIAGRAM

Activity diagrams show the workflow from a start point to the finish point detailing the many decision paths that exist in the progression of events contained in the activity.

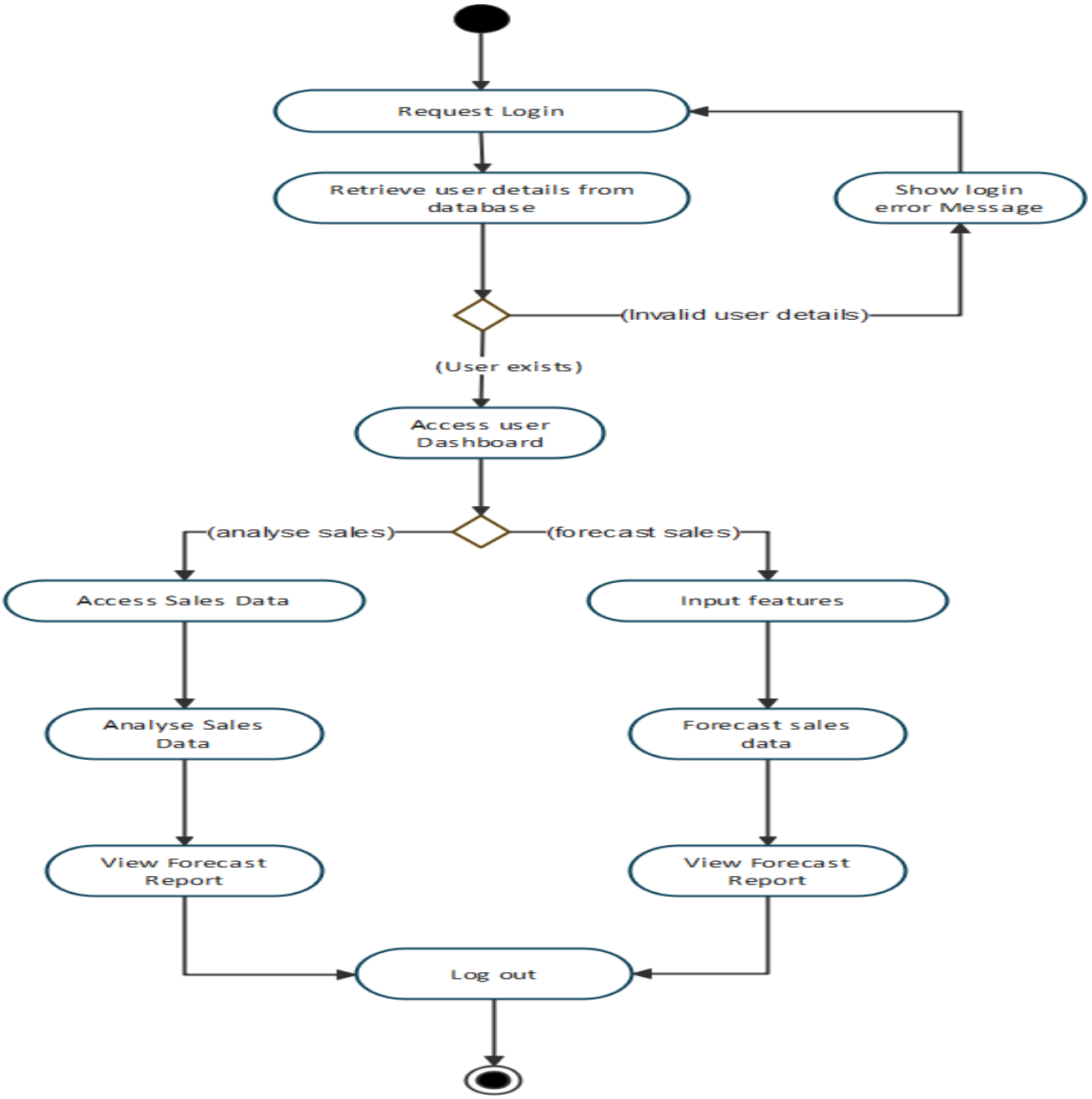


Figure 14:Activity Diagram

4.1.6 SEQUENCE DIAGRAM

This is an interaction diagram that details how operations are carried out -- what messages are sent and when.

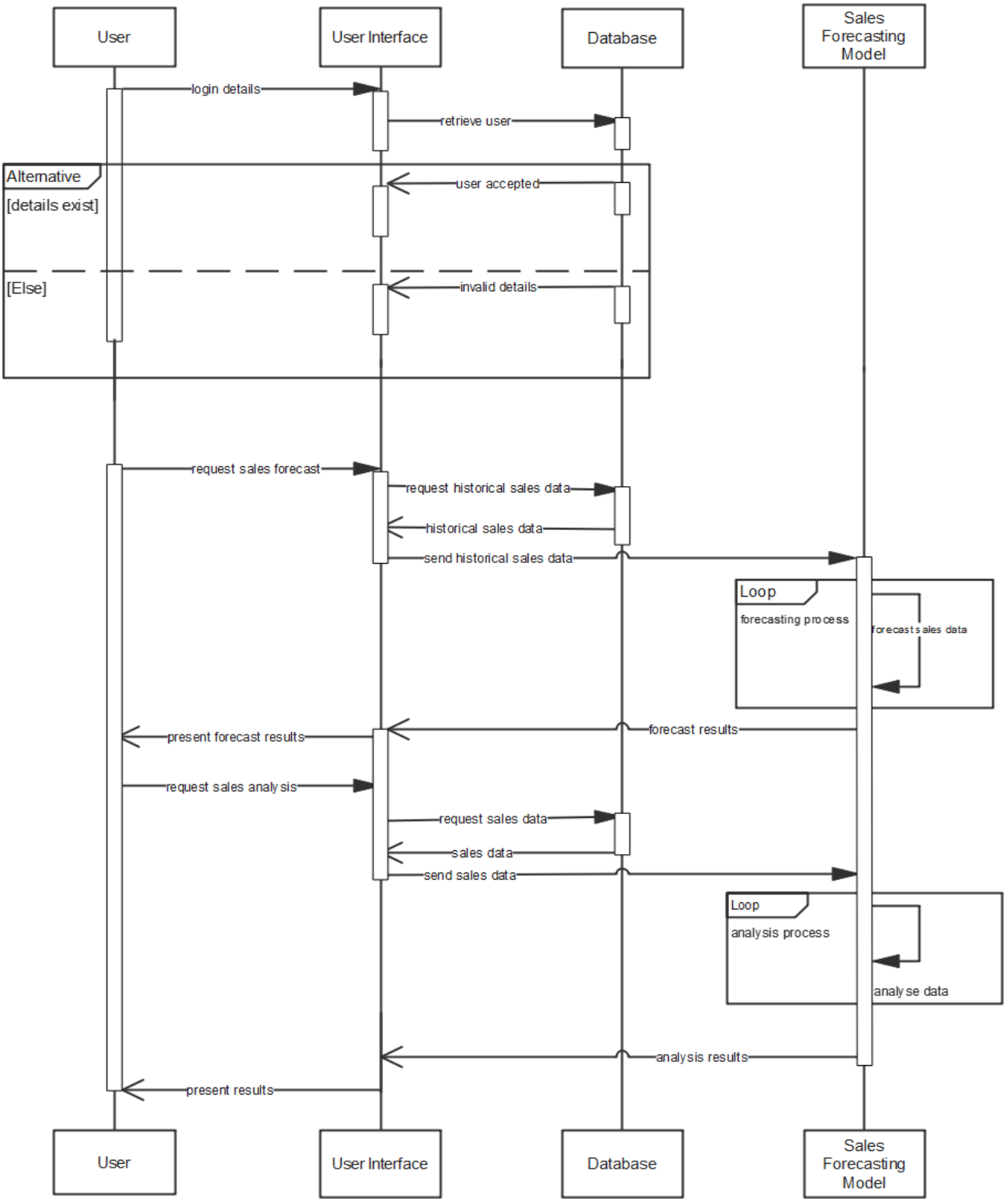


Figure 15: Sequence Diagram

4.1.7 CLASS DIAGRAM

This class diagram depicts the classes and their relationships within the sales forecasting system.

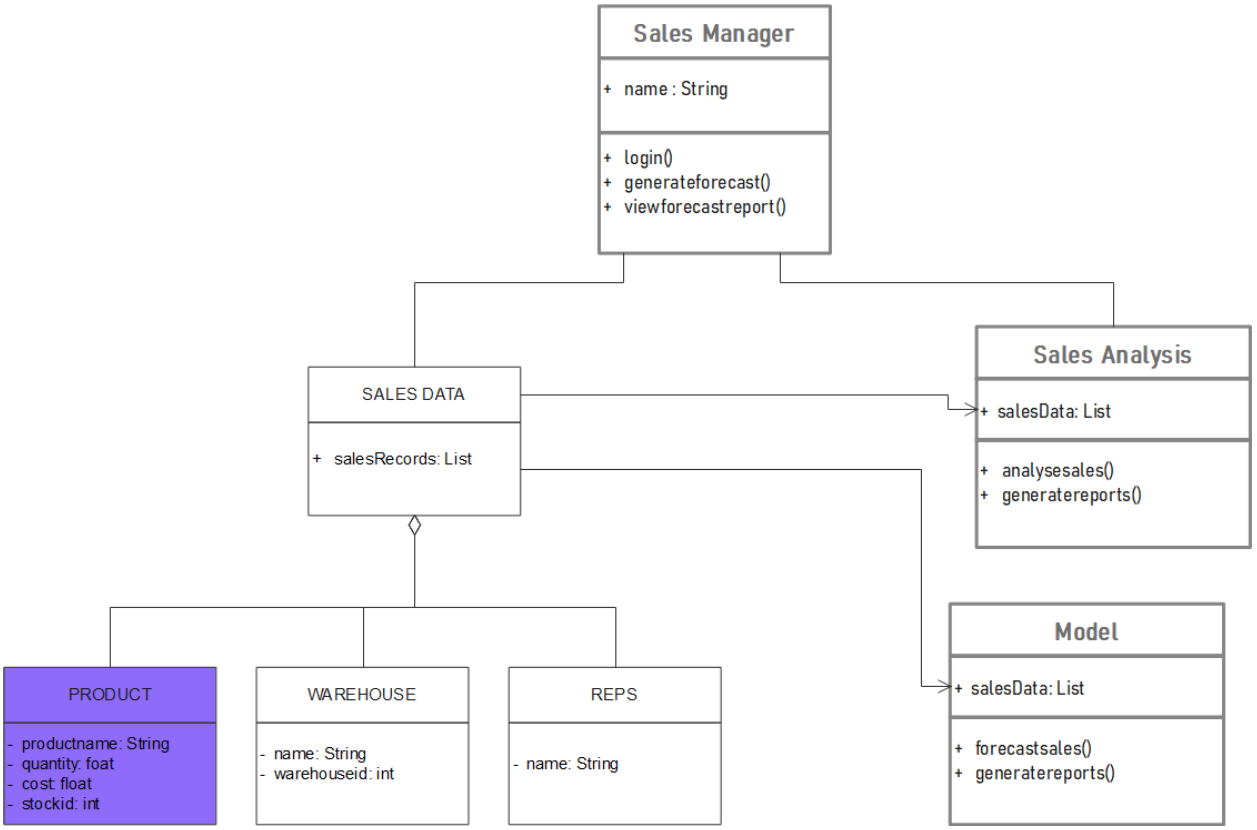


Figure 16: Class Diagram

4.1.8 ER DIAGRAM

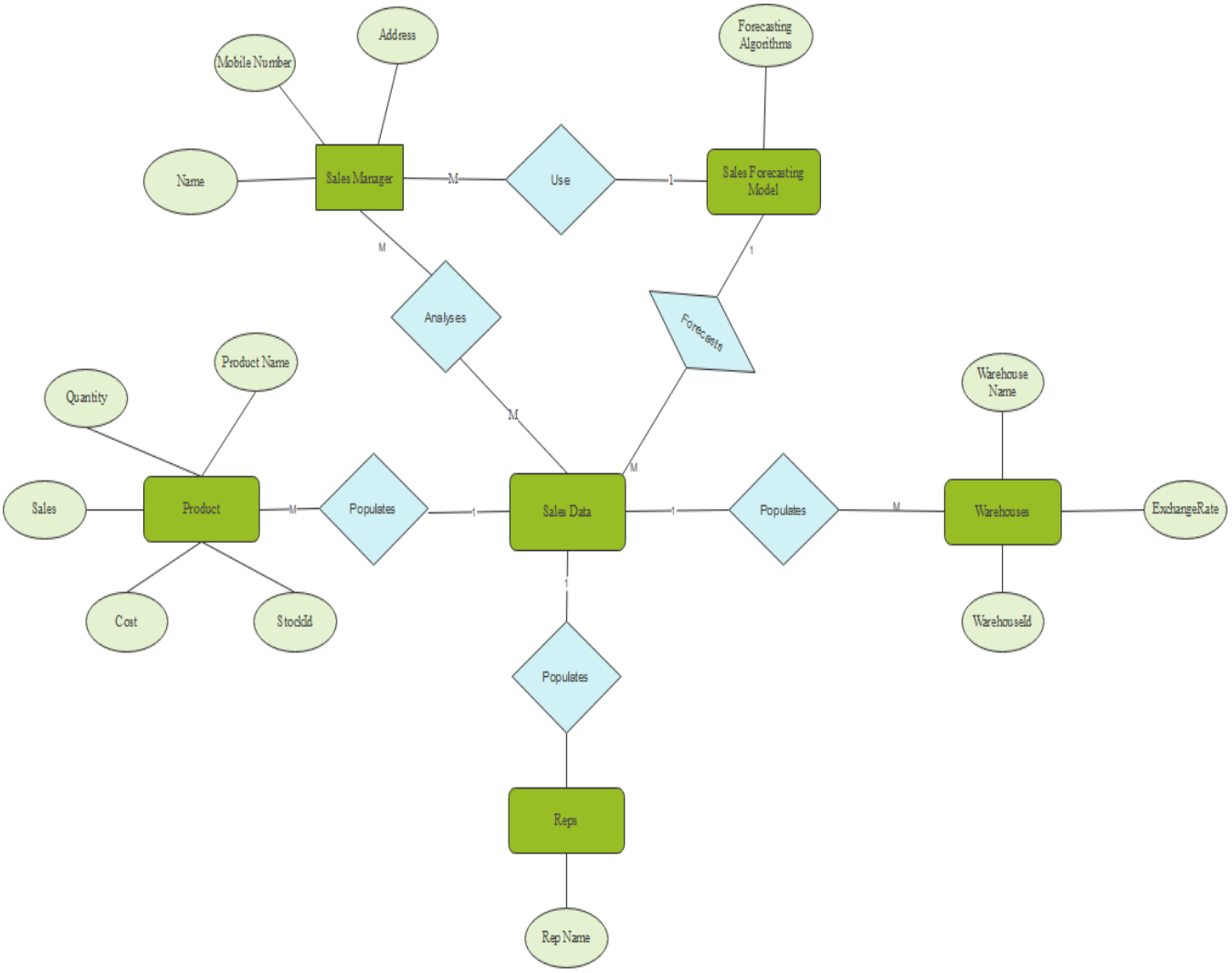


Figure 17: ER Diagram

4.1.9 PACKAGE DIAGRAM

A package diagram groups related classes into packages based on functionality.

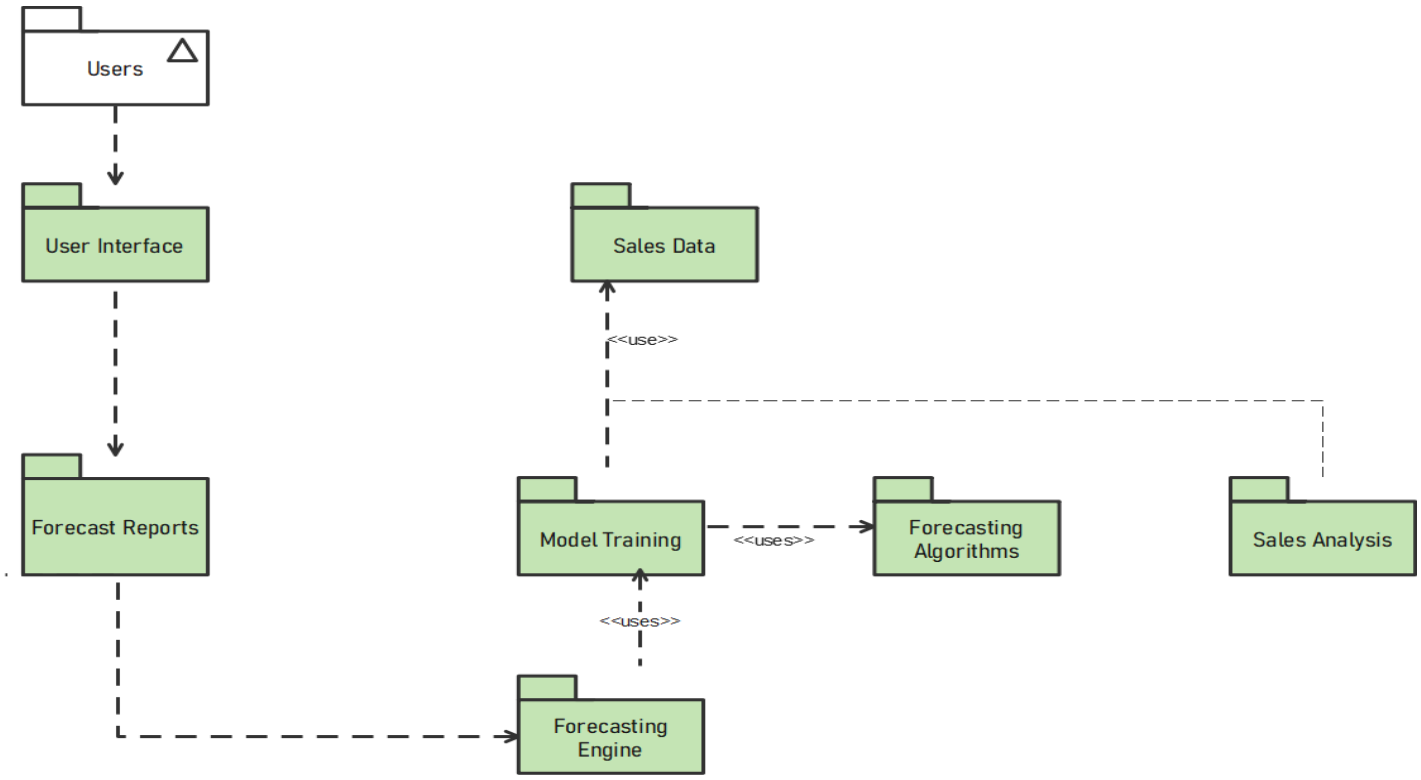


Figure 18: Package Diagram

4.1.10 ARCHITECTURE DIAGRAM

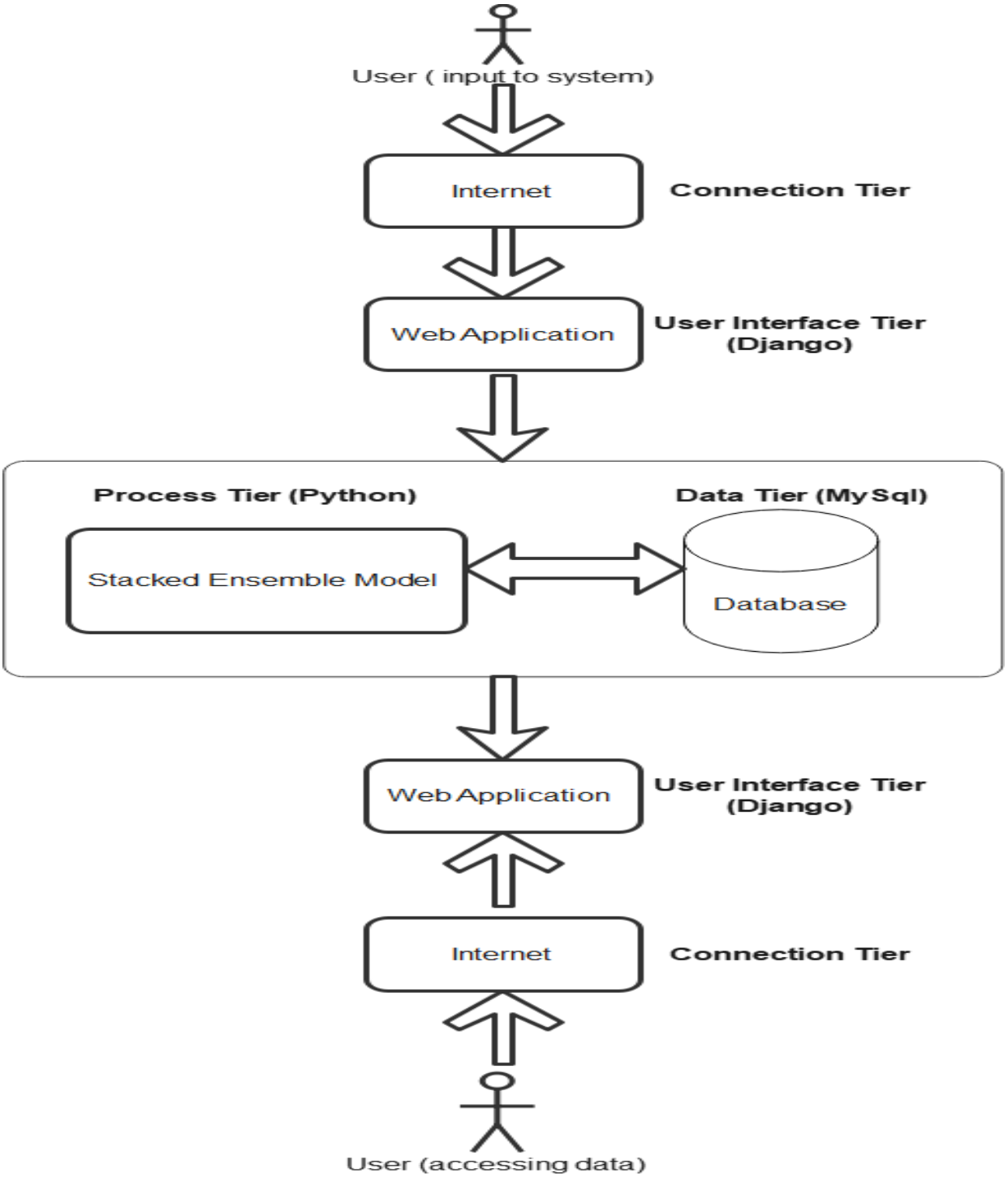


Figure 19: Architecture Diagram

4.2 INTERFACE DESIGN

4.2.1 LOGIN PAGE

This shows the log in page of the users.

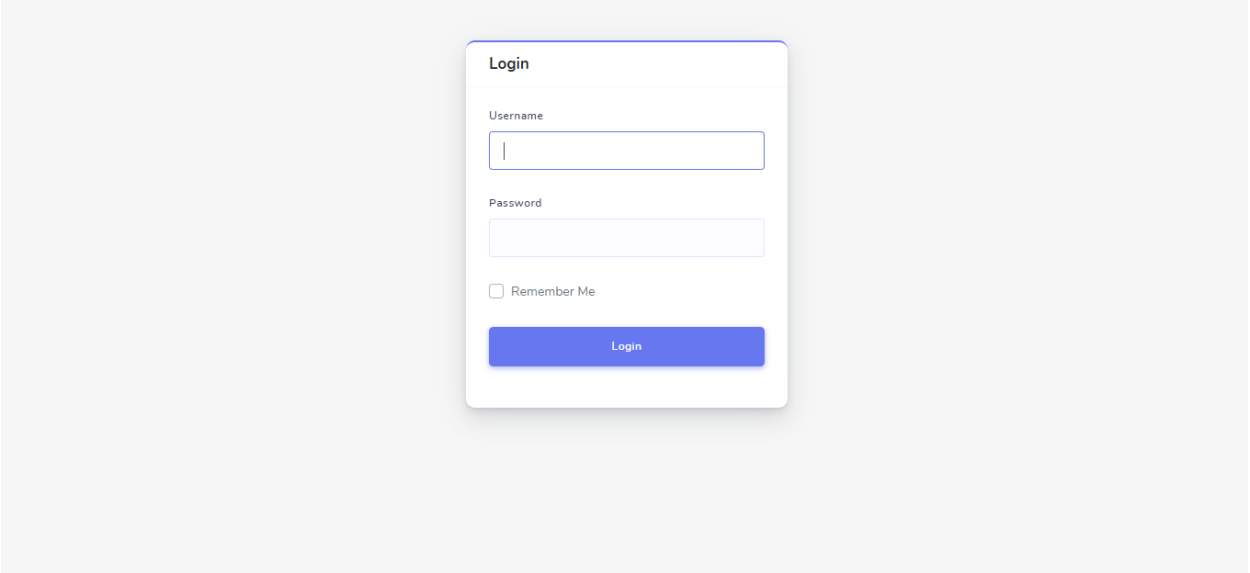


Figure 20: Login Page Design

4.2.2 MAIN DASHBOARD

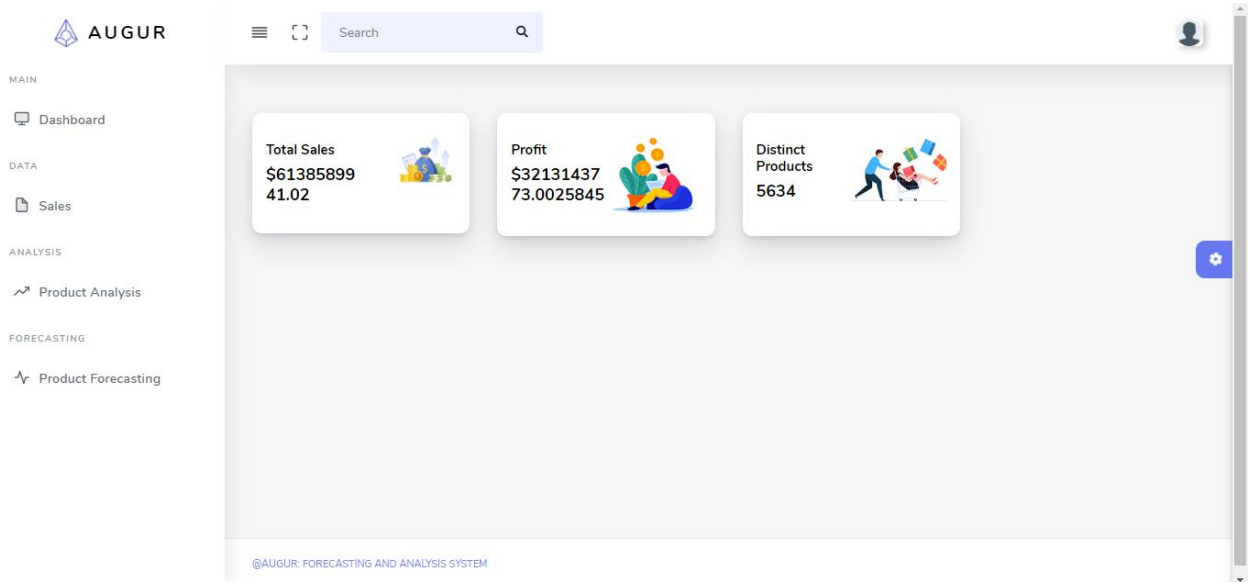


Figure 21: Dashboard Design

4.2.3 UPLOADING SALES DATA FILE

This is the form for uploading sales information for analysis.

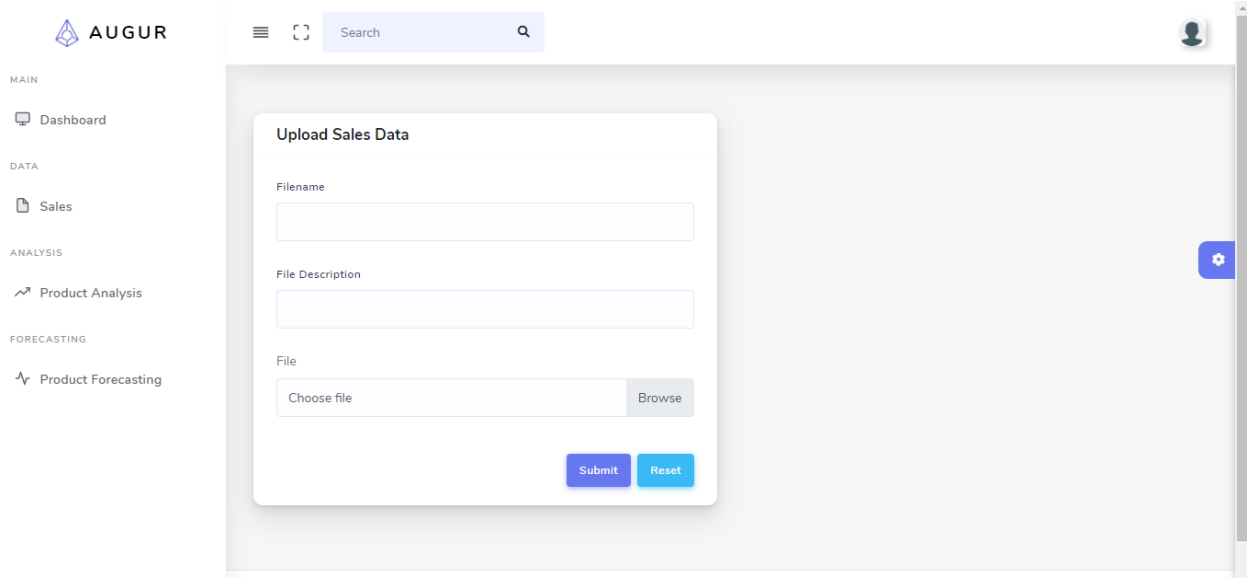


Figure 22: Upload Sales Data Form

4.2.4 PRODUCTS ANALYSIS VISUALIZATION

This shows the displaying of the results of the analysis of all the products.

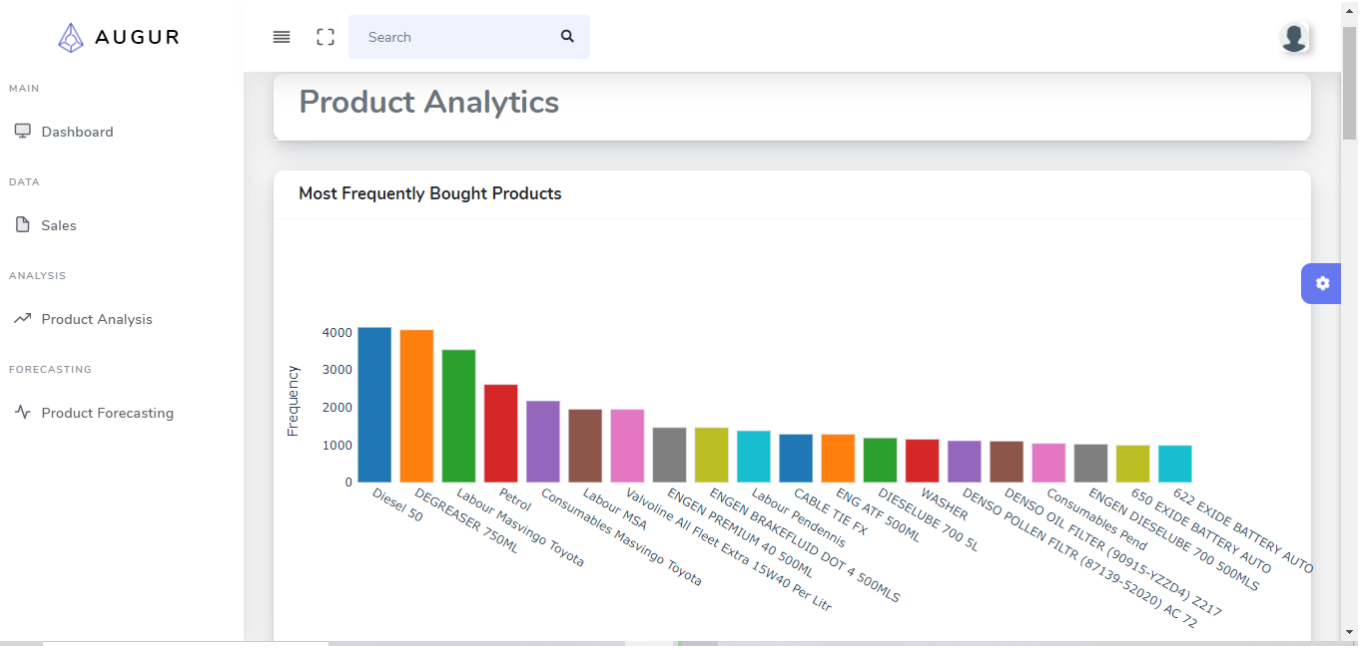


Figure 23: Products Analysis Results Visualization Design

4.2.5 SINGLE PRODUCT INPUT FORM

This shows the form where the user selects the product they want to analyze and what they want to analyze.

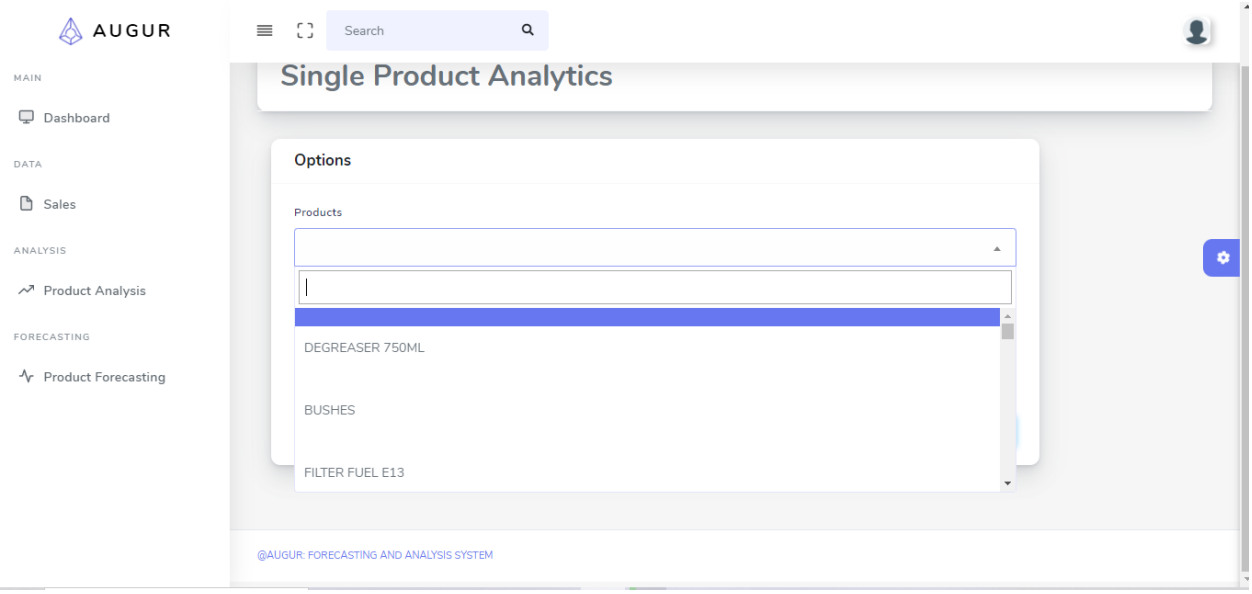


Figure 24: Choosing Product to Analyze

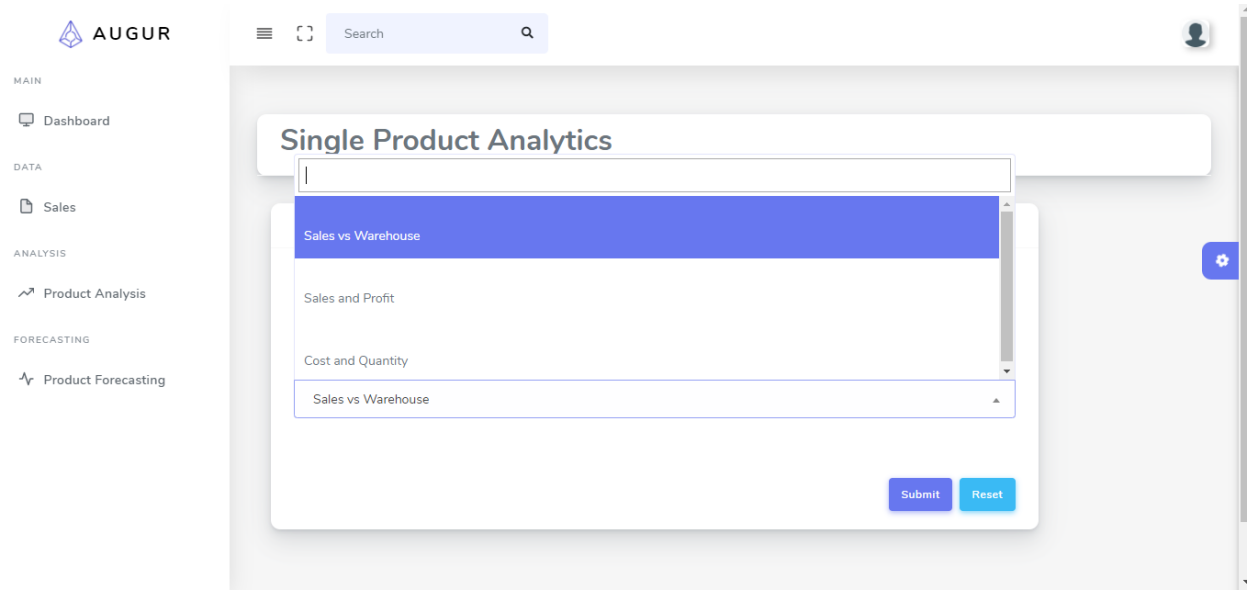


Figure 25: Choosing type of analysis

4.2.6 SINGLE PRODUCT ANALYSIS VISUALIZATION

This shows the visualization of the analysis of a single product selected by the user.

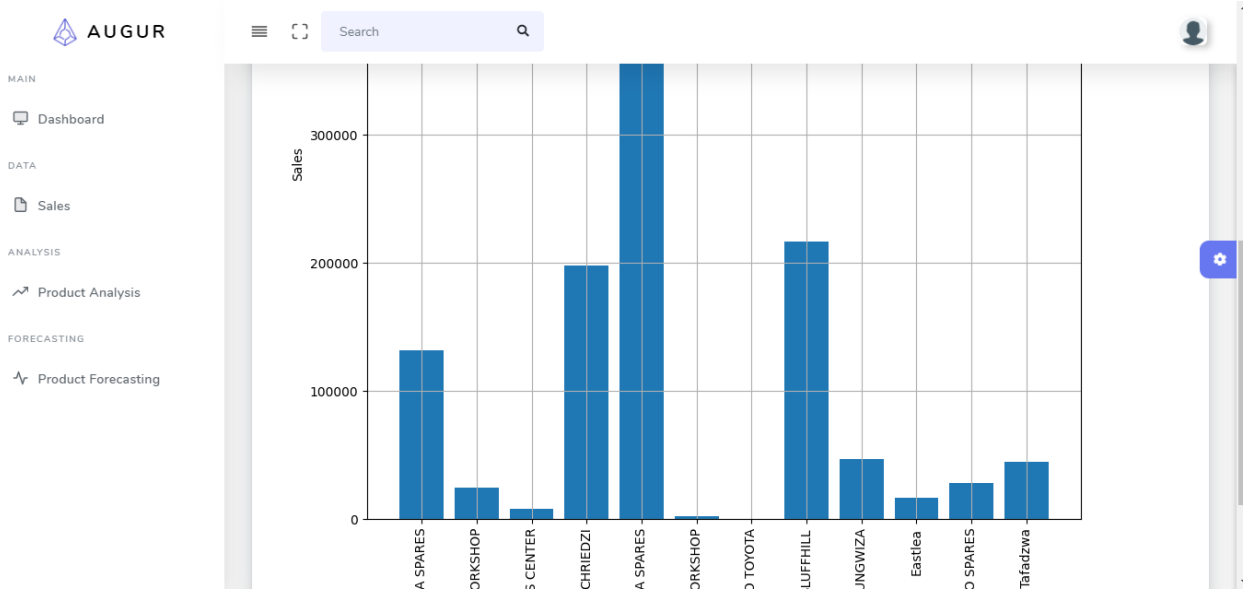


Figure 26: Single Product Analysis Results Visualization

4.2.7 PRODUCT BASED SALES FORECASTING INPUT FORM

This shows the input form for the user to input parameters for forecasting the sales of the selected product.

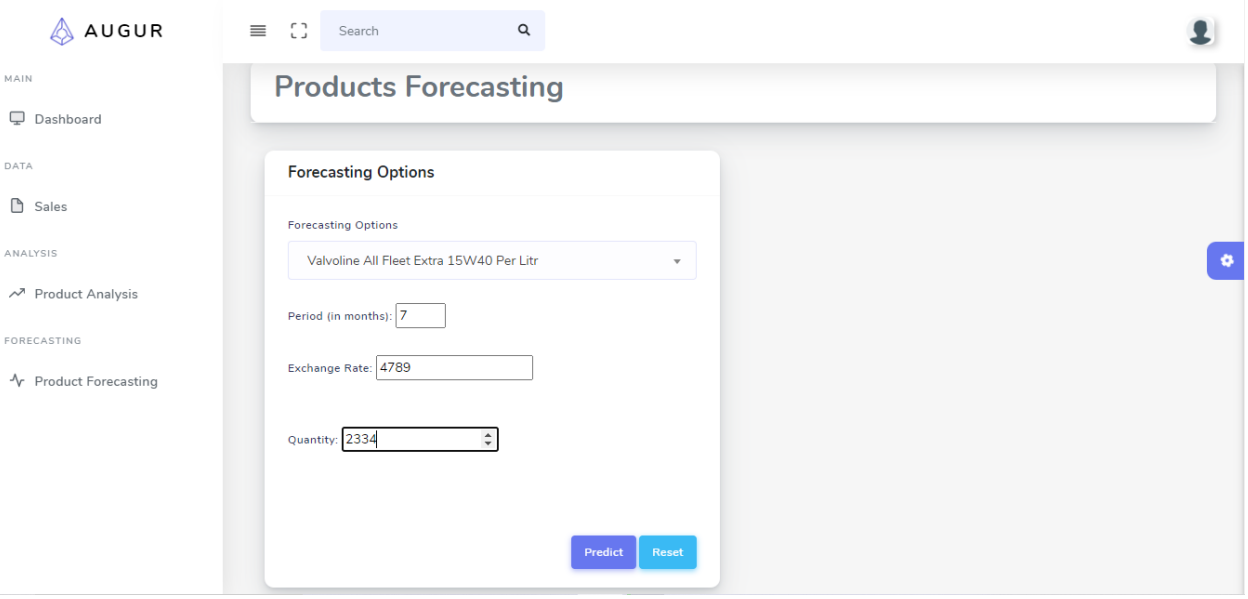


Figure 27: Product Forecasting Input Form

4.2.8 PRODUCT SALES FORECASTING VISUALIZATION

This shows the visualization of the forecasted sales.

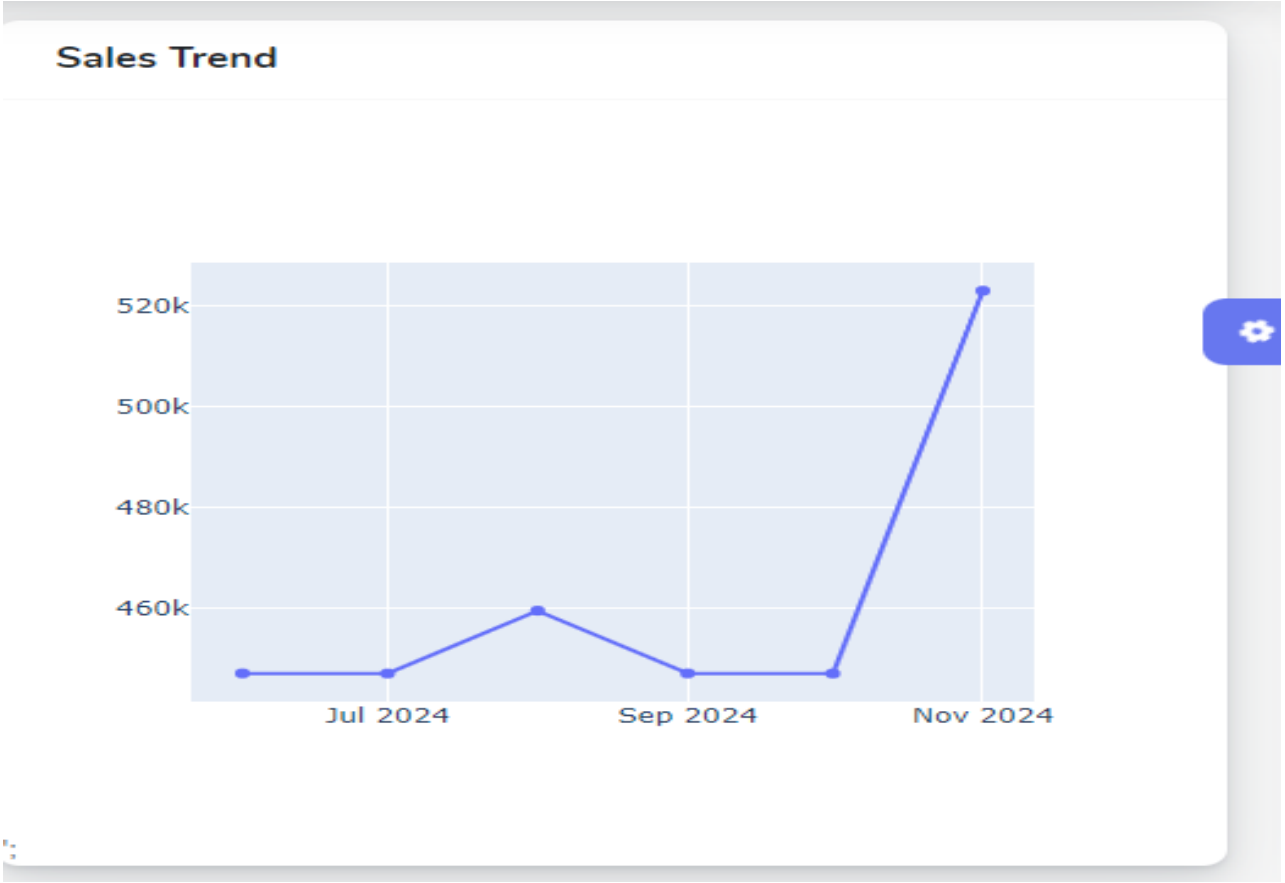


Figure 28: Product Forecasting Results

Exchange Rate:	RTGS 4789.0
Quantity:	2334.0 units
Date	Prediction
2024-06-01	447207.47
2024-07-01	447190.30
2024-08-01	459632.79
2024-09-01	447207.47
2024-10-01	447181.71
2024-11-01	523062.15

CHAPTER 5: IMPLEMENTATION AND TESTING

5.1 PSEUDO CODE

Code snippets of the major modules within the sales forecasting system.

5.1.1 DATA CLEANING FOR MODEL TRAINING

```
#DELETING DUPLICATE DATA
df = df.drop_duplicates()
```

```
#DELETING UNNECESSARY COLUMNS
df.drop(columns=['Id'], inplace = True)
```

```
#outlier detection
df_con= df.select_dtypes(include=[np.number])
plt.figure(figsize=[12,8])
sns.set(style="whitegrid")
sns.boxplot(x="variable", y="value", data=pd.melt(df_con), width=1)

plt.show()
```

```
#removing outliers
def remove_outlier(dataset, k=3.33):
    for col in dataset.columns:
        if dataset[col].dtype == "int64" or dataset[col].dtype == "float64":
            mean = dataset[col].mean()
            std = dataset[col].std()
            lower_bound = mean - k * std
            upper_bound = mean + k * std
            dataset = dataset[(dataset[col] > lower_bound) & (dataset[col] < upper_bound)]
    return dataset
```

```
ds = remove_outlier(df)
```

5.1.2 DATA PREPARATION FOR MODEL TRAINING

```
#appending month, year and month-year to the dataframe
ds['Year'] = ds['TxDate'].dt.year
ds['Month'] = ds['TxDate'].dt.month

ds['Month-Year'] = ds['TxDate'].dt.to_period('M')
```

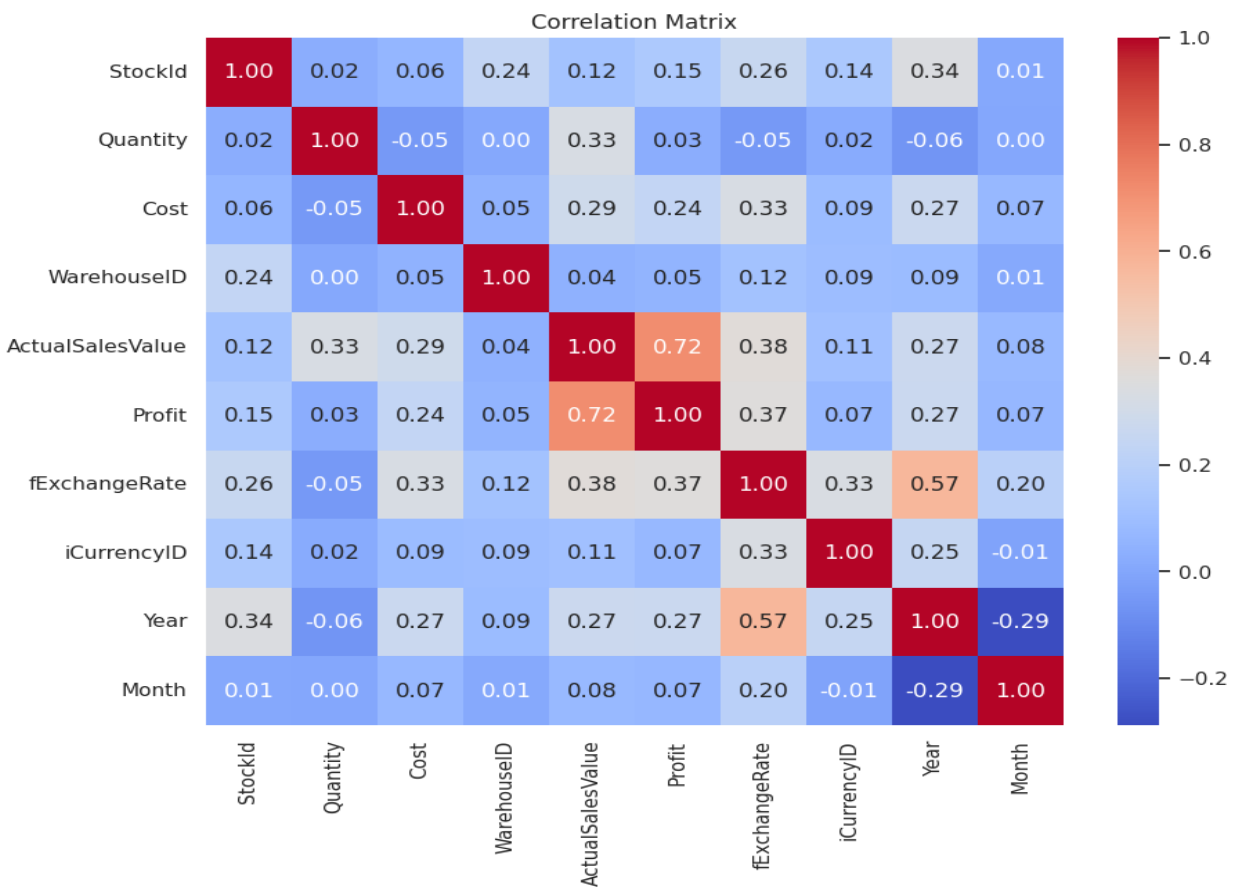
5.1.3 CORRELATION MATRIX

This shows the relationship between the numerical columns in the dataset.

```
#correlation
ds_numerical = ds.select_dtypes(include=[np.number])

# Compute correlation matrix
correlation_matrix = ds_numerical.corr()

# Visualize correlation matrix using a heatmap
plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Matrix')
plt.show()
```



The correlation matrix shows a close relationship between the target variable (ActualSalesValue) and the Quantity, Year, Profit and the Exchange Rate.

5.1.1 SELECTING PRODUCT TO FORECAST

This function selects the chosen model from the local path after the user has selected the product they want to forecast from the form in the interface.

```
def get_model_info(forecast_type):
 if forecast_type == 'consumables':
                    model_path = 'C:/Users/trevo/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast/models/consumables_model.pkl'
   product_name = 'Consumables Masvingo Toyota'
   features = ['fExchangeRate', 'Month', 'Year', 'Quantity']
   pass
 elif forecast_type == 'degreaser':
                    model_path = 'C:/Users/trevo/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast/models/degreaser_model.pkl'
   product_name = 'DEGREASER 750ML'
   features = ['fExchangeRate', 'Month', 'Year', 'Quantity']
   pass
 elif forecast_type == 'engenpremium':
                    model_path = 'C:/Users/trevo/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast/models/engine_premium_model.pkl'
   product_name = 'ENGEN PREMIUM 40 500ML'
   features = ['fExchangeRate', 'Month', 'Year', 'Quantity']
   pass
 elif forecast_type == 'valvoline':
                    model_path = 'C:/Users/trevo/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast/models/valvoline_model.pkl'
   product_name = 'Valvoline All Fleet Extra 15W40 Per Litr'
   features = ['fExchangeRate', 'Month', 'Year', 'Quantity']
   pass
 elif forecast_type == 'brakefluid':
                    model_path = 'C:/Users/trevo/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast/models/brakefluid_model.pkl'
   product_name = 'ENGEN BRAKEFLUID DOT 4 500MLS'
   features = ['fExchangeRate', 'Month', 'Year', 'Quantity']
   pass
 elif forecast_type == 'diesel':
   model_path = 'C:/Users/trevo/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast/models/diesel_model.pkl'
   product_name = 'Diesel 50'
   features = ['fExchangeRate', 'Month', 'Year', 'Quantity']
   pass
 elif forecast_type == 'petrol':
   model_path = 'C:/Users/trevo/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast/models/petrol_model.pkl'
   product_name = 'Petrol'
   features = ['fExchangeRate', 'Month', 'Year', 'Quantity']
   pass
 else:
   raise ValueError(f"Invalid forecast type: {forecast_type}")
 return model_path, product_name, features
```

5.1.2 LOADING SELECTED MODEL

This function loads the model from the local path and creates instances of the algorithms used to train the model into instances.

```
def load_model(model_path):
  with open(model_path, 'rb') as f:
    trained_models = pickle.load(f)
    lgbm_model = trained_models.get('lgbm_model')
    sarima_model = trained_models.get('sarima_model')
    sarima_model_fit = trained_models.get('sarima_model_fit')
    meta_learner = trained_models.get('meta_learner')
    if any(model is None for model in [lgbm_model, sarima_model, sarima_model_fit, meta_learner]):
      raise ValueError("Models not found in the loaded dictionary")
    return model_path, lgbm_model, sarima_model, sarima_model_fit, meta_learner
```

5.1.3 MAKING THE PREDICTION

This function makes the predictions and returns the predictions and the dates for the predicted values.

```
def make_predictions(forecast_type, product_name, period, fexchange_rate, quantity):

  model_info = get_model_info(forecast_type)
  model_path = model_info[0]
  product_name = model_info[1]
  features = model_info[2]
  # Load the trained models
  loaded_models = load_model(model_path)
  # Access elements of the returned tuple using indices
  lgbm_model = loaded_models[1]
  sarima_model = loaded_models[2]
  sarima_model_fit = loaded_models[3]
  meta_learner = loaded_models[4]

  current_date = datetime.datetime.now()
  start_date = current_date
  end_date = current_date + pd.DateOffset(months=period)
  # Generate list of dates for all days in the period
  dates = pd.date_range(start=start_date, end=end_date, freq='MS')

  features = []
  lgbm_predictions = []
  sarima_predictions = []

  for date in dates:
    month = date.month
    year = date.year

    month_data = Series({'Month': month, 'Year': year, 'fExchangeRate': fexchange_rate, 'Quantity': quantity})
```

```
    # Append features to the list
    features.append(month_data.values)

  # Convert features list to a NumPy array
  features_array = np.array(features)

  # Make predictions with both models for each month
  lgbm_predictions = lgbm_model.predict(features_array)
  sarima_predictions = sarima_model_fit.forecast(steps=len(features_array))

  # Combine features and predictions for each month
  stacked_inputs = np.column_stack((features_array, lgbm_predictions, sarima_predictions))

  # Predict with the meta-learner
  final_predictions = meta_learner.predict(stacked_inputs)

  dates_str = [d.strftime("%Y-%m-%d") for d in dates]

  generate_pdf_report(product_name, forecast_type, fexchange_rate, quantity, period, dates_str, final_predictions)

  return final_predictions, dates_str, product_name
```

5.1.4 DISPLAYING THE RESULTS

This view displays the predictions made by the models to the frontend.

```
def predict_sales(request):
    if request.method == 'POST':
        form = ForecastForm(request.POST)
        if form.is_valid():
            #Get form inputs

            forecast_type = request.POST.get('forecast_type')
            product_name = request.POST.get('product_name')
            period = int(request.POST.get('period'))
            fexchange_rate = float(request.POST.get('fexchange_rate'))
            quantity = float(request.POST.get('quantity'))

            # Make predictions
              predictions, months, product_name = make_predictions(forecast_type, product_name, period, fexchange_rate,
quantity)

            df = pd.DataFrame({'Date': pd.to_datetime(months), 'Predictions': predictions})
            fig = go.Figure([go.Scatter(x=df['Date'], y=df['Predictions'])])

            # Render the template
                return render(request, 'forecastingresults.html', {'plot': fig.to_html(full_html=False), 'predictions':predictions,
'months':months, 'product':product_name, 'period': period })
```

```
        else:
            return render(request, 'forecasting.html')

    else:
        return render(request, 'forecasting.html')
```

5.1.5 ANALYSIS OF SINGLE PRODUCTS

This view displays the sales analytics of a single product using already defined functions of graph types.

```
def generate_graph(request):
    if request.method == 'POST':
        form = SingleProductInputForm(request.POST, product_choices=get_product_choices())
        if form.is_valid():
            product_name = form.cleaned_data['product_choices']
            graph_type = form.cleaned_data['graph_choices']
            # Generate graph using the selected product name and graph type
            sales_data = SalesData.objects.last()
            if sales_data:
                if graph_type == 'sales_warehouse':
                    graph = generate_sales_warehouse_graph(sales_data, product_name)
                    return render(request, 'singleproductanalysis.html', {'graph': graph})

                if graph_type == 'sales_and_profit':
                    graph = salesandprofitgraph(sales_data, product_name)
                    return render(request, 'singleproductanalysis.html', {'graph': graph})

                if graph_type == 'cost_and_quantity':
                    graph = costandquantitygraph(sales_data, product_name)
                    return render(request, 'singleproductanalysis.html', {'graph': graph})

            else:
                pass
            return render(request, 'singleproductanalysis.html', {'product_name': product_name, 'graph_type': graph_type})
    else:
        form = SingleProductInputForm(product_choices = get_product_choices())
    return render(request, 'addsingleproduct.html', {'form': form})
```

5.2 SOFTWARE TESTING

This section shows the testing and the results of the testing of some major functions and the entire sales forecasting system. The testing was carried out using the pytest library.

5.2.1 UNIT TESTING

This shows tests of individual functions.

TESTING THE GET MODEL FUNCTION

This is the function that gets the model from the file system when user selects a product to forecast from the interface.

TEST CASES

```
“”” test_get_model.py “”””

import pytest
from forecast.views import get_model_info

def test_get_model_info_consumables():
  forecast_type = "consumables"
  model_path, product_name, features = get_model_info(forecast_type)
  assert model_path == "C:/Users/trevo/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast/models/consumables.pkl"
  assert product_name == "Consumables Masvingo Toyota"
  assert features == ["fExchangeRate", "Month", "Year", "Quantity"]

def test_get_model_info_petrol():
  forecast_type = "petrol"
  model_path, product_name, features = get_model_info(forecast_type)
  assert model_path == "C:/Users/trevo/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast/models/petrol.pkl"
  assert product_name == 'Petrol'
  assert features == ["fExchangeRate", "Month", "Year", "Quantity"]

def test_get_model_info_degreaser():
  forecast_type = "degreaser"
  model_path, product_name, features = get_model_info(forecast_type)
  assert model_path == "C:/Users/trevo/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast/models/degreaser.pkl"
  assert product_name == "DEGREASER 750ML"
  assert features == ["fExchangeRate", "Month", "Year", "Quantity"]

def test_get_model_info_engenpremium():
  forecast_type = "engenpremium"
  model_path, product_name, features = get_model_info(forecast_type)
  assert model_path == "C:/Users/trevo/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast/models/enginepremium.pkl"
  assert product_name == 'ENGEN PREMIUM 40 500ML'
  assert features == ["fExchangeRate", "Month", "Year", "Quantity"]

def test_get_model_info_valvoline():
  forecast_type = "valvoline"
  model_path, product_name, features = get_model_info(forecast_type)
  assert model_path == "C:/Users/trevo/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast/models/valvoline.pkl"
  assert product_name == 'Valvoline All Fleet Extra 15W40 Per Litr'
  assert features == ["fExchangeRate", "Month", "Year", "Quantity"]


def test_get_model_info_diesel():
  forecast_type = "diesel"
  model_path, product_name, features = get_model_info(forecast_type)
```

```
  assert model_path == "C:/Users/trevo/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast/models/diesel.pkl"
  assert product_name == 'Diesel 50'
  assert features == ["fExchangeRate", "Month", "Year", "Quantity"]

def test_get_model_info_invalid():
  forecast_type = "invalid_type"
  with pytest.raises(ValueError):
    get_model_info(forecast_type)
```

TESTING THE LOAD MODEL FUNCTION

This is the function that loads the model of a product after a user selects a product to forecast from the interface.

TEST CASES

```
# test_load_model.py

import pytest
import pickle
from unittest.mock import mock_open, patch
from forecast.views import load_model

def test_load_model():
    fake_models = {
        'lgbm_model': 'fake_lgbm_model',
        'sarima_model': 'fake_sarima_model',
        'sarima_model_fit': 'fake_sarima_model_fit',
        'meta_learner': 'fake_meta_learner'
    }

    m = mock_open()
    with patch('builtins.open', m), \
         patch('pickle.load', return_value=fake_models):

        # Call the function with a fake path
        model_path, lgbm_model, sarima_model, sarima_model_fit, meta_learner = load_model('fake_path.pkl')

        # Assertions to verify the correct models are returned
        assert model_path == 'fake_path.pkl'
        assert lgbm_model == 'fake_lgbm_model'
        assert sarima_model == 'fake_sarima_model'
        assert sarima_model_fit == 'fake_sarima_model_fit'
        assert meta_learner == 'fake_meta_learner'

    # Test for the case when models are not found in the dictionary
    incomplete_models = {
        'lgbm_model': 'fake_lgbm_model',
        'sarima_model': None,
        'sarima_model_fit': 'fake_sarima_model_fit',
```

```
        'meta_learner': 'fake_meta_learner'
    }

    with patch('builtins.open', m), \
         patch('pickle.load', return_value=incomplete_models):

        with pytest.raises(ValueError, match="Models not found in the loaded dictionary"):
            load_model('fake_path.pkl')
```

TESTING THE MAKE PREDICTIONS FUNCTION

This is the function that makes the actual predictions of a product using the model.

TEST CASE

```
# test_make_predictions.py
import pytest
from datetime import datetime
from forecast.views import make_predictions

@pytest.mark.django_db
def test_make_predictions():
    forecast_type = 'consumables'
    product_name = 'Consumables Masvingo Toyota'
    period = 3
    fexchange_rate = 1.5
    quantity = 100

    predictions, dates_str, product_name = make_predictions(forecast_type, product_name, period, fexchange_rate, quantity)

    assert len(predictions) == period
    assert len(dates_str) == period
    assert product_name == 'Consumables Masvingo Toyota'
```

RESULTS

```
(hit400)
Taku@DESKTOP-9LQARCG MINGW64 ~/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast
$ pytest
========================================================= test session starts =========================================================
platform win32 -- Python 3.12.0, pytest-8.2.0, pluggy-1.5.0
django: version: 5.0.4, settings: salesforecast.settings (from ini)
rootdir: C:\Users\trevo\OneDrive\Documents\HIT400\MyProject - Copy\salesforecast
configfile: pytest.ini
plugins: cov-5.0.0, django-4.8.0, mock-3.14.0
collected 9 items

tests\test_make_predictions.py .                                                                                                  [ 11%]
tests\test_get_model.py .......                                                                                                   [ 88%]
tests\test_load_model.py .                                                                                                        [100%]

========================================================== warnings summary ===========================================================
..\..\MyProject\hit400\Lib\site-packages\reportlab\lib\rl_safe_eval.py:11
  C:\Users\trevo\OneDrive\Documents\HIT400\MyProject\hit400\Lib\site-packages\reportlab\lib\rl_safe_eval.py:11: DeprecationWarning:

  ast.NameConstant is deprecated and will be removed in Python 3.14; use ast.Constant instead

tests/test_make_predictions.py::test_make_predictions
  C:\Users\trevo\OneDrive\Documents\HIT400\MyProject\hit400\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:836: ValueWarning:

  No supported index is available. Prediction results will be given with an integer index beginning at `start`.

tests/test_make_predictions.py::test_make_predictions
  C:\Users\trevo\OneDrive\Documents\HIT400\MyProject\hit400\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:836: FutureWarning:

  No supported index is available. In the next version, calling this method in a model without a supported index will result in an ex
ception.

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
================================================= 9 passed, 3 warnings in 10.52s ==================================================
(hit400)
Taku@DESKTOP-9LQARCG MINGW64 ~/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast
```

Figure 29: Unit tests results

The results shows that all the unit tests passed with high test coverages.

5.2.2 INTEGRATION TESTING

This tests the integration of the functions to show that they are communicating and interacting correctly and efficiently.

TEST CASE

```
“”” test  sales  forecast.py “”””

import os
import pickle
import datetime
import pandas as pd
import numpy as np
from forecast.views import get_model_info , load_model , make_predictions

def test_get_model_info():
    model_path, product_name, features = get_model_info('consumables')

    assert os.path.exists(model_path)
    assert product_name == 'Consumables Masvingo Toyota'
```

```
    assert features == ['fExchangeRate', 'Month', 'Year', 'Quantity']

def test_load_model():
    model_path = 'C:/Users/trevo/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast/models/consumables.pkl'
    with open(model_path, 'rb') as f:
        trained_models = pickle.load(f)

    model_path, lgbm_model, sarima_model, sarima_model_fit, meta_learner = load_model(model_path)

    assert model_path == 'C:/Users/trevo/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast/models/consumables.pkl'
    assert isinstance(lgbm_model, object)
    assert isinstance(sarima_model, object)
    assert isinstance(sarima_model_fit, object)
    assert isinstance(meta_learner, object)

def test_make_predictions():
    forecast_type = 'consumables'
    product_name = 'Consumables Masvingo Toyota'
    period = 3
    fexchange_rate = 1.2
    quantity = 10

    final_predictions, dates_str, product_name = make_predictions(forecast_type, product_name, period, fexchange_rate, quantity)

    assert isinstance(final_predictions, np.ndarray)
    assert isinstance(dates_str, list)
    assert product_name == 'Consumables Masvingo Toyota'

def test_generate_pdf_report():
    product_name = 'Consumables Masvingo Toyota'
    forecast_type = 'consumables'
    fexchange_rate = 1.2
    quantity = 10
    period = 3
    dates_str = ['2022-01-01', '2022-02-01', '2022-03-01']
    final_predictions = np.array([1, 2, 3])


    pdf_folder = "predictions"
    pdf_filename = f"sales_forecast_{product_name}_{datetime.datetime.now().strftime('%Y-%m-%d_%H-%M-%S')}.pdf"
    pdf_path = os.path.join(pdf_folder, pdf_filename)

    assert os.path.exists(pdf_path)
```

RESULTS

```
Taku@DESKTOP-9LQARCG MINGW64 ~/OneDrive/Documents/HIT400/MyProject - Copy/salesforecast
$ pytest
========================================================= test session starts =========================================================
platform win32 -- Python 3.12.0, pytest-8.2.0, pluggy-1.5.0
django: version: 5.0.4, settings: salesforecast.settings (from ini)
rootdir: C:\Users\trevo\OneDrive\Documents\HIT400\MyProject - Copy\salesforecast
configfile: pytest.ini
plugins: cov-5.0.0, django-4.8.0, mock-3.14.0
collected 13 items

tests\test_make_predictions.py .                                                                                                  [  7%]
tests\test_get_model.py .......                                                                                                   [ 61%]
tests\test_load_model.py .                                                                                                        [ 69%]
tests\test_sales_forecast.py ....                                                                                                 [100%]

========================================================== warnings summary ===========================================================
..\..\MyProject\hit400\Lib\site-packages\reportlab\lib\rl_safe_eval.py:11
  C:\Users\trevo\OneDrive\Documents\HIT400\MyProject\hit400\Lib\site-packages\reportlab\lib\rl_safe_eval.py:11: DeprecationWarning:

  ast.NameConstant is deprecated and will be removed in Python 3.14; use ast.Constant instead

tests/test_make_predictions.py::test_make_predictions
tests/test_sales_forecast.py::test_make_predictions
  C:\Users\trevo\OneDrive\Documents\HIT400\MyProject\hit400\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:836: ValueWarning:

  No supported index is available. Prediction results will be given with an integer index beginning at `start`.

tests/test_make_predictions.py::test_make_predictions
tests/test_sales_forecast.py::test_make_predictions
  C:\Users\trevo\OneDrive\Documents\HIT400\MyProject\hit400\Lib\site-packages\statsmodels\tsa\base\tsa_model.py:836: FutureWarning:

  No supported index is available. In the next version, calling this method in a model without a supported index will result in an ex
ception.

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
================================================= 13 passed, 5 warnings in 11.34s =================================================
```

Figure 30: Integration test results

All the test cases passed showing that all the functions are communicating efficiently.

5.2.3 SYSTEM TESTING

5.3.1 LOGIN VALIDATION

ON FAILED LOGIN

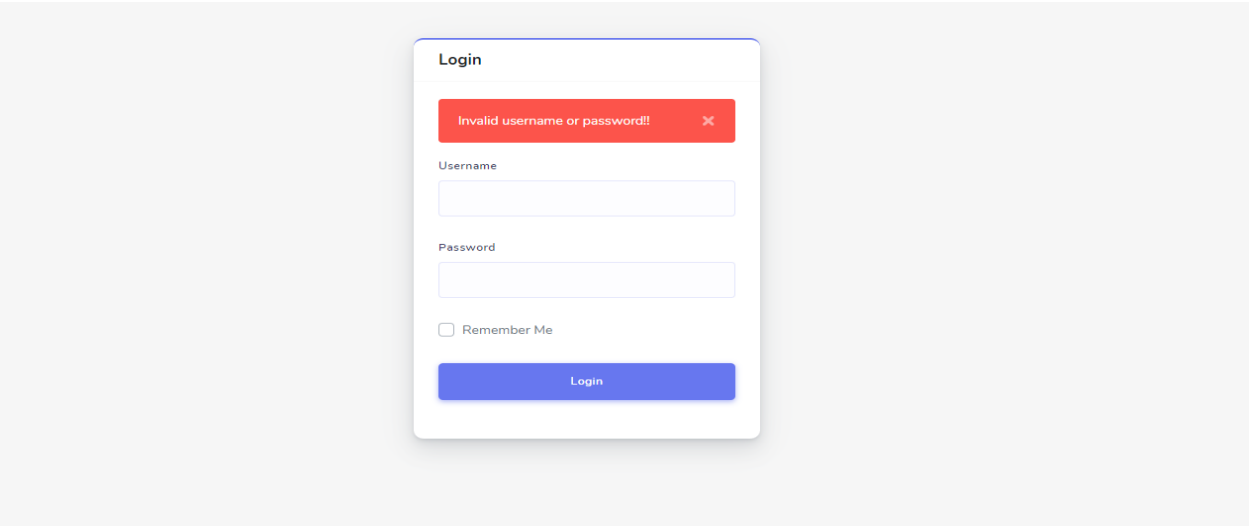


Figure 31: Login Failed

ON LOGIN SUCCESS

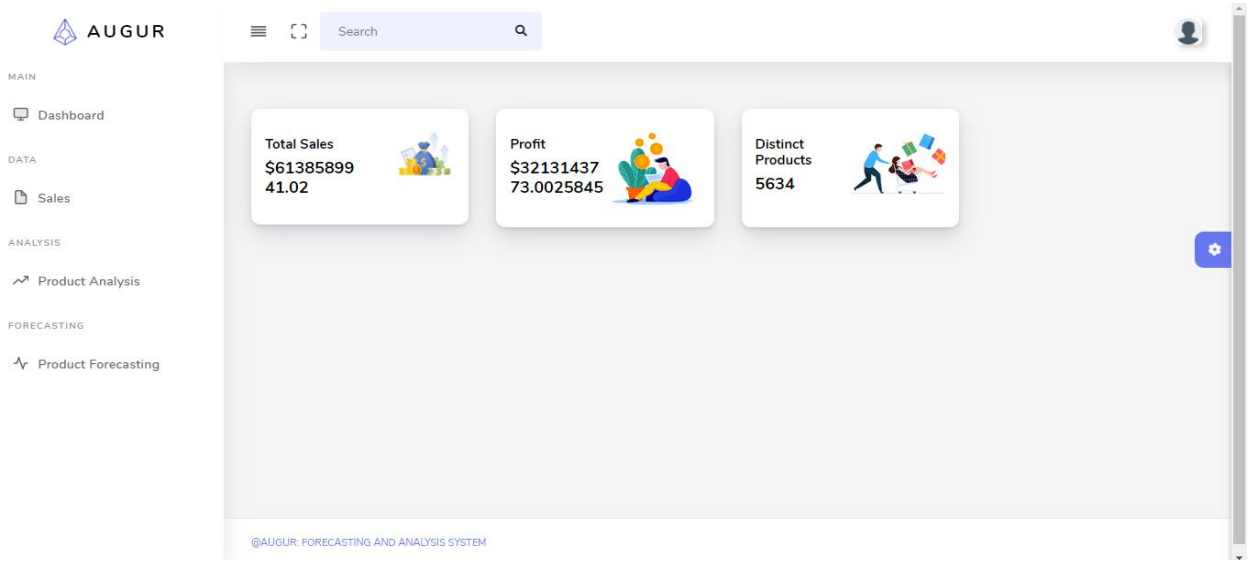


Figure 32: On login success

5.3.2 SINGLE PRODUCT ANALYSIS

ON FAILED SINGLE PRODUCT ANALYSIS

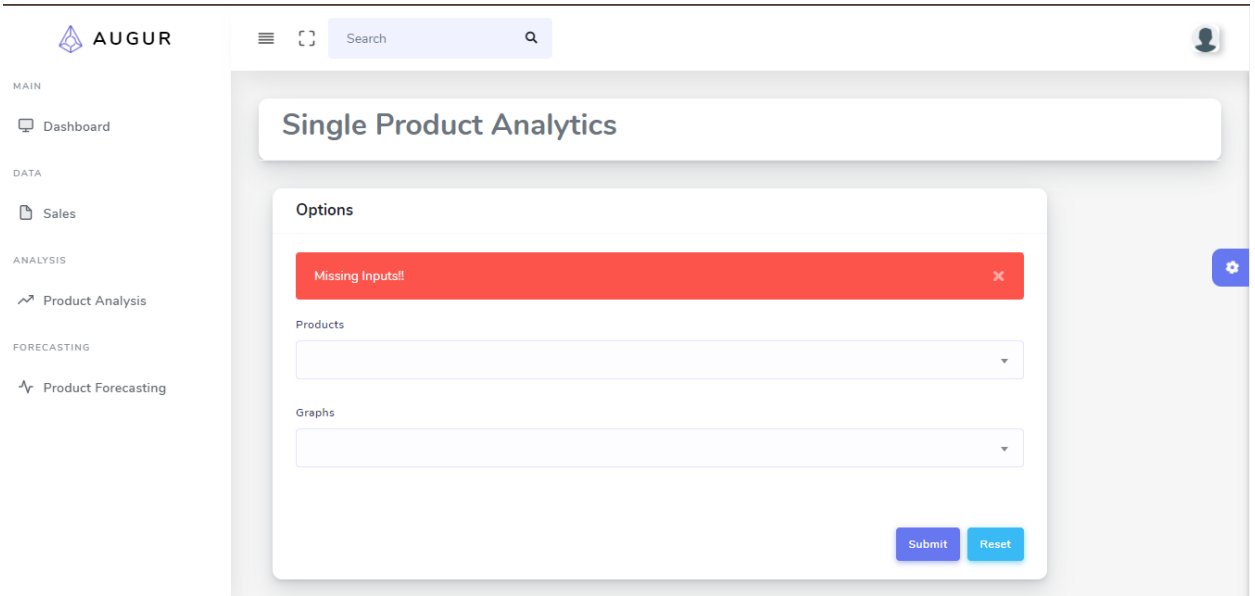


Figure 33: Failed single product analysis

ON SUCCESSFUL SINGLE PRODUCT ANALYSIS

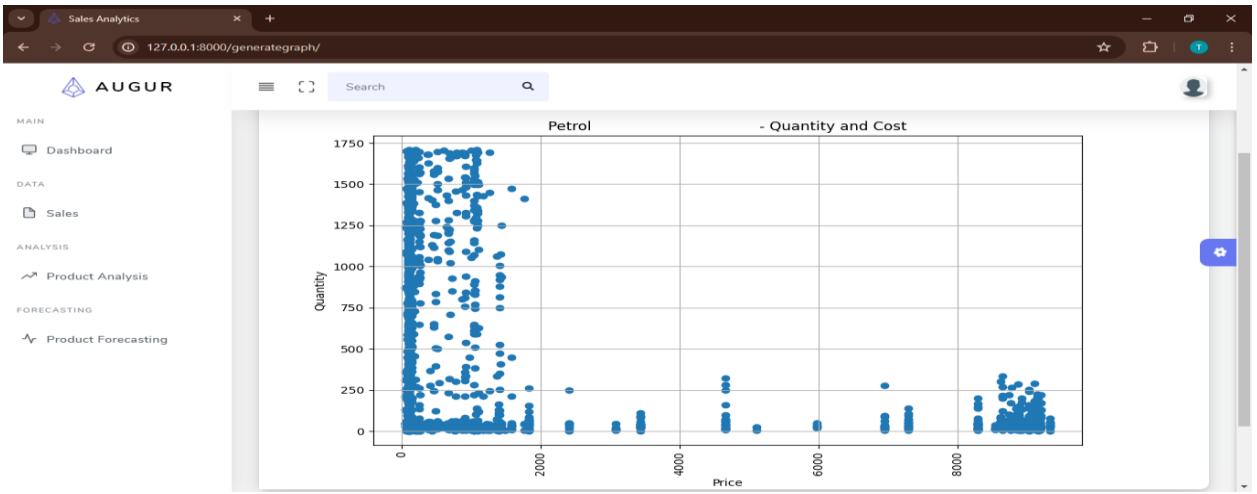


Figure 34: On successful single product analysis

5.3.3 PRODUCT FORECASTING

ON FAILED PRODUCT FORECASTING

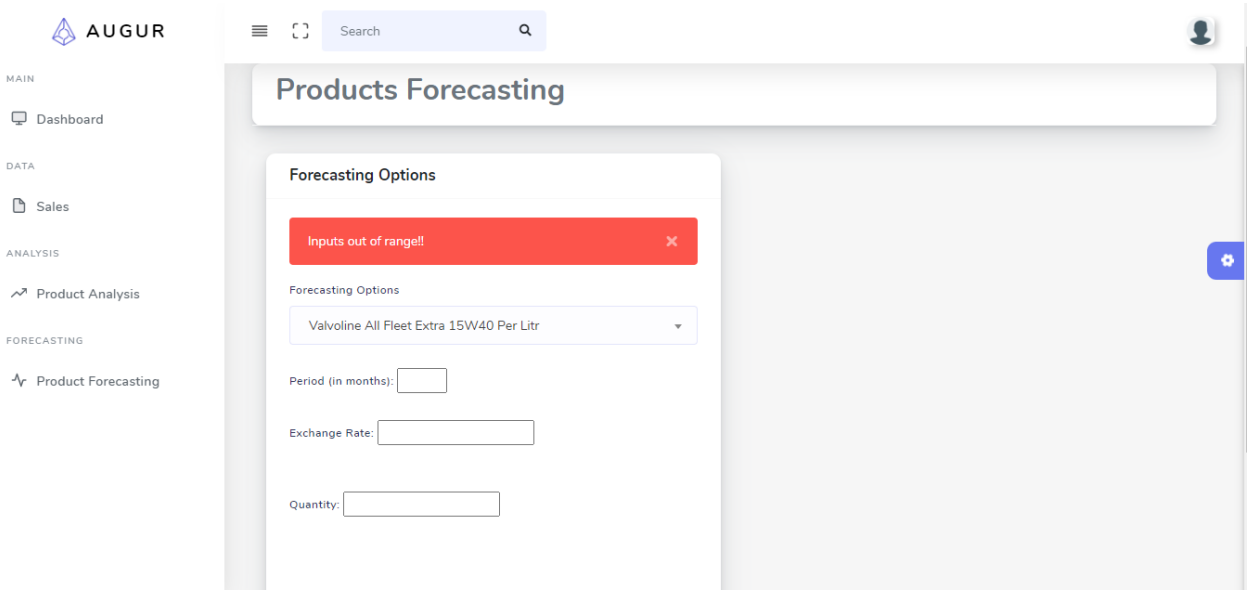


Figure 35: Failed product forecasting

ON SUCCESSFUL PRODUCT FORECASTING

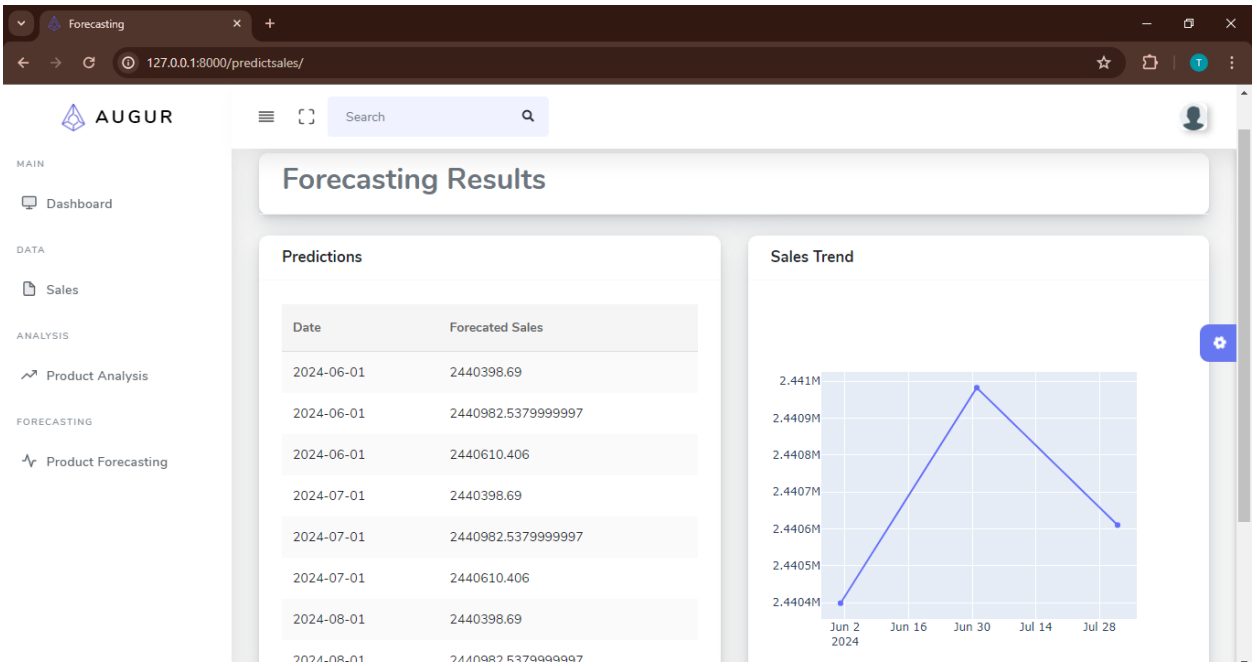


Figure 36: On successful product forecasting

5.2.3 ACCEPTANCE TESTING

RESPONSIVENESS OF USER INTERFACE

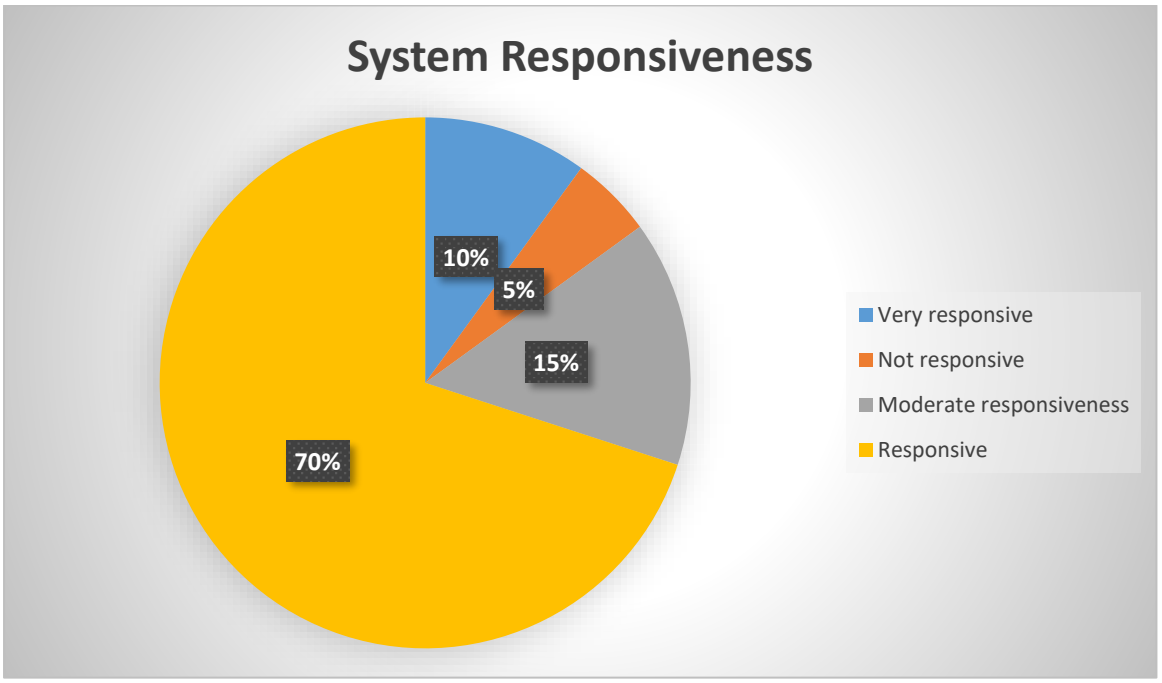


Figure 37: System responsiveness

The majority of the users found the system to be responsive while only a small proportion of the users had issues with the non-responsiveness of the system.

PERFORMANCE

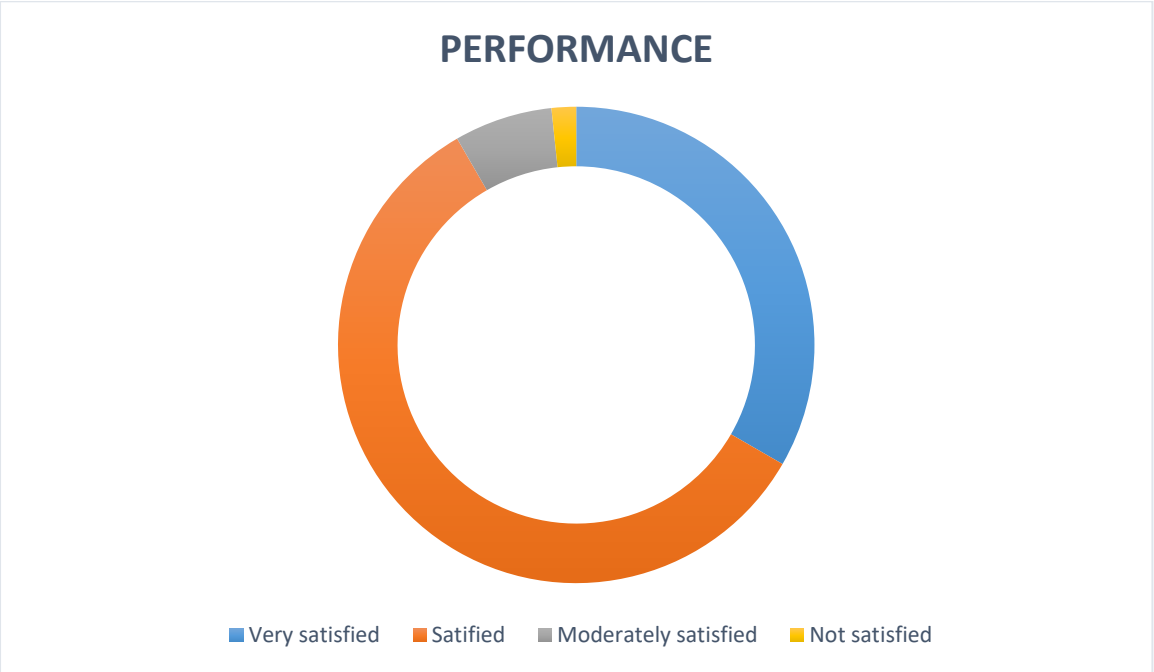


Figure 38: System Performance

The majority of the users where very satisfied with the speed of the system in carrying out tasks with only a very few users raising concerns about the performance of the system.

EASE OF USE

Figure 39: Ease of use of system

All the users gave feedback acknowledging that the system was very easy to use.

USER FRIENDLINESS

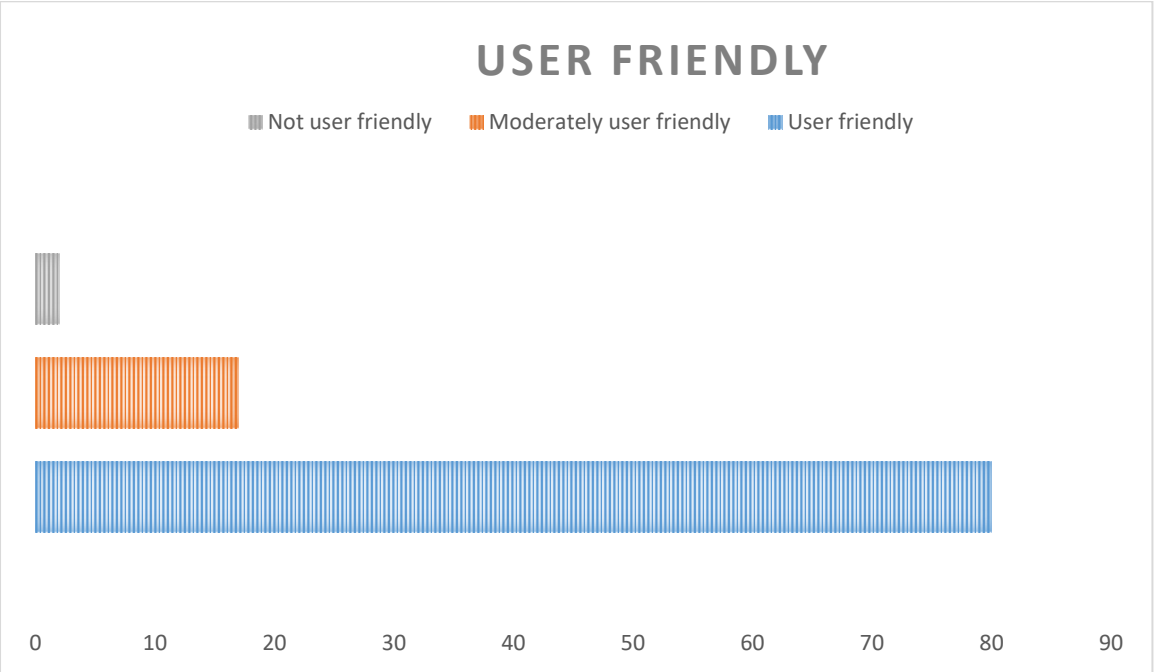


Figure 40: User friendliness of system

The majority of the users found the system to be user friendly while only a handful of the users found the system to not be user friendly.

In conclusion, the system was well accepted by the users in terms of its performance, user friendliness, responsiveness and ease of use with only a small percentage of users giving negative feedback about the performance and responsiveness of the system.

CHAPTER 6: CONCLUSIONS AND RECOMMENDATIONS

6.1 ANALYSIS OF RESULTS

EXCHANGE RATE DISTRIBUTION

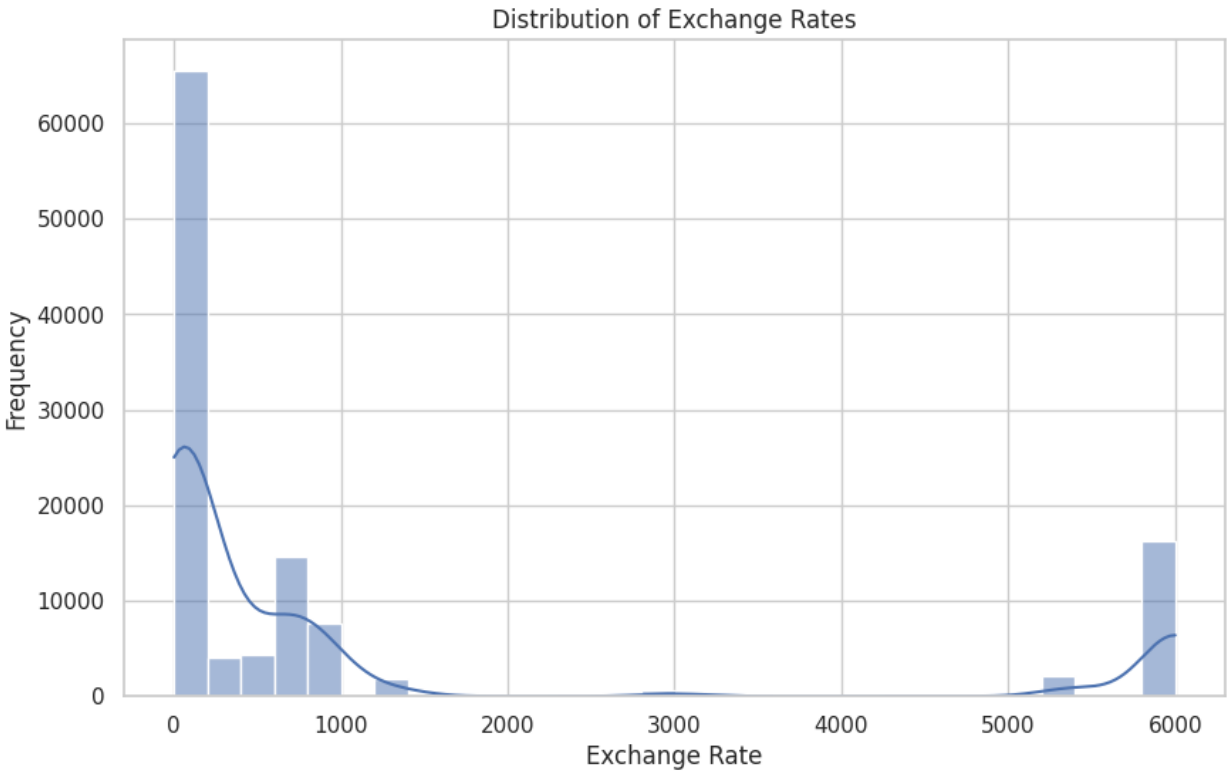


Figure 41: Exchange Rate Distribution

The above graph shows the distribution of the exchange rates of the RTGS currency as to the US Dollar currency. This seeks to establish the buying power of the RTGS currency and also to get an understanding of what sales values are very significant. A very high value of the RTGS value may be a small value of the US Dollar currency.

6.1.1 CONSUMABLES MASVINGO TOYOTA

Results for the training of the sales forecasting of Consumables Masvingo Toyota.

Actual Sales vs Predicted Sales

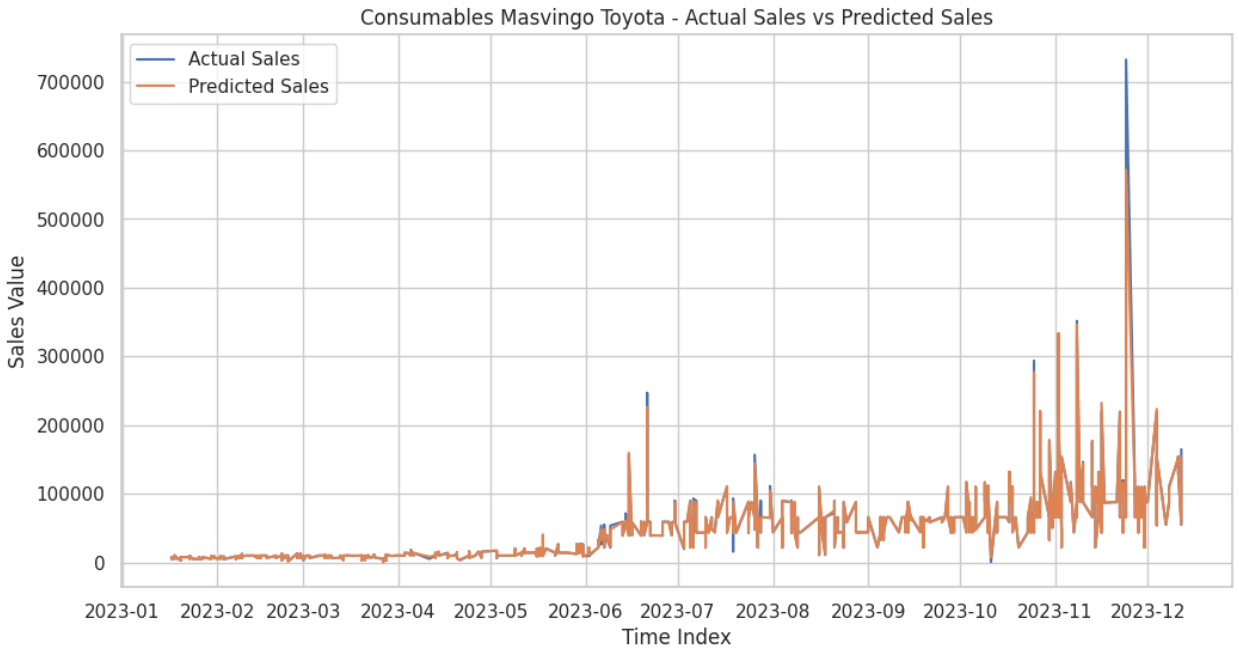


Figure 42: Actual sales vs Predicted sales for Consumables Masvingo Toyota

Performance Evaluation Metrics

Metric	Result
R-squared	0.9792098929916706
Mean Absolute Error (MAE)	1337.0251834112214
Root Mean Squared Error (RMSE)	8238.023535547341

Table 10: Evaluation metrics for Consumables Masvingo Toyota

The value of the R-Squared signifies high variance showing that almost 97.9% of the Consumables Masvingo Toyota sales can be explained by the model. This indicates that the model is accurate according to the R-Squared value.

The Mean Absolute Error (MAE) indicates that the average difference between the actual sales values and the predicted sales values is 1337.03 units. This suggests that the model is accurate as there's no huge difference between the predicted and the actual sales values.

The Root Mean Squared Error (RMSE) penalizes small differences between the predicted and the actual sales values and a very low value of 8238.02 units indicates that there's not much difference in the predicted and actual sales values.

Overally, the model was accurate in the forecasting of Consumables Masvingo Toyota sales values.

6.1.2 VALVOLINE

Results for the training of the sales forecasting of Valvoline.

Actual Sales vs Predicted Sales

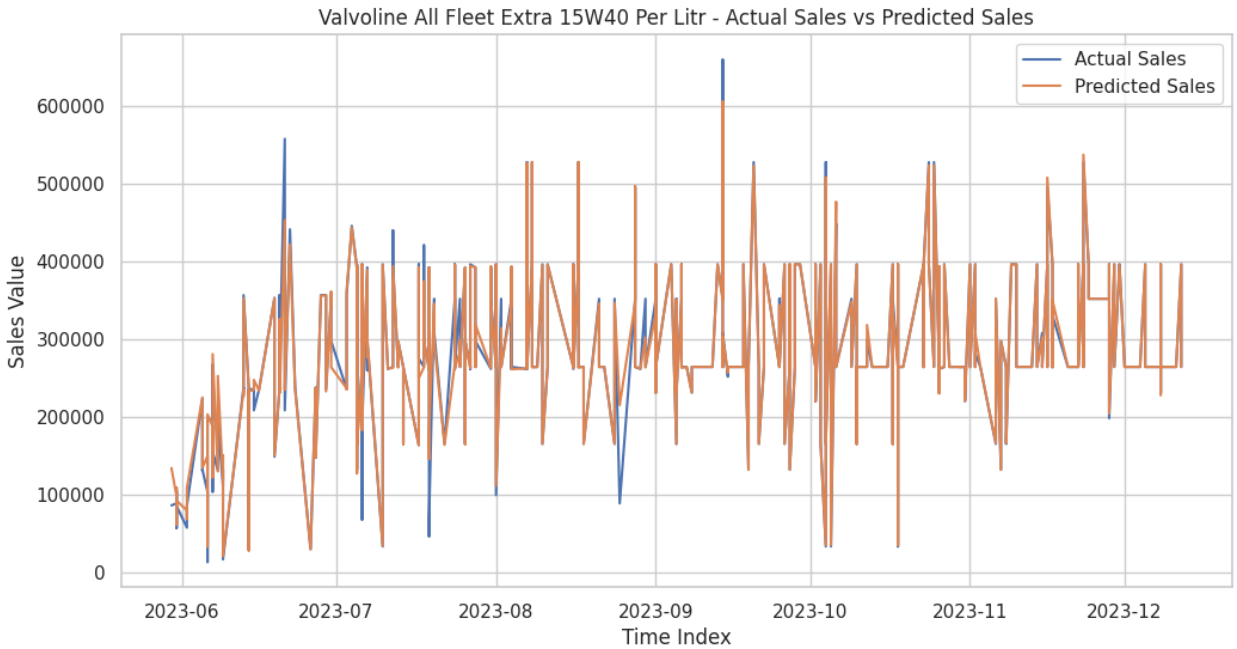


Figure 43: Actual sales vs Predicted sales for Valvoline

Performance Evaluation Metrics

Metric	Results
R-squared	0.9704573541969579
Mean Absolute Error (MAE)	5963.39436971438
Root Mean Squared Error (RMSE)	17334.74115945882

Table 11: Evaluation metrics for Valvoline

The value of the R-Squared signifies high variance showing that almost 97.04% of the Valvoline sales can be explained by the model. This indicates that the model is accurate according to the R-Squared value.

The Mean Absolute Error (MAE) indicates that the average difference between the actual sales values and the predicted sales values is 5963.39 units. This suggests that the model is accurate as there's no huge difference between the predicted and the actual sales values.

The Root Mean Squared Error (RMSE) penalizes small differences between the predicted and the actual sales values and a relatively low value of 17334.74 units indicates that there's a considerable difference in the predicted and actual sales values.

Overall, the model was accurate in the forecasting of Valvoline sales values.

6.1.3 ENGEN PREMIUM

Results for the training of the sales forecasting of Engen Premium.

Actual Sales vs Predicted Sales

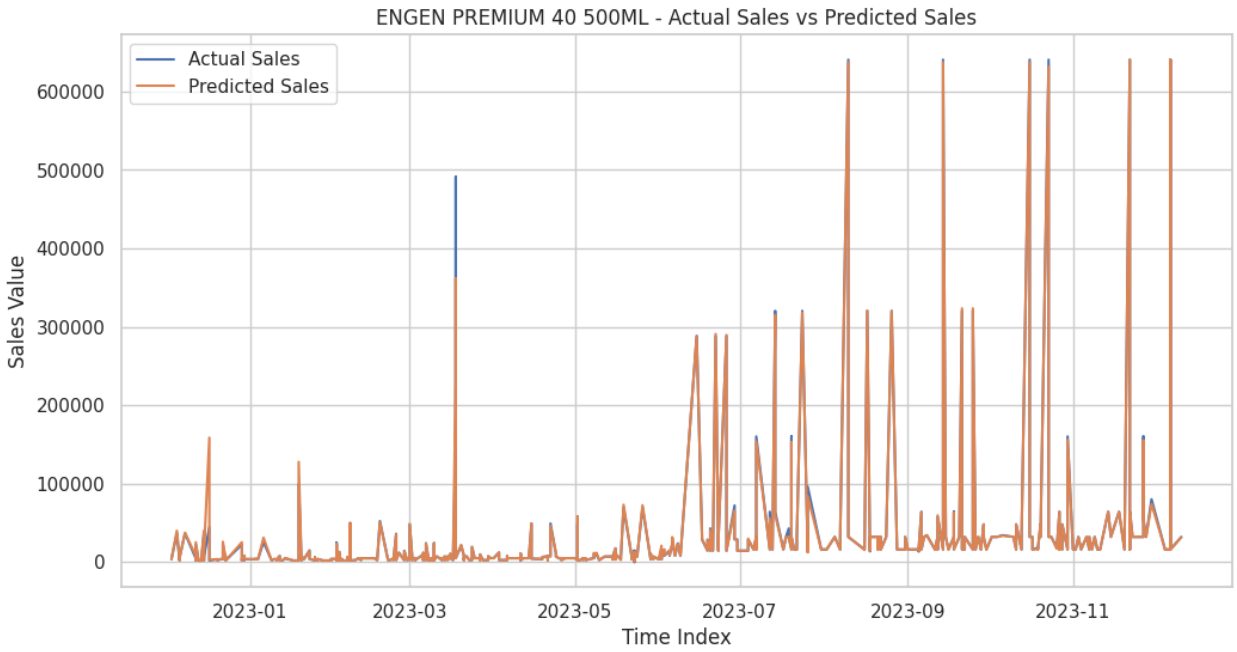


Figure 44: Actual Sales vs Predicted Sales for Engen Premium

Performance Evaluation Metrics

Evaluation Metric	Results
R-squared	0.9879823899128013
Mean Absolute Error (MAE)	1558.8108654110079
Root Mean Squared Error (RMSE)	11616.500714942213

Table 12: Evaluation metrics for Engen Premium

The value of the R-Squared signifies high variance showing that almost 98.8% of the Engen Premium sales can be explained by the model. This indicates that the model is accurate according to the R-Squared value.

The Mean Absolute Error (MAE) indicates that the average difference between the actual sales values and the predicted sales values is 1558.81 units. This suggests that the model is accurate as there's no huge difference between the predicted and the actual sales values.

The Root Mean Squared Error (RMSE) penalizes small differences between the predicted and the actual sales values and a relatively low value of 11616.50 units indicates that there's a considerable difference in the predicted and actual sales values.

Overall, the model was accurate in the forecasting of Engen Premium sales values.

6.1.4 ENGEN BRAKEFLUID

Results for the training of the sales forecasting of Engen Brake fluid.

Actual Sales vs Predicted Sales

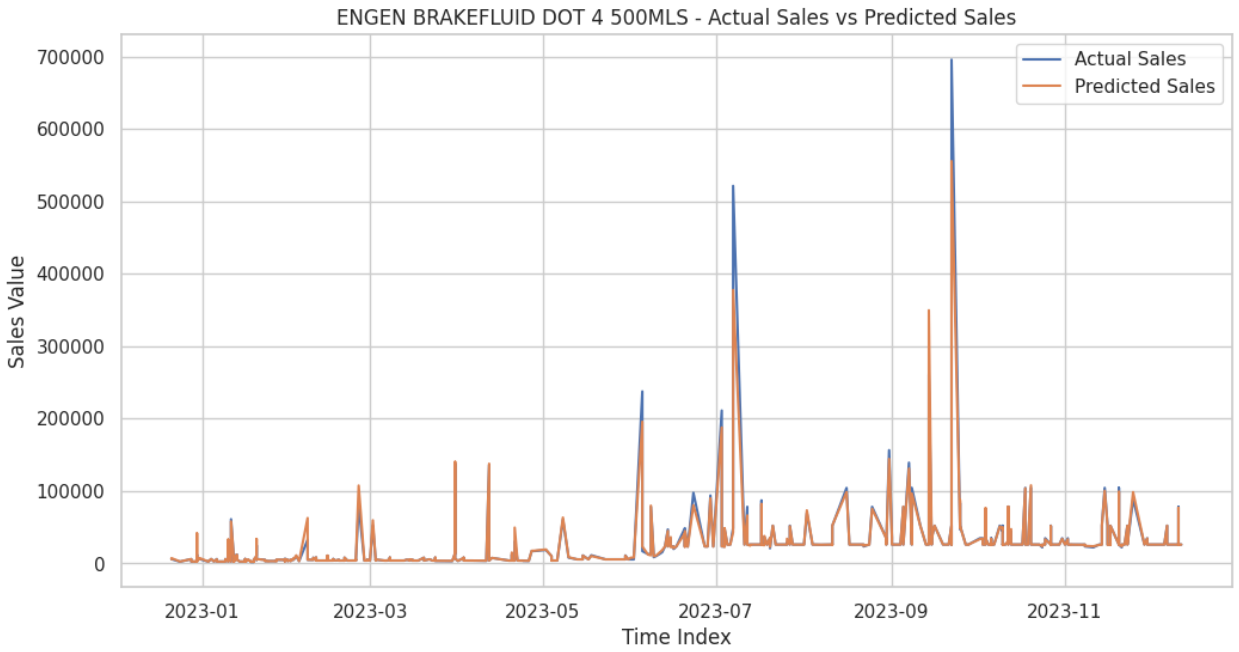


Figure 45: Actual Sales vs Predicted Sales for Engen Brake fluid

Performance Evaluation Metrics

Evaluation Metric	Results
R-squared	0.92001030989069
Mean Absolute Error (MAE)	3438.7315140893484
Root Mean Squared Error (RMSE)	16433.704513292636

Table 13: Evaluation metrics for Engen Brake fluid:

The value of the R-Squared signifies high variance showing that almost 92.0% of the Engen Brake fluid sales can be explained by the model. This indicates that the model is accurate according to the R-Squared value.

The Mean Absolute Error (MAE) indicates that the average difference between the actual sales values and the predicted sales values is 3438.73 units. This suggests that the model is accurate as there's no huge difference between the predicted and the actual sales values.

The Root Mean Squared Error (RMSE) penalizes small differences between the predicted and the actual sales values and a relatively low value of 16433.70 units indicates that there's a considerable difference in the predicted and actual sales values.

Overall, the model was accurate in the forecasting of Engen Brake fluid sales values.

6.1.5 DIESEL 50

Results for the training of the sales forecasting of Diesel 50.

Actual Sales vs Predicted Sales

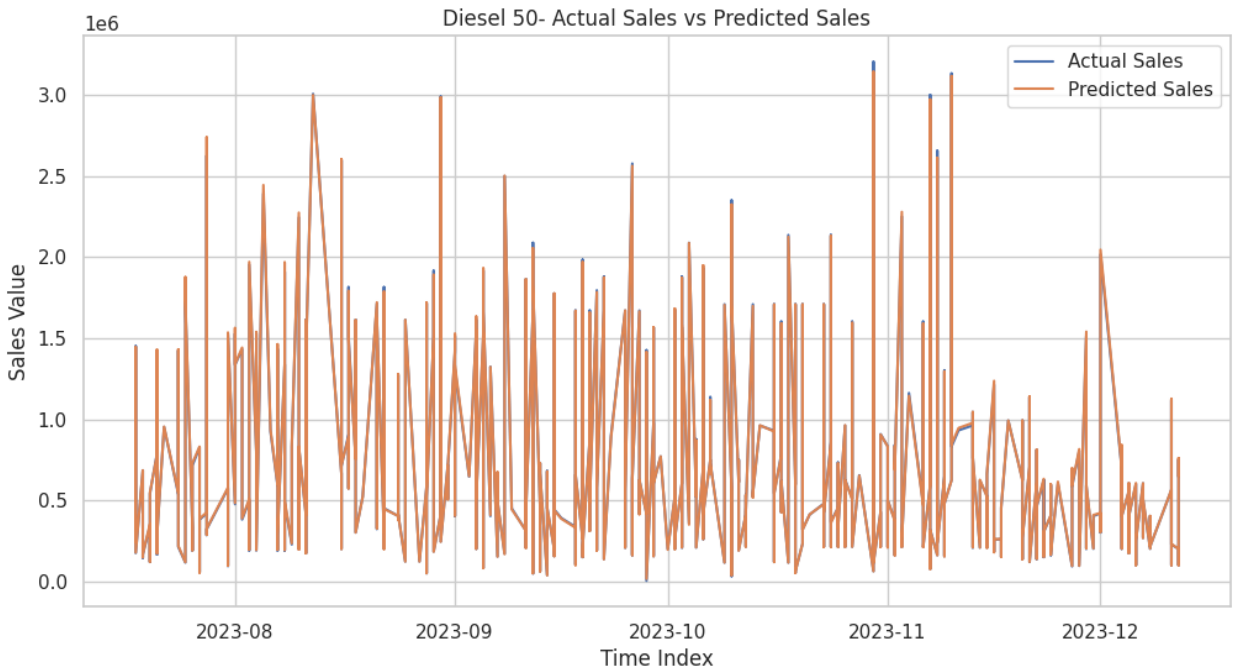


Figure 46: Actual Sales vs Predicted Sales for Diesel 50

Performance Evaluation Metrics

Evaluation Metric	Results
R-squared	0.9996737365324555
Mean Absolute Error (MAE)	5124.79497289158
Root Mean Squared Error (RMSE)	9859.15257943602

Table 14: Evaluation metrics for Diesel 50

The value of the R-Squared signifies high variance showing that almost 99.9% of the Diesel 50 sales can be explained by the model. This indicates that the model is accurate according to the R-Squared value.

The Mean Absolute Error (MAE) indicates that the average difference between the actual sales values and the predicted sales values is 5124.73 units. This suggests that the model is accurate as there's no huge difference between the predicted and the actual sales values.

The Root Mean Squared Error (RMSE) penalizes small differences between the predicted and the actual sales values and a relatively low value of 9859.15 units indicates that there's a considerable difference in the predicted and actual sales values.

Overall, the model was accurate in the forecasting of Diesel 50 sales values.

6.1.6 DEGREASER 750ML

Results for the training of the sales forecasting of Degreaser 750ML.

Actual Sales vs Predicted Sales

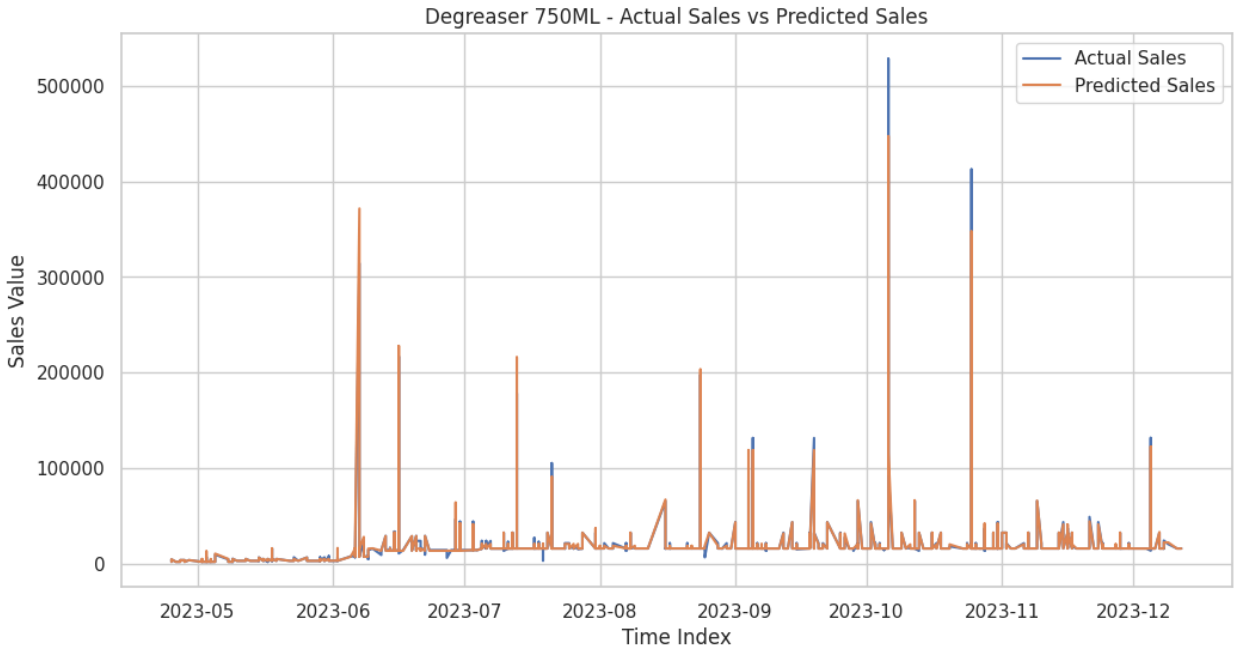


Figure 47: Actual Sales vs Predicted Sales for Degreaser 750ML

Performance Evaluation Metrics

Evaluation Metric	Results
R-squared	0.97171660456849
Mean Absolute Error (MAE)	796.6395550632978
Root Mean Squared Error (RMSE)	5101.363867328491

Table 15: : Evaluation metrics for Degreaser 750ML

The value of the R-Squared signifies high variance showing that almost 97.2% of the Degreaser 750ML sales can be explained by the model. This indicates that the model is accurate according to the R-Squared value.

The Mean Absolute Error (MAE) indicates that the average difference between the actual sales values and the predicted sales values is 796.63 units. This suggests that the model is accurate as there's no huge difference between the predicted and the actual sales values.

The Root Mean Squared Error (RMSE) penalizes small differences between the predicted and the actual sales values and a relatively low value of 5101.36 units indicates that there's a considerable difference in the predicted and actual sales values.

Overall, the model was accurate in the forecasting of Degreaser 750ML sales values.

6.2 RESULTS AND SUMMARY

From the results of the forecasting model provided above, it is evident that the stacked ensemble model utilizing the LightGBM and SARIMA algorithms as base learners and a meta-learner of Random Forest produced robust and accurate forecasts for the different products that were present in the dataset. The high variance values of around 90% indicates that the majority of the data in the dataset can be explained by the

model and also the relatively low Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) values indicate that there's a considerable difference between the predicted and the actual sales values thereby supporting the high accuracy and robustness of the model.

The completion of the web application and the integration of the stacked ensemble model produced significant results in the realization of the objectives of the system. The sales forecasting system was developed to help local retail businesses in Zimbabwe to make data-driven decisions on business related matters such as marketing strategies and inventory management. Some of the notable results of the sales forecasting system include the following: -

- Accurate forecasting of future product sales and trends
  The use of a robust stacked ensemble sales forecasting model enabled the prediction of future sales of products based on historical sales data. The stacked ensemble model's ability to capture complex relationships and trends within the data allowed for the prediction of future sales and trends of the products thereby giving insights into the profitability of the business in the future.
- Enhanced decision making
  The ability of the sales forecasting to predict future sales and sales trends can help businesses to make data-driven business decisions. Businesses will have insights into the future and they would be guided by the forecasts obtained to formulate marketing strategies and other important decisions on what measures to take to achieve high profitability figures.
- Enhanced inventory management
  The prediction of future product sales and product sales trends will enable businesses to have insights into the demand of the products in the future. This will allow businesses to determine the supply of the forecasted products. The business will have the ability to obtain the right supply of products for the right demand of the products determined from the sales forecasts.
- Enhanced current business understanding
  The analysis functionality of the system enables the deeper understanding of the current position of the sales of the company and other various aspects of the business. The system allows the analysis of individual products, warehouses, agents and overall sales enabling a deeper insight on the current trajectory of the business inasmuch as the sales of the business are concerned. This allows the business to make necessary measures in the present to keep the business profitable.

6.3 RECOMMENDATIONS

This section seeks to recommend additional methods and tasks to maintain the sales forecasting system.

- Hyperparameter tuning
  Performing thorough hyperparameter tuning for the LightGBM and SARIMA base models and also for the stacked ensemble model to optimize the performance of the model.
- Data preprocessing
  There needs to be continuous refinement of data preprocessing techniques such as removing missing values, removing duplicate entries, feature engineering and removing outliers in the data to improve the model's performance.
- Continuous monitoring
  The model's performance would require regular monitoring and also updating the model with new data to maintain the accuracy of the model in order to adapt to the changing market conditions.

6.4 FUTURE WORKS

The following future works are proposed in order to improve the accuracy of the sales forecasting model and also the to improve the sales forecasting system web application: -

- Real-time sales forecasting
  This involves creating sales forecasting models that can collect and analyze data in real-time thereby enabling businesses to react to market fluctuations, changes in the economic landscape, competitor advantages and changes in customer behavior more quickly enabling the business to always be ahead of the competitors.
- Explainable Artificial Intelligence
  This involves the development of a sales forecasting model that is interpretable. This enables the business to understand the rationale behind the predictions and help make informed decisions.
- Uncertainty quantification
  This involves implementing techniques to quantify the uncertainty of the model thereby providing a range of outcomes ensuring that the business is not misled by the system into making incorrect decisions.

REFERENCES

[1]. “Sales Prediction Analysis” by Naveem Kumar R et al. (2021)

[2]. “A Study On The Effect Of Sales Forecasting On The Enterprises” by Antonitte Vinoline, M. Deepika (2023).

[3]. “A Blending Model Combined DNN and LightGBM for Forecasting the Sales of Airline Tickets” by Xingyuan Tang et al. (n.d)

[4]." An Approach for Demand Forecasting in Steel Industries Using Ensemble Learning" by S. M. Taslim Uddin Raju et al. (2022)

[5]." Sales-forecasting of Retail Stores using Machine Learning Techniques" by Akshay Krishna et al. (2018)

[6]. “Machine-Learning Models for Sales Time Series Forecasting” by Bohdan M. Pavlyshenko (2018)

[7]. “Predicting sales using Machine Learning Techniques” by Sai Nikhil Boyapati and Ramesh Mummidi (2020)

[8]. “Spring onion seed demand forecasting using a hybrid Holt-Winters and support vector machine model” by Yihang Zhu et al. (2019).

[9]. “Predicting Walmart Sales, Exploratory Data Analysis, and Walmart Sales Dashboard” by Rashmi Jeswani (2021)

[10]. “Sales Forecasting of Truck Components using Neural networks” by Yeshwanth Reddy Gaddam (2020)

[11]. “Supply Chain Demand Forecasting and Price Optimisation Models with Substitution Effect” by Keun Hee Lee et al. (2023)

[12]. “Sales Forecasting Using Machine Learning Techniques” by Garud Akshada Anil et al. (2023)

[13]. “Inventory Mastery: Avoiding Overstocking and Understocking” by Mohammad Ali (2023)

[14]. “Development of a stacked ensemble model for forecasting and analyzing daily average $PM_{2.5}$ concentrations in Beijing, China” by Binxu Zhai and Jianguo Chen (2018)

[15] Exploratory Investigation of Sales Forecasting Process and Sales Forecasting System by Dnyanesh N. Sarang and Mohammad Laxmidhar (2006)

[16] Sales forecasting based on Ensemble Predictions by Dr. K. Alice et. al (n.d)

[17] Sales Prediction Using Machine Learning Algorithms by Purvika Bajaj et al (2020)

[18] Applying Machine Learning in Retail Demand Prediction—A Comparison of Tree-Based Ensembles and Long Short-Term Memory-Based Deep Learning by Mehran Nasseri et. al (2023)

[19] Inter Time Series Sales Forecasting by Mrs. Manisha Gahirwal and Vijayalakshmi M. (n.d)

[20] Time Series Sales Forecasting by James J. Pao and Danielle S. Sullivan (n.d)

[21] Sales Forecasting by J. Scott Armstrong (1999)

[22] Impact of Machine Learning on Product Sales Forecast by Besart Prebreza et. al (2023)

[23] Using Stacking Approaches for Machine Learning Models by Bohdan Pavlyshenko (2018)

[24] Hyperinflation and Its Impact on the Retail Sector in Zimbabwe by Godfrey Muchapreyi and Washington Mahiya (2019)

[25] Stacking Machine Learning Models to Forecast Hourly and Daily Electricity Consumption of Household Using Internet of Things by Alisha Banga et. al (2021)

[26] Forecasting Sales Using Various Time-Series and Machine Learning Methods by Ivar Padberg (2022)

[27] Stacking Model for Optimizing Subjective Well-Being Predictions Based on the CGSS Database by Na Ke et. al (2021)

APPENDICES

DATA COLLECTION METHODS

APPENDIX A

Interview Questions

1. How is sales forecasting done today?
2. Who is doing it?
3. Are there problems in the current forecasting?
4. Which tools are used?
5. Which data is used in forecasting?
6. Which timeframes are forecasted?
7. Are the forecasts reliable?
8. Are forecasts monitored and evaluated?

APPENDIX B

Questionnaire Questions

1. Business systems

- Do you have internal or external accounting in use? Are you using accounting data in forecasting?

……..

- Are you using Customer Relationship Management (CRM) in forecasting?

……..

- Are you using Enterprise Resource Planning (ERP) in forecasting?

…….

- Are you using sales tools data in forecasting?

…….

- Are you using Business Intelligence (BI) tools in forecasting?

…….

2. Sales Process and Systems

- Do you follow sales revenue and profitability per service or per product?

……..

- What kind of sales process, do you have in use?

……..

- How is market demand estimated? Which are the sources you are using?

……..

3. Sales Forecasting

- Are you doing sales forecasting? Who is doing forecasts? Are the current forecasts reliable?

……
……
……

- Which or what forecasting tools you are using?

……

- What kind of data are used in forecasting?

……

- Who needs and uses forecasts?

……

- Are there forecasts for the different time horizons, such as for less than a week, 1-3 months, 3-12 months, longer than 12 months. Who needs forecasts in a different time frame?

……
……

4. Business Challenges

- What are the current problems during the sales process?

……
……
……
……
……

- What are the current problems of sales forecasting?

……
……
……
……

5. Follow-up Questions

- What are the benefits of the improved sales forecasting?

……
……

……
……

- What kind of improvements in sales forecasting would you like to have?

……
……
……
……
……

SALES FORECASTING USING A STACKED ENSEMBLE MODEL

Takudzwa Trevor Ganda

Department of Software Engineering, School of Information Sciences and

Technology Harare Institute of Technology, Harare, Zimbabwe

h200148x@hit.ac.zw

***Abstract*— Zimbabwean retail businesses face a challenging economic landscape characterized by volatility and a heavy reliance on imports. Intuition-based planning methods prevalent in the sector often lead to significant losses due to overstocking or understocking. I propose a sales forecasting system that uses a stacked ensemble machine learning model to forecast future sales to address this challenge. The system leverages historical sales data to identify trends, seasonality, and patterns. Stacked ensemble models combine the strengths of multiple models, leading to improved generalizability and robustness. By providing accurate sales predictions, this system empowers businesses to anticipate demand, identify growth opportunities, and mitigate risks associated with economic volatility. This research contributes to the field of sales forecasting by demonstrating the effectiveness of stacked ensemble models in a real-world, economically volatile environment, particularly for Zimbabwean retail businesses.**

Keywords: sales forecasting, ensemble model, stacking, performance metrics

I. INTRODUCTION

The Zimbabwean retail industry despite its huge profitability in the country relies on instinct based decision making to plan market decisions to use. Despite its gross usage, it has its perks considering the currently volatile economic environment in the country which brings about unpredictability in the flow of the economic scale. As a result, many retail businesses suffer huge losses and even go bankrupt. To address this issue, this paper proposes a sales forecasting system using a stacked ensemble model to predict future sales based on historical sales data and the prevailing market and economic conditions. The ensemble models are trained on historical sales and data and the predictions from each of the algorithms are combined through stacking to produce a more robust forecasting model. The model is then integrated to a web application for interactive usability by the user. The web application then accepts user input for the current sales and features that affect sales then the system processes the input and forecasts future values for the specified duration.

Sales forecasting plays a critical role in the success of any retail business, allowing for informed decision-making regarding inventory management and marketing strategies [1]. Traditional forecasting methods based on intuition are inefficient in capturing complex relationships in sales data particularly in economic volatile markets thereby providing less accurate and less reliable predictions [12]. This is especially true for the Zimbabwean retail sector which faces challenges like import dependency and fluctuating exchange rates [3]. To address these limitations, advancements in machine learning techniques offer promising solutions.

Ensemble learning techniques which combine forecasts from multiple algorithms have emerged as powerful tools for improving the accuracy of sales forecasts [4]. Stacking which involves training a meta learner model on the forecasts from other base algorithms as new features has emerged as a very powerful method to improve robustness of the forecasts [5]. This approach leverages the strengths of various models to potentially capture diverse patterns within data and mitigate weaknesses of individual models [5].

In sales forecasting, ensemble models and stacking have gained popularity for their ability to generate more accurate and robust forecasts. By integrating various forecasting algorithms such as time series and regression, ensemble models can leverage the strengths of each approach to capture different aspects of the sales data.

Considering these developments, this paper proposes a sales forecasting system that uses a stacked ensemble model to forecast future sales and future sales trends accurately leveraging the power of ensemble models and the robustness of the models after stacking them.st. The web platform will provide an interactive platform for users to get valuable insights into the future using the power of machine learning which will help in effective decision making, formulating effective marketing strategies and effective inventory management.

II. PROBLEM STATEMENT

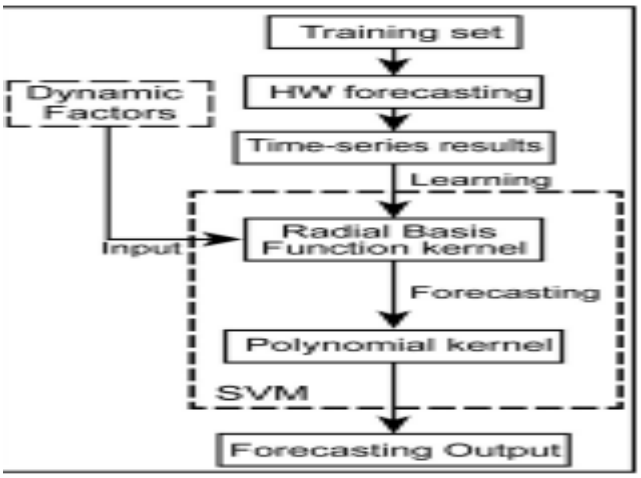
The Zimbabwean retail industry, a significant contributor to the national economy, suffers from substantial financial losses due to inefficiencies in inventory management and marketing strategies. These inefficiencies largely stem from a reliance on intuition-based decision making in a highly economically

volatile environment. Traditional planning methods fail to account for critical factors like import dependency, fluctuating exchange rates, inflation, complex relationships in data and seasonal trends. Failure to account for these factors results in overstocking which leads to global losses of $471.9 billion [13] or understocking with a global cost $34.1 billion [13], hindering profitability.

This highlights the critical need for a robust and accurate sales forecasting system specifically tailored to the Zimbabwean retail sector. Such a system would empower businesses to make informed decisions, mitigate risks associated with economic volatility, and achieve sustainable growth.

III. RELATED WORK

[8] presented a hybrid forecasting model combining the Holt-Winters and the Support Vector Machine algorithms to forecast the demand of spring onion seeds. The forecasting was based on dynamic factors including historical sales, seed inventory, price and weather [8]. The use of hybrid forecasting method led to more accurate predictions [1]. The model was designed to combine the strengths of the Holt-Winters algorithm and Support Vector Machines. The hybrid model was trained on a dataset of historical spring onion seed demand to learn to capture the patterns and relationships between input variables and the demand for spring onion seeds. An external factor was also added as a feature that would affect the demand of the spring onion seeds having a long-term influence on the demand which was the minimum absolute temperature. The



performance metric used was the forecasting error which is the difference between the actual and the forecasted demand [8].

Figure 1: The control flow diagram for the training of the Hybrid model using Holt-Winters and Support Vector Machines for spring onion seed demand forecasting.

Dr. K. Alice Et al. (2023) carried out a forecast on the Walmart dataset using XGBoost, LGBM and Catboost algorithms in order to compare their performances.

The implementation of a hybrid model using all three of the algorithms using a voting regressor was also carried out to combine the strengths of all the algorithms and help remove biases of individual models. The XGBoost algorithm was chosen because it is more flexible, scalable and could be customized to fit the best hyperparameters. LGBM was chosen because it was the best in terms of memory efficiency overall efficiency. Catboost was expected to have the best results overally. The performance metrics used included the MAE, RMSE and R-squared. The Voting Regressor was evaluated using the MAE.

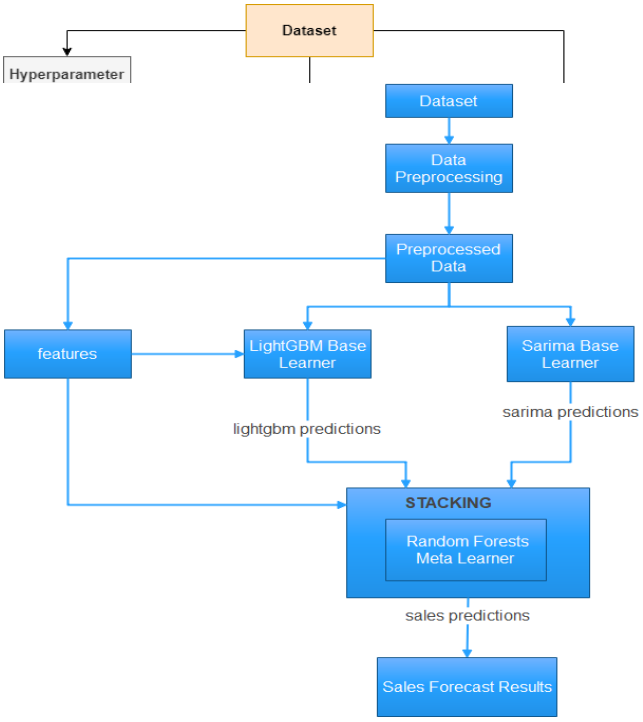


Figure 2: This shows the architecture of the hybrid ensemble model for the Walmart dataset.
Figure 3: This shows the architecture of the proposed stacked ensemble model.

IV. SOLUTION

To eradicate the problems faced by the retail industry in Zimbabwe in terms of making informed decisions on market strategies and improve their competitiveness in an economically

volatile environment, I propose the following objectives:

- To develop a web-based sales forecasting system that predicts future sales by utilizing a machine learning model trained on historical data.
- To develop a web-based sales forecasting system that predicts sales trends over a period of time using historical data.
- To develop a web-based sales forecasting system that analyzes sales data.

A. Features of the System

The proposed system will have the following features:

1. Web Dashboard platform: The system will have a web portal that would allow users to access the functionalities of the system.
2. Sales forecasting: The system will allow users to input forecasting details of products and generate future sales values and trends according to the time period they require.
3. Sales Analysis: The system will provide a sales analysis feature that will allow users to analyze various aspects of their sales information.

.

B. Benefits of the System

The proposed system offers the following benefits:

1. Enhanced decision making: Sales forecasts generated by the system will allow businesses to make informed decisions on marketing strategies based on future insight provided by the system.
2. Improved efficiency: The system will allow users to analyze large volumes of historical data and interpretability with the usage of data visualization tools
3. Competitive advantage: By leveraging the sales data analytics and accurate forecasting provided by the system, businesses can outperform competitors who rely on traditional prediction methods.

C. Solution Architecture

Client-Side: The web application is implemented using the Django framework.

Backend: The backend of the system is implemented using Python.

Model Integration: The stacked ensemble model for sales forecasting is integrated with the Django framework for usage through the web app as a pickle file.

V. RESULTS AND FUTURE WORKS

A. Results

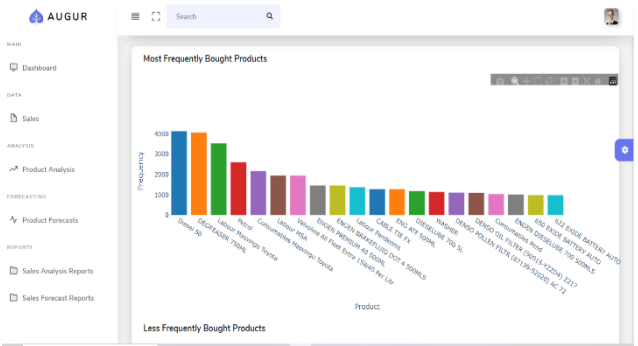


Figure 5: The interface analysis of sales data

TABLE 1: OBJECTIVES ACHIEVED

Objectives	Fully achieved	Partially achieved
To develop a web-based sales forecasting system that predicts future sales by utilizing machine learning model trained on historical data	✓	
To develop a web-based sales forecasting system that accurately forecasts sales trends over a period of time	✓	
To develop a web-based sales forecasting system that analyzes sales data	✓	

B. Future Works

Some potential areas of future work for the system could include:

1. Implementation of real-time data upload to the system to enhance forecasting accuracy and up to date data.
2. Integrating with CRM systems.
3. Usage of explainable AI to help users understand the rationale behind the sales forecasting results.

VI. CONCLUSION

In conclusion, the development of a web-based sales forecasting system using a stacked ensemble model holds immense potential for the Zimbabwean retail industry. Challenges faced by local businesses in optimizing inventory management and marketing strategies in an economically

volatile environment have highlighted the critical need for data-driven decision making. Traditional intuition-based planning methods have proved to lead to significant losses due to inefficiencies in addressing complexities of the market [12]. This system addresses the critical need for data driven insights by leveraging machine learning algorithms to generate robust and accurate sales forecasts. The system utilizes historical sales data, exchange rates and seasonality to come up with accurate sales forecasts. This integration of machine learning algorithms allows for the identification of patterns, complex relationships and seasonality in sales data which enables businesses to anticipate demand, identify growth opportunities and mitigate risks of inventory management effectively [14]. The development of a user-friendly web-application allows for the easy use of the system and the use of data visualization tools which would the easy interpretation of the results. Overally, this proposed sales forecasting system using a stacked ensemble model offers a promising solution for Zimbabwean retail businesses to thrive in an economically volatile and competitive environment.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who have contributed to the successful completion of this Sales forecasting project.

First and foremost, I would like to thank God Almighty, whose has helped me from the formulation of this idea through the development phase and the completion of this project.

I am also eternally grateful to my supervisor, Ms. Chibaya, for her priceless guidance, expert advice, and unwavering support throughout the project. Her dedication and profound insights have been instrumental to this project since the very start. She has been the light throughout the whole project showing all the loose threads, points to improve and even mental support to put me in the right attitude to carry out the project.

Furthermore, I would like to express my gratitude to the staff of the Software Engineering department for their assistance, reviews and insights throughout the duration of the project. The department's resources and facilities have been crucial in carrying out the research and development tasks.

I would like to thank my dear parents for fighting for me to have a quality education and also their encouragement and motivation. I would also like to thank the rest of my family and friends for their unwavering support, encouragement, and love. Their constant belief in my abilities has been a driving force in overcoming challenges and achieving success.

Finally, I would like to thank all the participants who volunteered their time and efforts to provide valuable data for this project. Their cooperation and contribution have been instrumental in the successful completion of this project.

REFERENCES

[1]. “Sales Prediction Analysis” by Naveem Kumar R et al. (2021)

[2]. “A Study On The Effect Of Sales Forecasting On The Enterprises” by Antonitte Vinoline, M. Deepika (2023).

[3]. The World Bank, "Zimbabwe Overview", https://www.worldbank.org/en/country/zimbabwe)

[4]. S. Lessmann, C. Stahlbock, "Ensemble learning for sales forecasting", International Journal of Forecasting, Volume 20, Issue 4, 2005, Pages 503-514, https://www.sciencedirect.com/science/article/abs/pii/S0968090X21003594

[5]. Zhou Zhi-hua, "Ensemble Methods: Foundations and Algorithms", Machine Learning and Cybernetics, 2012, 21st International Conference on, Volume 1, Pages 1-14, doi: 10.1109/ICMLC.2012.6323000

[6] Anandous K. El-Said, Yang Gao, Hsin-Yu Feng, "A survey of ensemble methods for water quality prediction", Environmental Modelling & Software, Volume 112, 2019, Pages 243-270,

[6]. “Machine-Learning Models for Sales Time Series Forecasting” by Bohdan M. Pavlyshenko (2018)

[7]. “Predicting sales using Machine Learning Techniques” by Sai Nikhil Boyapati and Ramesh Mummidi (2020)

[8]. “Spring onion seed demand forecasting using a hybrid Holt-Winters and support vector machine model” by Yihang Zhu et al. (2019).

[9]. “Predicting Walmart Sales, Exploratory Data Analysis, and Walmart Sales Dashboard” by Rashmi Jeswani (2021)

[10]. “Sales Forecasting of Truck Components using Neural networks” by Yeshwanth Reddy Gaddam (2020)

[11]. “Supply Chain Demand Forecasting and Price Optimisation Models with Substitution Effect” by Keun Hee Lee et al. (2023)

[12]. “Sales Forecasting Using Machine Learning Techniques” by Garud Akshada Anil et al. (2023)

[13]. “Inventory Mastery: Avoiding Overstocking and Understocking” by Mohammad Ali (2023)